

STARK POINTS AND p -ADIC ITERATED INTEGRALS ATTACHED TO MODULAR FORMS OF WEIGHT ONE

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To our families

ABSTRACT. Let E be an elliptic curve over $\mathbb{Q}$ and let $\varrho_{\flat}$ and $\varrho_{\sharp}$ be odd two-dimensional Artin representations for which $\varrho_{\flat} \otimes \varrho_{\sharp}$ is self-dual. The progress on modularity achieved in the last decades ensures the existence of normalised eigenforms f , g and h of respective weights 2, 1 and 1, giving rise to E , $\varrho_{\flat}$, and $\varrho_{\sharp}$ via the constructions of Eichler-Shimura and Deligne-Serre. When the Hasse-Weil-Artin L -series $L(E, \varrho_{\flat} \otimes \varrho_{\sharp}, s)$ has a double zero at the center, this article examines certain p -adic iterated integrals attached to the triple (f, g, h) , and proposes a formula for their value involving the formal group logarithms of global points on E —referred to as *Stark points*—which are defined over the number field cut out by $\varrho_{\flat} \otimes \varrho_{\sharp}$. This formula can be viewed as an elliptic curve analogue of Stark’s conjecture on units attached to weight one forms. It is proved when g and h are binary theta series attached to a common imaginary quadratic field in which p splits, by relating the arithmetic quantities that arise in it to elliptic units and Heegner points. Fast algorithms for computing p -adic iterated integrals based on Katz expansions of overconvergent modular forms are then exploited to gather numerical evidence in more exotic scenarios, encompassing Mordell-Weil groups over cyclotomic fields, ring class fields of real quadratic fields (a setting which may shed light on the theory of Stark-Heegner points attached to Shintani-type cycles on $\mathcal{H}_p \times \mathcal{H}$), and extensions of $\mathbb{Q}$ with Galois group a central extension of the dihedral group D_{2n} or of one of the exceptional subgroups A_4 , S_4 , and A_5 of $\mathbf{PGL}_2(\mathbb{C})$.

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INTRODUCTION

The purpose of this article is to formulate an analogue of the Stark conjecture on units in number fields in the setting of global points on elliptic curves. We prove this “elliptic Stark conjecture” in some cases and provide numerical evidence in its support in many others. It is offered as a more constructive alternative to the Birch and Swinnerton-Dyer (BSD) conjecture, since it often allows the efficient analytic computation of p -adic logarithms of global points. In contrast, the BSD conjecture (and its p -adic variant formulated by Mazur, Tate and Teitelbaum) gives the $(p$ -adic) heights of such points, from which it appears computationally more difficult to recover the points themselves [Si]. Our conjecture gives a unifying

perspective on the two currently known constructions of global points on elliptic curves over $\mathbb{Q}$ —Heegner points and the still conjectural Stark-Heegner points attached to real quadratic cycles on $\mathcal{H}_p \times \mathcal{H}$ —and suggests a possible global underpinning for the latter. It also yields conjectural constructions and explicit formulae, in situations of rank one and two, for global points over cyclotomic fields, abelian extensions of quadratic fields which are not necessarily anticyclotomic, and extensions of $\mathbb{Q}$ with Galois group a central extension of A_4 , S_4 or A_5 .

Let E be an elliptic curve over $\mathbb{Q}$ and let

$$\varrho : G_{\mathbb{Q}} \longrightarrow \mathrm{Aut}(V_{\varrho}) \simeq \mathbf{GL}_n(L), \quad \dim_L(V_{\varrho}) = n \geq 1$$

be an Artin representation—i.e., a continuous Galois representation factoring through a finite quotient $\mathrm{Gal}(H/\mathbb{Q})$ of $G_{\mathbb{Q}}$ —with coefficients in some finite extension $L \subset \mathbb{C}$ of $\mathbb{Q}$. The *Hasse-Weil-Artin L -series* $L(E, \varrho, s)$ of E twisted by ϱ is defined on the right half-plane $\mathrm{Re}(s) > 3/2$ by an absolutely convergent Euler product of degree $2n$ and is expected to admit an analytic continuation to the whole complex plane. Under this assumption, the *analytic rank* of the pair (E, ϱ) is defined to be

$$r_{\mathrm{an}}(E, \varrho) := \mathrm{ord}_{s=1} L(E, \varrho, s).$$

A natural Galois equivariant refinement of the Birch and Swinnerton-Dyer conjecture predicts that

$$r_{\mathrm{an}}(E, \varrho) \stackrel{?}{=} r(E, \varrho), \tag{1}$$

where $r(E, \varrho)$ is the *multiplicity* of V_{ϱ} as a subrepresentation of $E(H) \otimes L$ when ϱ is irreducible, defined for general ϱ by

$$r(E, \varrho) := \dim_L \mathrm{Hom}_{G_{\mathbb{Q}}}(V_{\varrho}, E(H) \otimes L).$$

Equality (1) has been proved when $r_{\mathrm{an}}(E, \varrho) = 0$ and (in roughly chronological order)

- I. ϱ is the odd self-dual two-dimensional Galois representation induced from a ring class character of an imaginary quadratic field, by a series of works building on the breakthroughs of Gross-Zagier and Kolyvagin;
- II. ϱ is a Dirichlet character, by the work of Kato [Kato];
- III. ϱ is an odd irreducible two-dimensional Galois representation satisfying some mild restrictions, by the ongoing work [BDR2];
- IV. ϱ is an irreducible constituent of the tensor product $\varrho_b \otimes \varrho_{\sharp}$ of two odd irreducible two-dimensional Galois representations which is self-dual and satisfies some other mild restrictions, by [DR2].

The setting where $r_{\mathrm{an}}(E, \varrho) > 0$ raises the challenge of constructing non-torsion elements in

$$E(H)_L^{\varrho} := \sum_{\varphi} \varphi(V_{\varrho}), \quad \text{where } \varphi \text{ runs over a basis of } \mathrm{Hom}_{G_{\mathbb{Q}}}(V_{\varrho}, E(H) \otimes L).$$

Devising strategies for constructing such global points is arguably the most fundamental issue surrounding the Birch and Swinnerton-Dyer conjecture. Yet very little is known in this direction beyond the setting alluded to in item I above, where the methods of Gross, Zagier and Kolyvagin can also be exploited to establish (1) when $r_{\mathrm{an}}(E, \varrho) = 1$, by using Heegner points (as recalled in Ch. 3 below) to construct the predicted non-trivial elements of $E(H)_L^{\varrho}$.

As explained in the survey [BCDDPR], the techniques used to analyze settings II, III and IV can be organised under the common rubric of “Euler systems of Garrett-Rankin-Selberg type”. The present work considers the same settings, but in the less well-understood scenario of “analytic rank ≤ 2 .” More precisely, it will be assumed that ϱ is an irreducible constituent of the tensor product $\varrho_b \otimes \varrho_{\sharp}$, where ϱ_b and $\varrho_{\sharp}$ are odd, two-dimensional, but not necessarily irreducible, Artin representations satisfying the self-duality condition

$$\chi := \det(\varrho_b)^{-1} = \det(\varrho_{\sharp}),$$

and that $r_{\text{an}}(E, \varrho) = 1$ or 2 . The primary goal of this paper is to propose a conjectural p -adic analytic formula relating global points in $E(H)_L^{\varrho}$ to “ p -adic iterated integrals” attached to an appropriate triple of modular forms, and to prove it in some cases.

The following general notations and background are required to formulate the “elliptic Stark conjecture” (Conjecture ES) below. For any positive integer N , any Dirichlet character $\eta : (\mathbb{Z}/N\mathbb{Z})^{\times} \rightarrow \bar{\mathbb{Q}}^{\times}$ and any field F , write $S_k(N, \eta)_F \subseteq M_k(N, \eta)_F$ for the spaces of classical cusp forms and modular forms of weight k , level N and character η with Fourier coefficients in F . (When η is suppressed from the notation, it is to be understood that η is the trivial character.)

Thanks to Wiles’ proof of the Shimura-Taniyama conjecture, the elliptic curve E is known to be *modular*: there is a normalised weight two newform $f \in S_2(N_f)_{\mathbb{Q}}$, where N_f is the conductor of E , satisfying

$$L(f, s) = L(E, s).$$

The work of Khare and Wintenberger [KW] on Serre’s modularity conjecture likewise implies that any odd, two-dimensional Artin representation arises from a classical (cuspidal or Eisenstein) normalized newform of weight one. More precisely, denoting by N_g and N_h the Artin conductors of ϱ_g and ϱ_h , there are weight one newforms $g \in M_1(N_g, \chi^{-1})_L$ and $h \in M_1(N_h, \chi)_L$ satisfying

$$L(g, s) = L(\varrho_g, s), \quad L(h, s) = L(\varrho_h, s).$$

Henceforth, it will be convenient to use ϱ_g and ϱ_h to designate ϱ_g and ϱ_h , to write

$$\varrho_{gh} := \varrho_g \otimes \varrho_h : \text{Gal}(H/\mathbb{Q}) \longrightarrow \mathbf{SL}_4(L),$$

and to denote by V_g , V_h and V_{gh} the L -vector spaces underlying ϱ_g , ϱ_h and ϱ_{gh} , respectively.

When either g or h is an Eisenstein series, the Rankin-Selberg method yields the analytic continuation and functional equation for the L -function $L(E, \varrho_{gh}, s) = L(f \otimes g \otimes h, s)$ relating its values at s and $2 - s$. This has been extended to the general case where both g and h are cusp forms, by a result of Garrett. The self-duality condition on ϱ_{gh} implies that the root number $\varepsilon(E, \varrho_{gh})$ that appears in this functional equation is ± 1 , and determines the parity of $r_{\text{an}}(E, \varrho_{gh})$. Furthermore, it can be written as a product of local root numbers

$$\varepsilon(E, \varrho_{gh}) = \prod_{v|N_{fgh} \cdot \infty} \varepsilon_v(E, \varrho_{gh}), \quad (\text{where } N_{fgh} := \text{lcm}(N_f, N_g, N_h)),$$

and the local sign $\varepsilon_{\infty}(E, \varrho_{gh})$ is always equal to 1. The following hypothesis on the local root numbers $\varepsilon_v(E, \varrho_{gh})$ at the non-archimedean places, which is discussed in further detail in Ch. 2, shall be made throughout this article:

Hypothesis A. (Local sign hypothesis) For all finite places $v|N_{fgh}$, the local signs $\varepsilon_v(E, \varrho_{gh})$ are equal to $+1$. In particular, the same is true of the global root number, and hence $L(E, \varrho_{gh}, s)$ vanishes to *even order* at $s = 1$.

Under Hypothesis A, the method of Garrett-Rankin-Selberg relates the central critical value $L(f \otimes g \otimes h, 1)$ to the values of the trilinear form

$$I : S_2(N_{fgh})_{\mathbb{C}} \times M_1(N_{fgh}, \chi^{-1})_{\mathbb{C}} \times M_1(N_{fgh}, \chi)_{\mathbb{C}} \longrightarrow \mathbb{C}$$

defined by

$$I(\check{f}, \check{g}, \check{h}) := \langle \check{f}, \check{g}\check{h} \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the Petersson scalar product on $S_2(N_{fgh})_{\mathbb{C}} \times M_2(N_{fgh})_{\mathbb{C}}$. More precisely, for any integer N , let $\mathbb{T}_N$ denote the Hecke algebra generated by the good Hecke operators T_{ℓ} with $\ell \nmid N$. If M is any $\mathbb{T}_N$ -module and $\phi \in M$ is a simultaneous eigenvector for the action

of $\mathbb{T}_N$, satisfying $T_\ell \phi = a_\ell(\phi)\phi$ for all $\ell \nmid N$, let $I_\phi \subset \mathbb{T}_N$ be the ideal associated to this system of Hecke eigenvalues and denote by

$$M[\phi] = \text{Ker}(I_\phi) = \{m \in M : T_\ell m = a_\ell(\phi)m \text{ for all } \ell \nmid N\} \quad (2)$$

the corresponding eigenspace. By the work [HaKu] of Harris-Kudla, the restriction

$$I_{fgh} : S_2(N_{fgh})[f] \times M_1(N_{fgh}, \chi^{-1})[g] \times M_1(N_{fgh}, \chi)[h] \longrightarrow \mathbb{C} \quad (3)$$

of I to the product of isotypic components for f , g and h is identically zero if and only if the central critical value $L(f \otimes g \otimes h, 1)$ vanishes. Since this article is solely interested in the setting where ϱ_{gh} admits irreducible constituents ϱ for which $L(E, \varrho, 1) = 0$, the following hypothesis is also made throughout:

Hypothesis B. (Global vanishing hypothesis). The L -function $L(E, \varrho_{gh}, s)$ vanishes at $s = 1$, and hence the trilinear form I_{fgh} in (3) is identically zero.

Under Hypotheses A and B, the L -function $L(E, \varrho_{gh}, s)$ vanishes to order at least 2 at $s = 1$. When $L''(E, \varrho_{gh}, 1) \neq 0$, the elliptic Stark conjecture of this article expresses the values of certain p -adic iterated integrals attached to the triple (f, g, h) in terms of the formal group logarithms of global points in the ϱ_{gh} -isotypic subspace $E(H)_L^{\varrho_{gh}}$ of $E(H) \otimes L$.

The definition of these p -adic iterated integrals is based on the theory of p -adic and overconvergent modular forms. Let p be a prime number and let $\mathbb{C}_p$ denote the completion of $\bar{\mathbb{Q}}_p$. Assume that p is chosen in such a way that

$$\text{ord}_p(N_f) \leq 1 \text{ and } p \nmid N_g N_h, \text{ and set } N := \text{the prime-to-} p \text{ part of } N_{fgh}. \quad (4)$$

(This hypothesis is made primarily for ease of exposition, and would eventually be interesting to relax.)

The symbols $S_k^{(p)}(N, \chi) \subset M_k^{(p)}(N, \chi)$ refer to the spaces of p -adic cusp forms (resp. modular forms) of weight k , level N and character χ with coefficients in $\mathbb{C}_p$, while $S_k^{\text{oc}}(N, \chi) \subset M_k^{\text{oc}}(N, \chi)$ stand for the corresponding subspaces of overconvergent modular forms. The relations between these various spaces are summarised in the following sequence of natural inclusions of $\mathbb{C}_p$ -vector spaces

$$\begin{array}{ccccc} S_k(Np, \chi)_{\mathbb{C}_p} & \hookrightarrow & S_k^{\text{oc}}(N, \chi) & \hookrightarrow & S_k^{(p)}(N, \chi) \\ \downarrow & & \downarrow & & \downarrow \\ M_k(Np, \chi)_{\mathbb{C}_p} & \hookrightarrow & M_k^{\text{oc}}(N, \chi) & \hookrightarrow & M_k^{(p)}(N, \chi). \end{array}$$

These spaces are all endowed with an action of the Hecke algebra $\mathbb{T}_N^{(p)}$ generated by the Hecke operators T_ℓ with $\ell \nmid Np$ and by the Hecke operator U_p . Let $e_{\text{ord}} := \lim_n U_p^{n!}$ be Hida's ordinary idempotent and write $M^{\text{ord}} = e_{\text{ord}} \cdot M$, for any $L_p[U_p]$ -module M . While the spaces $S_k^{\text{oc}}(N, \chi)$ and $M_k^{\text{oc}}(N, \chi)$ are infinite-dimensional over $\mathbb{C}_p$, their ordinary parts $S_k^{\text{oc,ord}}(N, \chi)$ and $M_k^{\text{oc,ord}}(N, \chi)$ are known to be finite-dimensional. In fact, Coleman's classicality theorem asserts that, for all $k \geq 2$, the natural inclusion

$$M_k^{\text{ord}}(Np, \chi)_{\mathbb{C}_p} \hookrightarrow M_k^{\text{oc,ord}}(N, \chi)$$

is an isomorphism.

Serre's differential operator

$$d = q \frac{d}{dq} : M_k^{(p)}(N, \chi) \longrightarrow M_{k+2}^{(p)}(N, \chi)$$

on p -adic modular forms sends overconvergent forms of weight 0 to overconvergent forms of weight 2. For any $\check{f} \in S_2^{\text{oc}}(N)$, the overconvergent modular function

$$\check{F} := d^{-1}\check{f} := \lim_{t \rightarrow -1} d^t \check{f} \in S_0^{\text{oc}}(N)$$

is called the *overconvergent primitive* of $\check{f}$. The Fourier expansions of $\check{f}$ and $\check{F}$ are related by

$$\check{f}(q) = \sum_{n \geq 1} a_n(\check{f})q^n, \quad \check{F}(q) = \sum_{p \nmid n} \frac{a_n(\check{f})}{n} q^n.$$

For any $k \geq 2$, the p -adic iterated integral attached to the data

$$(\check{f}, \check{\gamma}, \check{h}) \in S_2(Np)_L \times M_k^{\text{ord}}(Np, \chi)_L^\vee \times M_k(Np, \chi)_L$$

(in which the superscript $\vee$ denotes the L -linear dual) is defined to be

$$\int_{\check{\gamma}} \check{f} \cdot \check{h} := \check{\gamma}(e_{\text{ord}}(\check{F} \cdot \check{h})) \in \mathbb{C}_p. \quad (5)$$

Expressions of this sort with $k \geq 2$ play an important role in [DR1] and [DR2] where they are related (when $\check{f}$, $\check{\gamma}$ and $\check{h}$ are cuspidal eigenvectors) to the special values of Garrett-Hida triple product p -adic L -functions and to the images of certain *generalised Gross-Schoen diagonal cycles* under the p -adic Abel-Jacobi map. When $\check{\gamma}$ and/or $\check{h}$ arise from weight two Eisenstein series, these expressions are also related to p -adic regulators of *Beilinson elements* in the K -groups $K_2(X_1(N))$ or in $K_1(X_1(N)^2)$ (cf. [BD2] and [BDR1], [BDR2] respectively).

On the computational side, using a method for computing with overconvergent modular forms using Katz expansions developed in [La1], the article [La2] describes an algorithm for the efficient numerical evaluation of these p -adic iterated integrals, and uses it to calculate certain *Chow-Heegner points* on the elliptic curve E which were first defined and studied by Shouwu Zhang. These global points, which arise when $k \geq 2$ and $\check{\gamma}$ and $\check{h}$ are eigenvectors for $\mathbb{T}_N$ with the same eigenvalues, have a well-understood geometric provenance, and can also be calculated by complex analytic means following the strategy described in [DRS] and [DDLRL].

The setting $k = 1$ which is germane to the present work is significantly more mysterious: the p -adic iterated integrals (and indeed, the modular forms g and h themselves) cease to enjoy geometric interpretations in terms of the cohomology of Kuga-Sato varieties. On a more technical level, the space $M_1^{\text{oc,ord}}(N, \chi)$, while still finite-dimensional, may contain the space $M_1^{\text{ord}}(Np, \chi)_{\mathbb{C}_p}$ as a proper subspace. Fix a root α_g of the Hecke polynomial

$$x^2 - a_p(g)x + \chi^{-1}(p) = (x - \alpha_g)(x - \beta_g)$$

attached to g at p , and let $g_\alpha \in M_1(Np, \chi^{-1})_L$ denote the corresponding p -stabilisation. It is an eigenform for the Hecke algebra $\mathbb{T}_N^{(p)}$ satisfying

$$U_p g_\alpha = \alpha_g \cdot g_\alpha.$$

Let $g_\alpha^* = g_\alpha \otimes \chi$ denote the newform of level N_g corresponding to the twist of g_α by χ , the inverse of its nebentype character.

If ϕ is a simultaneous eigenvector for either the Hecke algebra $\mathbb{T}_N^{(p)}$ or $\mathbb{T}_{Np}$, then, extending the definitions in (2), let

$$M[\phi] := \text{Ker}(I_\phi), \quad M[[\phi]] = \cup_{n \geq 1} \text{Ker}(I_\phi^n)$$

denote the eigenspace and generalized eigenspace attached to its system of eigenvalues. The following assumption is a crucial substitute for Coleman's classicality theorem when $k = 1$:

Hypothesis C (Classicality property for g_α). If g is a cusp form, the natural inclusion

$$M_1(Np, \chi)_{\mathbb{C}_p}[g_\alpha^*] \hookrightarrow S_1^{\text{oc,ord}}(N, \chi)[[g_\alpha^*]]$$

is an isomorphism of $\mathbb{C}_p$ -vector spaces. If g is an Eisenstein series, the analogous inclusion maps $M_1(Np, \chi)_{\mathbb{C}_p}[g_\alpha^*]$ to the cuspidal subspace $S_1^{\text{oc,ord}}(N, \chi)[g_\alpha^*]$ and induces an isomorphism

$$M_1(Np, \chi)_{\mathbb{C}_p}[g_\alpha^*] \simeq S_1^{\text{oc,ord}}(N, \chi)[[g_\alpha^*]].$$

Hypothesis C is often satisfied in practice. Section 1.1 gives explicit conditions on g_α , arising from work of Cho-Vatsal [CV] and Bellaïche-Dimitrov [BeDi], for this hypothesis to hold when g is a cusp form which is *regular* at p i.e., with $\alpha_g \neq \beta_g$ (for g a cusp form we do not expect Hypothesis C to hold in the irregular case). Hypothesis C makes it possible to adapt the definition of the p -adic iterated integrals to define a $\mathbb{C}_p$ -valued trilinear form

$$I'_p : S_2(Np)_L[f] \times M_1(Np, \chi)_L^\vee[g_\alpha] \times M_1(Np, \chi)_L[h] \longrightarrow \mathbb{C}_p \quad (6)$$

by the simple rule

$$I'_p(\check{f}, \check{\gamma}, \check{h}) = \int_{\check{\gamma}} \check{f} \cdot \check{h} \quad := \quad \check{\gamma}(e_{g_\alpha^*} e_{\text{ord}}(\check{F} \cdot \check{h})),$$

where

$$e_{g_\alpha^*} : S_1^{\text{oc,ord}}(N, \chi) \longrightarrow S_1^{\text{oc,ord}}(N, \chi)[[g_\alpha^*]] = M_1(Np, \chi)_{\mathbb{C}_p}[g_\alpha^*] \quad (7)$$

is the projector onto the generalized eigenspace attached to g_α^* for the action of the Hecke algebra $\mathbb{T}_N^{(p)}$. Note the crucial role played by Hypothesis C in the definition of this trilinear form via the equality in (7).

Since the prime p is unramified in H , there is an embedding $H \longrightarrow \mathbb{Q}_p^{\text{u-r}}$ into the maximal unramified extension of $\mathbb{Q}_p$. Fix such an embedding and denote by $\sigma_p \in \text{Gal}(H/\mathbb{Q})$ the associated Frobenius element at p . This element acts semisimply on the Artin representations V_g (resp. V_h) with eigenvalues α_g and β_g (resp. α_h and β_h), and determines a decomposition $V_g = V_g^{\alpha_g} \oplus V_g^{\beta_g}$ into one-dimensional σ_p -eigenspaces when $\alpha_g \neq \beta_g$. If F is any finite extension of $\mathbb{Q}_p$, with ring of integers $\mathcal{O}_F$, let

$$\log_p : \mathcal{O}_F^\times \longrightarrow F$$

be the usual p -adic logarithm, and let

$$\log_{E,p} : E(F) \longrightarrow F$$

denote the canonical extension to $E(F)$ of the p -adic formal group logarithm associated to a Néron differential on E . Via the chosen embedding of H into $\mathbb{C}_p$ the maps $\log_p$ and $\log_{E,p}$ give rise to homomorphisms

$$\log_p : (\mathcal{O}_H)_L^\times \longrightarrow H_p \otimes L, \quad \log_{E,p} : E(H)_L \longrightarrow H_p \otimes L.$$

Hypotheses A and B together imply that $r_{\text{an}}(E, \varrho_{gh}) \geq 2$, and of course the same is expected of the algebraic rank. When $r(E, \varrho_{gh}) = 2$, let (Φ_1, Φ_2) be an L -vector space basis for $\text{hom}_{G_{\mathbb{Q}}}(V_{gh}, E(H)_L)$. The choice of the U_p eigenvalue α_g (under the assumption that $\alpha_g \neq \beta_g$) likewise determines a canonical two-dimensional subspace of V_{gh} by the rule

$$V_{gh}^{g_\alpha} := V_g^{\alpha_g} \otimes V_h.$$

When $\alpha_g = \beta_g$ we assume g is Eisenstein (for otherwise Hypothesis C is expected to fail), and in this case define $V_{gh}^{g_\alpha} := W \otimes V_h$ where W is any line in V_g which is not an eigenspace for the global Galois action; that is, not one of the two one-dimensional irreducible summands of V_g . Fix an L -vector space basis (v_1, v_2) for $V_{gh}^{g_\alpha}$. The bases (Φ_1, Φ_2) and (v_1, v_2) give rise to the 2×2 regulator matrix

$$R_{g_\alpha}(E, \varrho_{gh}) := \begin{pmatrix} \log_{E,p}(\Phi_1(v_1)) & \log_{E,p}(\Phi_1(v_2)) \\ \log_{E,p}(\Phi_2(v_1)) & \log_{E,p}(\Phi_2(v_2)) \end{pmatrix},$$

which is independent of the choice of basis (Φ_1, Φ_2) (resp., of the basis (v_1, v_2)) up to left (resp., right) multiplication by a matrix in $\mathbf{GL}_2(L)$ (and in the irregular case independent of the choice of W). The determinant of $R_{g_\alpha}(E, \varrho_{gh})$ is therefore well defined, up to multiplication by $L^\times$. This determinant belongs to $\mathbb{Q}_p^{\text{u-r}} \otimes L$, and σ_p acts on it with the same eigenvalue as on $\wedge^2 V_{gh}^{\alpha_g}$, namely, $\alpha_g \alpha_h \alpha_g \beta_h = \alpha_g / \beta_g$.

Let $\text{Ad}_g := \text{hom}^0(V_g, V_g)$ denote the three-dimensional *adjoint representation* attached to ϱ_g , on which the Frobenius element σ_p acts with eigenvalues 1, α_g / β_g , and β_g / α_g . Section 1.2 attaches to the eigenform g_α a Gross-Stark unit $u_{g_\alpha} \in (\mathcal{O}_{H_g}[1/p]^\times)_L^{\text{Ad}_g}$, where H_g is the number field cut out by Ad_g . When g is a cusp form, the invariant u_{g_α} is an actual Stark unit, and it is often determined up to scaling (for instance, when $\alpha_g / \beta_g \neq \pm 1$) by the twin requirements that

$$u_{g_\alpha} \text{ belongs to } (\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g}, \quad \sigma_p(u_{g_\alpha}) = \frac{\alpha_g}{\beta_g} u_{g_\alpha}.$$

A more detailed description of u_{g_α} , encompassing the more subtle case where g is an Eisenstein series, is given in Section 1.2. The unit u_{g_α} is analogous to the Stark units of [St2] and [DuLi] whose logarithms encode the Petersson scalar product of g_α with itself. Section 1.2 explains the striking connection between the existence of such a unit and Hypothesis C.

The main conjecture of the present work is

Conjecture ES: (Elliptic Stark Conjecture). *Assume that hypotheses A, B, and C are satisfied, so that in particular $r_{\text{an}}(E, \varrho_{gh}) \geq 2$. If $r(E, \varrho_{gh}) > 2$, the trilinear form I'_p of (6) is identically zero. Otherwise, there exist test vectors*

$$(\check{f}, \check{\gamma}, \check{h}) \in S_2(Np)_L[f] \times M_1(Np, \chi)_L^\vee[g_\alpha] \times M_1(Np, \chi)_L[h]$$

for which

$$\int_{\check{\gamma}} \check{f} \cdot \check{h} = \frac{\det(R_{g_\alpha}(E, \varrho_{gh}))}{\log_p(u_{g_\alpha})}. \quad (8)$$

Note that both the numerator and denominator on the right-hand side of (8) belong to the same eigenspace for σ_p , with eigenvalue α_g / β_g , and hence that the ratio belongs to $L_p \subset H_p \otimes L$, consistent with the fact that this is clearly true of the left-hand side.

The condition $r(E, \varrho_{gh}) = 2$ implies that either:

- (1) the Artin representation ϱ_{gh} admits a decomposition

$$\varrho_{gh} = \varrho_1 \oplus \varrho_2, \quad \text{with} \quad r(E, \varrho_1) = 2, \quad r(E, \varrho_2) = 0, \quad (9)$$

where ϱ_1 is an irreducible constituent of ϱ_{gh} and ϱ_2 is possibly trivial, or reducible;

- (2) the Artin representation ϱ_{gh} admits a decomposition

$$\varrho_{gh} = \varrho_1 \oplus \varrho_2 \oplus \varrho_3, \quad \text{with} \quad r(E, \varrho_1) = r(E, \varrho_2) = 1, \quad r(E, \varrho_3) = 0, \quad (10)$$

where ϱ_1 and ϱ_2 are irreducible constituents of ϱ_{gh} and ϱ_3 is possibly trivial or reducible.

These scenarios will be referred to respectively as the *rank 2* and *rank (1,1) settings* for the elliptic Stark conjecture. The apparent rarity of double zeroes for primitive L -functions at central critical points suggests that rank 2 settings may only occur sporadically, while a systematic supply of rank (1,1) settings arises when the constituents ϱ_1 and ϱ_2 in (10) are self-dual and the signs in the functional equations for $L(E, \varrho_1, s)$ and $L(E, \varrho_2, s)$ are both equal to -1 . This explains why more of the experiments reported on below have tested Conjecture ES in the rank (1,1) setting (although enough data has been gathered to lend convincing evidence for Conjecture ES in the rank 2 scenario as well).

Another attractive feature of the rank (1,1) setting is that it often leads to explicit formulae for the p -adic logarithms of global points in $E(H)_L^{\varrho_1}$ and $E(H)_L^{\varrho_2}$. More precisely, the regulator

$R_{g_\alpha}(E, \varrho_{gh})$ is said to be *factorisable* if, after eventually interchanging α_h and β_h and/or ϱ_1 and ϱ_2 , the Frobenius σ_p acting on V_{ϱ_1} admits $\alpha_g\alpha_h$ but not $\alpha_g\beta_h$ as an eigenvalue. Relative to a basis $(\Phi_1, \Phi_2) \in \text{Hom}_{G_{\mathbb{Q}}}(V_{\varrho_1}, E(H)_L) \times \text{Hom}_{G_{\mathbb{Q}}}(V_{\varrho_2}, E(H)_L)$ and to a σ_p -eigenbasis (v_1, v_2) for $V_{gh}^{g_\alpha}$ satisfying

$$\sigma_p v_1 = \alpha_g \alpha_h v_1, \quad \sigma_p v_2 = \alpha_g \beta_h v_2,$$

the regulator matrix R_{g_α} takes on the simple form

$$R_{g_\alpha}(E, \varrho_{gh}) = \begin{pmatrix} \log_{E,p} P_1 & 0 \\ \log_{E,p} Q & \log_{E,p} P_2 \end{pmatrix},$$

where the points P_j belong to $E(H)_L^{\varrho_j}$ and are uniquely determined (up to $L^\times$) by the requirement that

$$\sigma_p P_1 = \alpha_g \alpha_h P_1, \quad \sigma_p P_2 = \alpha_g \beta_h P_2. \quad (11)$$

Conjecture ES then admits the simpler formulation

$$\int_{\check{\gamma}} \check{f} \cdot \check{h} = \frac{\log_{E,p}(P_1) \cdot \log_{E,p}(P_2)}{\log_p(u_{g_\alpha})}.$$

The prototype for such a formula, discovered experimentally and subsequently proved, is equation (44) in Remark 3.4.

Turning to a more detailed overview of the contents of the paper, Chapter 1 gives explicit criteria for Hypothesis C which may be of independent interest, and defines the Stark unit u_{g_α} , showing that Hypothesis C bears an intimate connection to the existence of this Stark unit.

Building on the results of Harris-Kudla and Ichino on trilinear periods and Jacquet's conjecture, Chapter 2 recasts the trilinear form I'_p as the value of a Garrett-Hida triple product p -adic L -function at a point lying outside its region of classical interpolation. This connection with p -adic L -functions justifies viewing Conjecture ES as an elliptic curve analogue of the (p -adic) Stark conjectures. It also suggests that Conjecture ES might be regarded as a p -adic limit of Jacquet's conjecture on trilinear periods, prompting a possible strategy for proving it in some cases and suggesting that it might be viewed as a deep manifestation of the "Artin formalism" for certain p -adic L -series.

The rest of the paper is devoted to giving both theoretical and experimental evidence in support of Conjecture ES.

Chapters 3 and 4 treat the case where ϱ_g and ϱ_h are induced from characters ψ_g and ψ_h of the same quadratic field K , and with inverse central characters. The representation

$$V_{gh} = V_{\psi_1} \oplus V_{\psi_2}$$

then decomposes as the direct sum of a pair of representations induced from ring class characters ψ_1 and ψ_2 of K .

Chapter 3 focusses on the setting where K is imaginary quadratic. Building on the results of Chapter 2 and the precise formulae of Watson [Wa] and Woodbury [Wo1], Conjecture ES is proved in Section 3.2 under the assumption that the prime p splits in K and a Heegner hypothesis is satisfied. The main ingredients in the proof are the p -adic Gross-Zagier formula of [BDP1] expressing Heegner points in terms of values of a Katz-style p -adic Rankin L -series, the relation established in [DR1] between p -adic iterated integrals and Garrett-Hida p -adic L -series, and a subtle relation between these two types of L -series. It is in the non-trivial comparison between periods used in their definitions that a value of Katz's two variable p -adic L -function makes an (a priori unexpected) appearance. The unit u_{g_α} can be expressed in terms of elliptic units in this case, and emerges from Katz's p -adic analogue of the Kronecker limit formula.

Katz’s approach to p -adic L -functions is based on the evaluation of p -adic modular forms at CM points when they lie on the ordinary locus of the modular curve, i.e., when p is *split* in K . This approach breaks down entirely when p is inert in the quadratic imaginary field. Section 3.3 gives numerical evidence in support of Conjecture ES in the inert case, both in the rank $(1, 1)$ setting—where the theory of elliptic units and Heegner points can still be used to produce the global objects that should enter into the right-hand side of (8)—and in the more mysterious rank 2 setting.

Chapter 4 attempts a parallel treatment when ϱ_g and ϱ_h are induced from characters (of mixed signature) of the same *real* quadratic field K . In some cases, the global points P_1 and P_2 can then be expressed, at least conjecturally, in terms of the Stark-Heegner points of [Dar] and [Gre] attached to “real quadratic cycles on $\mathcal{H}_p \times \mathcal{H}$ ”. The resulting relation between the Stark points of the present work and these Stark-Heegner points might be amenable to proof via the methods of Section 3.2. The prospect of tying the real quadratic cycles on $\mathcal{H}_p \times \mathcal{H}$ to global (but not necessarily motivic) cohomology classes in the Selmer groups of E over ring class fields of real quadratic fields represents an encouraging first step towards better understanding their global nature.

Chapters 5, 6, and 7 consider the remaining interesting cases, all of which appear to lie beyond the reach of previously available theory or conjectural construction.

Chapter 5 reports on a number of experiments in the setting where h is an exotic weight one form, whose associated Galois representation has projective image A_4 , S_4 or A_5 , and $g = h^*$ is its dual. The representation V_{gh} then breaks up as

$$V_{gh} = L \oplus \text{Ad}_g.$$

In the rank $(1, 1)$ (resp. the rank 2) setting, Conjecture ES leads to a p -adic analytic construction of a global point (resp. of a canonical Selmer class) on E over the field cut out by Ad_g , whose logarithm describes the value of the associated p -adic iterated integral.

Chapter 6 takes up the setting where $g \in S_1(N, \bar{\chi})$ is a cusp form and $h := E_1(1, \chi) \in M_1(N, \chi)$ is the weight one Eisenstein series attached to the pair $(1, \chi)$ of Dirichlet characters. A novelty of this setting is that

$$V_{gh} = V_g \oplus V_{g^*}$$

decomposes into a direct sum of two representations which are not self-dual in general, and the regulator matrix $R_{g_\alpha}(E, \varrho_{gh})$ involves points in the isotypic parts for these non self-dual representations.

Chapter 7 concludes with the most “degenerate case” of the setting described in Ch. 6, in which both $g := E_1(1, \chi^{-1})$ and $h := E_1(\chi_1, \chi_2)$ are Eisenstein series, with $\chi_1 \chi_2 = \chi$. Because

$$V_{gh} = \chi_1 \oplus \bar{\chi}_1 \oplus \chi_2 \oplus \bar{\chi}_2$$

is then a direct sum of one-dimensional representations, the regulator term in Conjecture ES involves the logarithms of points defined over the abelian extension of $\mathbb{Q}$ cut out by either χ_1 or χ_2 . The scenario in which these Dirichlet characters are non-quadratic is particularly intriguing. According to predictions of David, Fearnley and Kisilevsky [DFK], the occurrence of global points over cyclic extensions of degree greater than 2 is quite sporadic; the L -series that control them have non-real coefficients, and their functional equation gives no indication as to the parity of their order of vanishing at the center. Although ostensibly rare, global points over cyclic cubic extensions do occur in practice. Fairly extensive tables of pairs (E, χ) with χ cubic for which $r_{\text{an}}(E, \chi) > 0$ have been drawn up by the authors of [DFK], and the computationally more challenging task of verifying that $r(E, \chi) > 0$ for such pairs is carried out—albeit in more limited ranges—in [Co]. Section 7.2 describes a few experimental verifications of Conjecture ES involving the p -adic logarithms of such cyclic cubic points.

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1. OVERCONVERGENT WEIGHT ONE FORMS AND GROSS-STARK UNITS

1.1. Classical and overconvergent modular forms of weight one. This section is devoted to giving explicit conditions guaranteeing that the classical weight one eigenform g_α satisfies the classicality condition (Hypothesis C) described in the introduction.

This condition turns out to be easier to analyze when the eigenform g and the prime p satisfy the following further hypothesis:

Definition 1.1. A classical eigenform g of weight one is said to be *regular* at p if $\alpha_g \neq \beta_g$, i.e., the Frobenius element σ_p acts on V_g with distinct eigenvalues.

When g is regular at p , the choice of α_g determines the associated p -stabilisation g_α on which U_p acts with this eigenvalue, as well as the one-dimensional α_g -eigenspace for σ_p acting on V_g .

One of the main results of this section is the following concrete criterion for Hypothesis C to be satisfied when g is a cusp form, which follows directly from a result of Cho-Vatsal and Bellaïche-Dimitrov.

Proposition 1.2. *Let $g \in S_1(N, \chi^{-1})$ be a cusp form of weight one which is regular at p , and let g_α denote one of its p -stabilisations. Then the natural inclusion*

$$S_1(Np, \chi)_{\mathbb{C}_p}[g_\alpha^*] \hookrightarrow S_1^{\text{oc,ord}}(N, \chi)[[g_\alpha^*]]$$

is an isomorphism of $\mathbb{C}_p$ -vector spaces if and only if

$$\rho_g \text{ is not induced from a character of a real quadratic field in which } p \text{ splits.} \quad (12)$$

Proof. Let $\mathbb{T}_k(N)$ denote the $\mathbb{Z}_p$ -algebra generated by the Hecke operators T_ℓ (with $\ell \nmid N$) and U_q (with $q|N$), together with the diamond operators, acting on the space $S_k(N)$ of classical modular forms on $\Gamma_1(N)$. When $k \geq 2$, there is a natural perfect duality

$$\mathbb{T}_k(N) \times S_k(N) \longrightarrow \mathbb{Z}_p \quad (13)$$

defined by $\langle T, f \rangle := a_1(Tf)$.

Hida's Λ -adic Hecke algebra is defined to be

$$\mathbb{T}(N) = \varprojlim_{\leftarrow, t} (e_{\text{ord}} \mathbb{T}_2(Np^t)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p,$$

where the inverse limit is taken with respect to the natural projection maps. The algebra $\mathbb{T}(N)$ has a natural structure of a $\Lambda \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ -module arising from the action of the diamond operators, and it can also be viewed as a module over the $\Lambda \otimes \mathbb{Q}_p$ -subalgebra $\mathbb{T}^0(N)$ of $\mathbb{T}(N)$ generated

by the “good” Hecke operators T_n with $(n, pN) = 1$ and by U_p . Let $\varphi_k : \Lambda \otimes \mathbb{Q}_p \rightarrow \mathbb{Q}_p$ denote the “weight k specialisation map” sending a group-like element $\langle a \rangle \in 1 + p\mathbb{Z}_p \subset \Lambda^\times$ to a^{k-1} , and let $\wp_k$ denote its kernel. The pairing of (13) induces a perfect duality

$$((\mathbb{T}(N)/\wp_k) \otimes_{\mathbb{Q}_p} \mathbb{C}_p) \times S_k^{\text{oc,ord}}(N) \rightarrow \mathbb{C}_p.$$

In particular, the space $S_1^{\text{oc,ord}}(N)$ is naturally identified with the vector space of $\mathbb{C}_p$ -linear homomorphisms $\text{hom}(\mathbb{T}(N)/\wp_1, \mathbb{C}_p)$. Under this identification, the classical weight one form $g_\alpha \in S_1^{\text{oc,ord}}(N)$, which is a normalised eigenvector for the good Hecke operators T_n with $(n, Np) = 1$ and for U_p , corresponds to an algebra homomorphism

$$\varphi_{g_\alpha} \in \text{hom}_{\text{alg}}(\mathbb{T}^0(N)/\wp_1, \mathbb{C}_p).$$

Let $\wp_{g_\alpha} \subset \mathbb{T}^0(N)$ be the kernel of φ_{g_α} . Its image under the “projection map to weight space”

$$\text{hom}_{\text{alg}}(\mathbb{T}^0(N), \mathbb{C}_p) \rightarrow \text{hom}_{\text{alg}}(\Lambda, \mathbb{C}_p)$$

induced by restriction is equal to φ_1 , i.e., $\wp_{g_\alpha}$ contains the ideal $\wp_1 \mathbb{T}^0(N)$.

The eigenspace $S_1^{\text{oc,ord}}(N)[g_\alpha^*]$ that arises in the statement of Proposition 1.2 can be identified with the dual of $\mathbb{T}(N)/\wp_{g_\alpha}$:

$$S_1^{\text{oc,ord}}(N)[g_\alpha^*] = \text{hom}(\mathbb{T}(N)/\wp_{g_\alpha}, \mathbb{C}_p).$$

Furthermore,

$$S_1^{\text{oc,ord}}(N)[[g_\alpha^*]] = \text{hom}(\mathbb{T}(N)_{\wp_{g_\alpha}}/\wp_1, \mathbb{C}_p),$$

where

$$\mathbb{T}^0(N)_{\wp_{g_\alpha}} := \varprojlim_{\leftarrow, n} \mathbb{T}^0(N)/\wp_{g_\alpha}^n, \quad \mathbb{T}(N)_{\wp_{g_\alpha}} := \varprojlim_{\leftarrow, n} \mathbb{T}(N)/\wp_{g_\alpha}^n$$

denote the completions at the maximal ideal $\wp_{g_\alpha}$ of $\mathbb{T}^0(N) \otimes \mathbb{C}_p$.

Hence the dual of the natural inclusion $S_1^{\text{oc,ord}}(N)[g_\alpha^*] \hookrightarrow S_1^{\text{oc,ord}}(N)[[g_\alpha^*]]$ is the natural surjection

$$\mathbb{T}(N)_{\wp_{g_\alpha}}/\wp_1 \rightarrow \mathbb{T}(N)/\wp_{g_\alpha}. \quad (14)$$

The ideal $\wp_1 \mathbb{T}^0(N)_{\wp_{g_\alpha}}$ is equal to a positive power $\wp_{g_\alpha}^t$ (with $t \geq 1$) of the maximal ideal $\wp_{g_\alpha}$ in the local ring $\mathbb{T}^0(N)_{\wp_{g_\alpha}} \otimes \mathbb{C}_p$. The map (14) is an isomorphism if and only if $t = 1$, i.e., if and only if the point of $\text{hom}(\mathbb{T}^0(N)_{\wp_g}, \mathbb{C}_p)$ attached to φ_{g_α} is smooth and étale over weight space. Under the regularity assumption on g at p , Theorem 1.1. of [BeDi] implies that this étaleness condition is satisfied precisely when (12) holds. The proposition follows. $\square$

We now turn to Hypothesis C in the case where $g = E_1(\chi_1, \chi_2)$ is the weight one Eisenstein series associated to the pair of characters χ_1 and χ_2 , admitting the following (not necessarily distinct) stabilisations:

$$g_1(q) := g(q) - \chi_2(p)g(q^p), \quad g_2(q) := g(q) - \chi_1(p)g(q^p).$$

Assume that χ_1 and χ_2 are primitive with relatively prime conductors N_1 and N_2 , and set $\chi := \chi_1 \chi_2$ and $N := N_1 N_2$.

Proposition 1.3. *For $j = 1, 2$, the image of the natural inclusion*

$$M_1(Np, \chi)[g_j] \hookrightarrow M_1^{\text{oc,ord}}(N, \chi)[[g_j]] \quad (15)$$

is contained in $S_1^{\text{oc,ord}}(N, \chi)$ if and only if $\chi_1(p) = \chi_2(p)$, i.e., the p -stabilisations g_1 and g_2 coincide.

Proof. Let $X(N; p) = X_1(N) \times_{X(1)} X_0(p)$ denote the modular curve associated to the congruence group $\Gamma_1(N) \cap \Gamma_0(p)$ and let $X_1(N)_{\mathbb{C}_p}^{\text{ord}}$ denote the complement of the supersingular residue discs in $X_1(N)(\mathbb{C}_p)$, regarded as an affinoid domain. For a suitable wide open neighbourhood $\mathcal{W}$ of $X_1(N)_{\mathbb{C}_p}^{\text{ord}}$, let

$$s : \mathcal{W} \longrightarrow X(N; p)_{\mathbb{C}_p}$$

be the rigid analytic section of the natural projection $X(N; p) \longrightarrow X_1(N)$, defined by sending the point attached to an ordinary pair $(E, P_N)_{/\mathbb{C}_p} \in \mathcal{W}$ to the (moduli of the) triple (E, P_N, C_{can}) , where P_N is a point of order N on E and C_{can} denotes the canonical subgroup of order p . The inclusion (15) is described geometrically in terms of s , as follows: letting $\underline{\omega}$ denote the sheaf over $X(N; p)$ with logarithmic poles at the cusps discussed in [BDP1, Sec. 1.1], a classical weight one form ϕ in the source of (15) may be viewed as a section of $\underline{\omega}$ over $X(N; p)$, and the map of (15) sends it to its pull-back $s^*(\phi) \in H^0(\mathcal{W}, \underline{\omega})$, naturally identified with an overconvergent modular form of level N .

The non-zero constant terms of the classical form $g = E_1(\chi_1, \chi_2)$ are concentrated at the cusps of $X_1(N)$ attached to the “Katz test objects”

$$c_{ab} := (\mathbb{G}_m/q^{\mathbb{Z}}, \zeta_{N_1}^a q^{b/N_2}), \quad d_{ab} := (\mathbb{G}_m/q^{\mathbb{Z}}, q^{a/N_1} \zeta_{N_2}^b),$$

where $\mathbb{G}_m/q^{\mathbb{Z}}$ is the Tate generalised elliptic curve over $\text{Spec}(\mathbb{C}_p[[q]])$, and ζ_{N_1} and ζ_{N_2} are fixed primitive N_1 and N_2 -th roots of unity; the indices (a, b) range over $(\mathbb{Z}/N_1\mathbb{Z})^\times \times (\mathbb{Z}/N_2\mathbb{Z})^\times$. If g is any weight one modular form on $\Gamma_0(N)$ with character χ , then the modular form g_p on $X(N; p)$ whose q -expansion is equal to $g(q^p)$ at the cusp $\infty = (\mathbb{G}_m/q^{\mathbb{Z}}, \zeta_N, \mu_p)$ is defined on test objects of the form (E, P_N, C_p) , with $C_p \subset E$ of order p , by the rule

$$g_p((E, P_N, C_p)) := \chi^{-1}(p) p^{-1} \varphi_{C_p}^*(g(E/C_p, P_N + C_p)),$$

where $\varphi_{C_p} : E \longrightarrow E/C_p$ is the natural quotient morphism. This can be seen by checking this identity on the test object $(\mathbb{G}_m/q^{\mathbb{Z}}, \zeta_N, \mu_p)$ and invoking the q -expansion principle. Let $g_{ab}(q)$ and $g'_{ab}(q) \in \mathbb{C}_p[[q^{1/N}]]$ denote the q -expansions of g at the cusps c_{ab} and d_{ab} respectively, and let x_{ab} and y_{ab} denote the associated leading terms. The q -expansion of g_p , viewed as a p -adic modular form, at the cusps c_{ab} and d_{ab} is then given by the rule

$$g_p(\mathbb{G}_m/q^{\mathbb{Z}}, \zeta_{N_1}^a q^{b/N_2}) = \chi^{-1}(p) p^{-1} \varphi^*(g(\mathbb{G}_m/q^{p\mathbb{Z}}, \zeta_{N_1}^{pa} q^{pb/N_2})) = \chi_2^{-1}(p) g_{ab}(q^p) \frac{dt}{t}, \quad (16)$$

$$g_p(\mathbb{G}_m/q^{\mathbb{Z}}, q^{a/N_1} \zeta_{N_2}^b) = \chi^{-1}(p) p^{-1} \varphi^*(g(\mathbb{G}_m/q^{p\mathbb{Z}}, q^{pa/N_1} \zeta_{N_2}^{pb})) = \chi_1^{-1}(p) g'_{ab}(q^p) \frac{dt}{t}. \quad (17)$$

Here we have used the fact the canonical subgroup $\mu_p \subset \mathbb{G}_m/q^{\mathbb{Z}}$ of the Tate elliptic curve is the kernel of the map $\varphi : \mathbb{G}_m/q^{\mathbb{Z}} \longrightarrow \mathbb{G}_m/q^{p\mathbb{Z}}$ sending t to t^p . Equations (16) and (17) yield the constant terms of the forms g , g_p , g_1 , and g_2 at the cusps c_{ab} and d_{ab} , as summarised in the table below:

	c_{ab}	d_{ab}
g	x_{ab}	y_{ab}
g_p	$\chi_2^{-1}(p) x_{ab}$	$\chi_1^{-1}(p) y_{ab}$
g_1	0	$(1 - \chi_2/\chi_1(p)) y_{ab}$
g_2	$(1 - \chi_1/\chi_2(p)) x_{ab}$	0

The proposition now follows from the fact that the non-zero constant terms of the p -stabilised overconvergent forms g_1 and g_2 are concentrated at the cusps c_{ab} and d_{ab} , viewed as points of $\mathcal{W}$. $\square$

Experimental evidence suggests that in fact $\chi_1(p) = \chi_2(p)$ may be a necessary *and sufficient* condition for Hypothesis C itself to hold in the Eisenstein setting. In any case, because of Hypothesis C and Proposition 1.3, we shall henceforth assume that, if g is not a cusp form,

then it is an Eisenstein series of the form $E_1(\chi_1, \chi_2)$ with $\chi_1(p) = \chi_2(p)$, admitting therefore a unique p -stabilisation, which is a p -adic cusp form.

1.2. The Gross-Stark unit attached to a weight one form. The aim of this section is to associate to the p -stabilised eigenform g_α a global element $u_{g_\alpha} \in H^\times \otimes L$ (often well-defined up to scaling by $L^\times$), whose p -adic logarithm arises in Conjecture ES. This element belongs to the Ad_g -isotypic subspace of $H^\times \otimes L$, where

$$\text{Ad}_g := \text{End}_0(V_g)$$

is the *adjoint representation* attached to V_g , the 3-dimensional Artin representation consisting of trace zero L -linear endomorphisms of V_g on which $G_\mathbb{Q}$ acts by conjugation via ϱ_g .

In defining u_{g_α} , it shall be convenient to replace the field H of the introduction by the smaller field H_g cut out by Ad_g , and L by any field over which the representation Ad_g can be defined. To give a rough feeling for the nature of u_{g_α} , let us mention at the outset that

$$L \cdot u_{g_\alpha} \subset \begin{cases} (\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} & \text{if } g \text{ is cuspidal,} \\ (\mathcal{O}_{H_g}[1/p]^\times)_L^{\text{Ad}_g} & \text{if } g \text{ is Eisenstein,} \end{cases} \quad (18)$$

and

the Frobenius element σ_p acts on u_{g_α} with eigenvalue α_g/β_g .

As explained below, these requirements often characterize u_{g_α} uniquely up to scaling by $L^\times$.

Write $\sigma_\infty \in G := \text{Gal}(H_g/\mathbb{Q})$ for the involution given by complex conjugation induced by the fixed choice of complex embedding of $H_g \subset \bar{\mathbb{Q}}$. The representation Ad_g decomposes as a direct sum of $+$ and $-$ eigenspaces for the action of σ_∞ , denoted Ad_g^+ and Ad_g^- . These eigenspaces are of dimensions 1 and 2 respectively:

$$\dim_L \text{Ad}_g^+ = 1, \quad \dim_L \text{Ad}_g^- = 2. \quad (19)$$

As suggested by (18), the nature of the invariant u_{g_α} depends on whether ϱ_g is reducible or irreducible, and each case will be dealt with in turn.

If ϱ_g is irreducible, then g is a *cuspidal* eigenform of weight one. The definition of u_{g_α} rests on the following proposition:

Proposition 1.4. (1) *The L -vector space $\text{Hom}_{G_\mathbb{Q}}(\text{Ad}_g, \mathcal{O}_{H_g}^\times \otimes L)$ is one-dimensional.*
 (2) *The Artin L -function $L(\text{Ad}_g, s)$ has a simple zero at $s = 0$.*

Proof. The finite group $G = \text{Gal}(H_g/\mathbb{Q})$ acts faithfully on Ad_g and is identified with the image of the projective representation attached to V_g . Note that $G_\infty = \langle \sigma_\infty \rangle$ is of order two since Ad_g is not totally even. Dirichlet's unit theorem determines the structure of $\mathcal{O}_{H_g}^\times \otimes L$ as an $L[G]$ -module:

$$\mathcal{O}_{H_g}^\times \otimes L = (\text{Ind}_{G_\infty}^G L) - L,$$

where the two occurrences of L on the right refer to trivial representations (of G_∞ and G respectively). The irreducibility of V_g implies that the representation Ad_g contains no G -invariant vectors, by Schur's lemma, and hence

$$\text{hom}_G(\text{Ad}_g, \mathcal{O}_{H_g}^\times \otimes L) = \text{hom}_G(\text{Ad}_g, \text{Ind}_{G_\infty}^G L).$$

By Frobenius reciprocity,

$$\text{hom}_G(\text{Ad}_g, \text{Ind}_{G_\infty}^G L) = \text{hom}_{G_\infty}(\text{Ad}_g, L) = \text{hom}(\text{Ad}_g^+, L).$$

The first part of the proposition now follows from (19). The second statement follows from the shape of the Γ -factors in the functional equation of $L(\text{Ad}_g, s)$ determined by (19). $\square$

Choose a basis φ for the one-dimensional vector space $\text{hom}_G(\text{Ad}_g, \mathcal{O}_{H_g}^\times \otimes L)$, and let

$$(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} := \varphi(\text{Ad}_g) \subset \mathcal{O}_{H_g}^\times \otimes L.$$

The $L[G]$ -module $(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g}$ is of dimension ≤ 3 over L and is abstractly isomorphic to the unique irreducible subrepresentation of Ad_g containing Ad_g^+ .

Recall the prime number p chosen in the introduction. The frobenius element σ_p acts on Ad_g with the (not necessarily different) eigenvalues 1, α_g/β_g and β_g/α_g . Define

$$U_{g_\alpha} := \left\{ u \in (\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} \text{ such that } \sigma_p(u) = \frac{\alpha_g}{\beta_g} u \right\}.$$

Lemma 1.5. *Assume that V_g is regular. If ϱ_g is induced from a character of a real quadratic field in which p splits, then $U_{g_\alpha} = 0$. Otherwise, $\dim_L(U_{g_\alpha}) \geq 1$, with equality if either*

- (1) $\alpha_g \neq -\beta_g$, i.e., Ad_g is regular, or,
- (2) ϱ_g is induced from a character of a quadratic field in which p is inert.

Proof. This lemma is proved by a case by case analysis, according to the following possibilities for V_g :

- a) V_g has imaginary dihedral projective image;
- b) V_g has real dihedral projective image;
- c) g is an exotic weight one form with projective image A_4 , S_4 or A_5 .

a) *Imaginary dihedral forms.* Assume that the representation V_g is induced from a character ψ_g of an imaginary quadratic field K . Then

$$\text{Ad}_g = L(\chi_K) \oplus V_\psi, \tag{20}$$

where χ_K is the quadratic Dirichlet character associated to K and $V_\psi := \text{Ind}_K^\mathbb{Q} \psi$ is the two-dimensional representation induced from the ring class character $\psi := \psi_g/\psi'_g$ of K . Since σ_∞ acts with eigenvalue -1 on the first factor $L(\chi_K)$, it follows that $(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g}$ is a subrepresentation of V_ψ .

If V_ψ is irreducible, then

$$(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} \simeq V_\psi.$$

The eigenvalues of σ_p on V_ψ are α_g/β_g and β_g/α_g if $\chi_K(p) = 1$, and 1 and $-1 = \alpha_g/\beta_g$ if $\chi_K(p) = -1$. In either case, $\dim_L(U_{g_\alpha}) \geq 1$, with equality if either $\alpha_g/\beta_g \neq -1$ or if p is inert in K .

If V_ψ is reducible, the character ψ is necessarily quadratic, and V_ψ further breaks up as a sum of one-dimensional representations attached to odd and even Dirichlet quadratic characters χ^- and χ^+ , respectively, satisfying $\chi^+\chi^- = \chi_K$. Because of this,

$$(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} = (\mathcal{O}_{H_g}^\times)_L^{\chi^+} \simeq L(\chi^+)$$

is isomorphic to the one-dimensional representation associated to χ^+ . The space $U_{g_\alpha} \subset (\mathcal{O}_{H_g}^\times)_L^{\chi^+}$ is therefore of dimension at most one, with equality if and only if $\alpha_g/\beta_g = \chi^+(p)$. This equality always holds when $p = \wp\bar{\wp}$ splits in K , since in that case

$$\alpha_g/\beta_g = \psi(\wp) = \psi(\bar{\wp}) = \chi^+(p) = \chi^-(p).$$

(Note that the regularity assumption $\alpha_g \neq \beta_g$ that is made on g forces all these expressions to be equal to -1 rather than 1.) If p is inert in K , then σ_p acts on V_ψ with eigenvalues 1 and -1 , and hence $\alpha_g/\beta_g = \beta_g/\alpha_g = -1$. In that case, either $\chi^+(p) = -1$ and $\dim(U_{g_\alpha}) = 1$, or p splits in the real quadratic field K^+ cut out by χ^+ , and $U_{g_\alpha} = 0$. In this last scenario, the representations V_ψ and V_g are also induced from a character of the real quadratic field K^+ , in which p splits.

b) Real dihedral forms. If the representation V_g is induced from a character ψ_g of signature $(+, -)$ of a real quadratic field K , then the character $\psi = \psi_g/\psi'_g$ is a totally odd ring class character of K , and cuts out an abelian extension of K of dihedral type. The decomposition (20) continues to hold, but now, since σ_∞ acts with eigenvalue 1 on $L(\chi_K)$, it follows that

$$(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} \simeq L(\chi_K) \quad (21)$$

in all cases. If p splits in K , then $\chi_K(p) = 1 \neq \alpha_g/\beta_g$, and hence $U_{g_\alpha} = 0$. If p is inert in K , then $\chi_K(p) = \alpha_g/\beta_g = -1$, and hence U_{g_α} is also one-dimensional, and generated over L by a fundamental unit of the real quadratic field K .

c) Exotic forms. Assume to conclude that the projective image G of the representation V_g is isomorphic to A_4 , S_4 or A_5 . Since these groups admit no faithful irreducible representations of dimension < 3 , it follows that Ad_g is irreducible, and hence

$$(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g} \simeq \text{Ad}_g.$$

The lemma follows directly in this case, since σ_p acts on Ad_g with eigenvalues 1, α_g/β_g , and β_g/α_g . $\square$

Remark 1.6. Note that condition (12) which was shown in Ch. 1.1 to imply Hypothesis C for g_α is precisely what guarantees the non-triviality of the group U_{g_α} .

Remark 1.7. The scenarios considered in (a) and (b) in the above proof are not mutually exclusive, and can arise simultaneously in certain cases where ψ_g is quadratic and ϱ_g has image isomorphic to the dihedral group of order 8: some of the numerical examples below focus on such settings.

Definition 1.8. A non-zero element u_{g_α} of U_{g_α} is called a *Stark unit attached to g_α* .

Remark 1.9. Lemma 1.5 furnishes us with many concrete instances where the unit u_{g_α} is well-defined up to multiplication by $L^\times$. This happens, for instance, when ϱ_g is induced from a character of a real quadratic field K in which p is inert, when the vector space U_{g_α} is always generated by the fundamental unit of K . The case where p splits in K is more mysterious, since $U_{g_\alpha} = 0$, and a natural definition of u_{g_α} does not seem to be available, consistent with the failure of Hypothesis C.

More generally, $\dim_L(U_{g_\alpha}) > 1$ occurs precisely when $\alpha_g/\beta_g = \pm 1$. In the special case where V_g is induced from a character of an imaginary quadratic field in which p splits, there is nonetheless a natural candidate for u_{g_α} which is well-defined up to scaling by $L^\times$: cf. Sec. 3.2 for more details.

Remark 1.10. Several results in the literature relate the Petersson norm of g to $L'(\text{Ad}_g, 0)$, which in turn is expected to admit an elementary expression as the logarithm of units in the subspace of $(\mathcal{O}_{H_g}^\times)_L^{\text{Ad}_g}$ on which $\sigma_\infty = 1$. See for example §6 of [St1], or Prop. 4.1 of [DuLi]. It is in this sense that the p -adic logarithm of u_{g_α} might be viewed as a p -adic avatar of $\langle g_\alpha, g_\alpha \rangle$.

Consider now the case where ϱ_g is reducible, i.e., g corresponds to the Eisenstein series $E_1(\chi_1, \chi_2) \in M_1(N, \chi)$, where χ_1 and χ_2 are odd and even Dirichlet characters satisfying (as motivated in Section 1.1) the condition

$$\chi_1(p) = \chi_2(p). \quad (22)$$

In that case,

$$V_g = L(\chi_1) \oplus L(\chi_2), \quad \text{Ad}_g = L \oplus L(\eta) \oplus L(\eta^{-1}), \quad \text{where } \eta := \chi_1/\chi_2.$$

Condition (22) implies that $\eta(p) = 1$ and hence p splits completely in the abelian extension $H_\eta/\mathbb{Q}$ cut out by η .

To define u_{g_α} , choose a prime ideal $\wp$ of $\mathcal{O}_{H_\eta}$ over p inducing the embedding $H_\eta \hookrightarrow \mathbb{C}_p$ chosen at the outset, and let $u_\wp \in \mathcal{O}_{H_\eta}[1/p]^\times$ be a generator of the principal ideal $\wp^h$, where h is the class number of H_η :

$$\wp^h = (u_\wp), \quad u_\wp \in (\mathcal{O}_{H_\eta}[1/\wp])^\times \pmod{\mathcal{O}_{H_\eta}^\times}.$$

The unit u_{g_α} is defined as follows (in additive notation):

Definition 1.11. Let

$$e_\eta := \frac{1}{\#G} \sum_{\sigma \in G} \eta^{-1}(\sigma) \sigma, \quad e_{\eta^{-1}} := \frac{1}{\#G} \sum_{\sigma \in G} \eta(\sigma) \sigma,$$

be the usual idempotents in $L[G]$ giving the projection onto the η and η^{-1} -isotypic component for the action of G . Set

$$u_{g_\alpha} = e_\eta u_\wp + e_{\eta^{-1}} u_\wp.$$

The unit u_{g_α} is a linear combination of *Gross-Stark units* attached to the characters η and η^{-1} , whose p -adic logarithms

$$\log_\wp(u_{g_\alpha}) = \frac{1}{\#G} \sum_{\sigma \in \text{Gal}(H_\eta/\mathbb{Q})} (\eta(\sigma) + \eta^{-1}(\sigma)) \log_\wp(\sigma(u_\wp))$$

account for the derivative at 0 of the Kubota-Leopoldt p -adic L -function of η . In this formula, $\log_\wp$ denotes Iwasawa's logarithm satisfying $\log_\wp(p) = 0$, composed with the p -adic embedding of H_η attached to $\wp$. It is not hard to see that

- (1) Although the unit $u_\wp$ is only well-defined modulo $\mathcal{O}_{H_\eta}^\times$, there is no resulting ambiguity in the unit u_{g_α} , since η and η^{-1} are odd characters and hence the idempotents e_η and $e_{\eta^{-1}}$ annihilate $\mathcal{O}_{H_\eta}^\times \otimes L$.
- (2) The unit $u_\wp$, and hence also the unit u_{g_α} , depends crucially (and not just by scaling by $L^\times$!) on the choice of the ideal $\wp$ above p , but the expression $\log_\wp u_{g_\alpha}$ is independent of this choice, up to multiplication by a root of unity.

2. ITERATED INTEGRALS AND THE GARRETT-HIDA p -ADIC L -FUNCTION

The main goal of this chapter is to recast the p -adic iterated integral that arose in the introduction as the value of a p -adic L -function at a point which lies outside its region of classical interpolation. This interpretation is crucial for the proof of Theorem 3.3 of Section 3.2, which provides the main piece of theoretical evidence for Conjecture ES in a setting where the p -adic trilinear form I'_p does not vanish.

We begin by briefly recalling a few key facts about Garrett's triple product classical L -function and the work of Harris-Kudla, Ichino, Watson and Woodbury on Jacquet's conjecture on trilinear periods.

Let (k, ℓ, m) be a triple of positive integers and let

$$g_k \in M_k(N_g, \chi_g), \quad f_\ell \in M_\ell(N_f, \chi_f), \quad h_m \in M_m(N_h, \chi_h)$$

be three newforms (assumed to be cuspidal if the weight is at least 2) satisfying $\chi_g \chi_f \chi_h = 1$. Note that this forces $k + \ell + m$ to be even. Set $N = \text{lcm}(N_f, N_g, N_h)$ and write $\mathbb{Q}(g_k, f_\ell, h_m)$ for the number field generated by the Fourier coefficients of the three forms.

Let $L(g_k \otimes f_\ell \otimes h_m, s)$ denote Garrett's triple product L -function associated to (g_k, f_ℓ, h_m) . It is a self-dual L -function defined by an Euler product of degree 8 and satisfying a functional equation with central critical point $c = \frac{k+\ell+m-2}{2}$ and root number

$$\varepsilon = \prod_{v|N\infty} \varepsilon_v, \quad \text{with} \quad \varepsilon_v(g_k \otimes f_\ell \otimes h_m) \in \{\pm 1\}.$$

The archimedean sign ε_∞ is equal to 1 if and only if the triple (k, ℓ, m) of weights is *unbalanced*, i.e., one of the weights is greater or equal to the sum of the other two. If $v \mid N$ is a finite place, the main result of [Pr] asserts that $\varepsilon_v(g_k \otimes f_\ell \otimes h_m) = +1$ if and only if there exists a $\mathbf{GL}_2(\mathbb{Q}_v)$ -invariant linear form in the tensor product $\pi_{f,v} \otimes \pi_{g,v} \otimes \pi_{h,v}$ of local admissible representations of $\mathbf{GL}_2(\mathbb{Q}_v)$ associated to the three eigenforms. This is the case if $v \nmid \gcd(N_f, N_g, N_h)$, or if one of $\pi_{f,v}$, $\pi_{g,v}$ or $\pi_{h,v}$ does not belong to the discrete series.

Motivated by the Ph.D thesis of D. Prasad, Jacquet formulated a conjecture suggesting a necessary and sufficient criterion for the non-vanishing of the central critical value $L(g_k \otimes f_\ell \otimes h_m, c)$ in terms of certain *trilinear periods*. This conjecture was proved by Harris and Kudla in [HaKu], and several authors have subsequently refined this result by providing explicit formulae relating the L -value to the periods.

In order to describe the precise relationships between L -values and trilinear periods, we *assume for the rest of this chapter* that

$$\varepsilon_v(g_k \otimes f_\ell \otimes h_m) = +1 \quad \text{for all } v \mid N\infty. \quad (23)$$

When $(k, \ell, m) = (1, 2, 1)$ and $(g_1, f_2, h_1) = (g, f, h)$ are the modular forms considered in the introduction, then $L(g_k \otimes f_\ell \otimes h_m, s) = L(E/\mathbb{Q}, \varrho_{gh}, s)$ is the Hasse-Weil-Artin L -function considered there, and (23) is just Hypothesis A. The above discussion yields sufficient conditions for Hypothesis A to hold, and shows that this is not a very stringent assumption, in practice. Condition (23) at $v = \infty$ implies that the triple (k, ℓ, m) is unbalanced; assume for the sake of definiteness that $k \geq \ell + m$ from now on.

As in the introduction, let

$$S_k(N, \chi_g)[g_k], \quad S_\ell(N, \chi_f)[f_\ell], \quad S_m(N, \chi_h)[h_m]$$

denote the eigenspaces corresponding to the three forms in level N with respect to the Hecke operators $\{T_\ell, \ell \nmid N\}$.

For any $w \geq 1$ and any Dirichlet character χ of conductor N , let $S_w^{\text{nh}}(N, \chi)$ denote the space of nearly holomorphic modular forms of weight w , level N and nebentype χ , and let

$$\delta = \frac{1}{2\pi i} \left(\frac{d}{dz} + \frac{w}{z - \bar{z}} \right) : S_w^{\text{nh}}(N, \chi) \longrightarrow S_{w+2}^{\text{nh}}(N, \chi)$$

denote the Shimura-Maass operator, as described for instance in [Hi2, Ch. X]. Let also

$$\langle \cdot, \cdot \rangle : S_w^{\text{nh}}(N, \chi) \times S_w^{\text{nh}}(N, \chi) \longrightarrow \mathbb{C}, \quad \langle f_1, f_2 \rangle = \int_{\Gamma_1(N) \backslash \mathfrak{H}} \overline{f_1(z)} f_2(z) y^w \frac{dx dy}{y^2},$$

denote the Petersson scalar product on these spaces.

Set $t = (k - \ell - m)/2 \geq 0$. A *test vector* for the triple (g_k, f_ℓ, h_m) is a triplet

$$(\check{g}_k, \check{f}_\ell, \check{h}_m) \in S_k(N, \chi_g)[g_k] \times S_\ell(N, \chi_f)[f_\ell] \times S_m(N, \chi_h)[h_m].$$

For such a test vector, recall that we denote $g_k^* = g_k \otimes \chi_g^{-1}$, and define the quantities

$$I(\check{g}_k, \check{f}_\ell, \check{h}_m) := \langle \check{g}_k^*, \delta^t(\check{f}_\ell) \cdot \check{h}_m \rangle$$

as well as

$$J(\check{g}_k, \check{f}_\ell, \check{h}_m) := I(\check{g}_k, \check{f}_\ell, \check{h}_m) / \langle g_k^*, g_k^* \rangle, \quad \omega(\check{g}_k, \check{f}_\ell, \check{h}_m) := I(\check{g}_k^*, \check{f}_\ell^*, \check{h}_m^*) / I(\check{g}_k, \check{f}_\ell, \check{h}_m). \quad (24)$$

The following result is a direct consequence of the fundamental work of Ichino [Ich]. The precise formulation given here is only valid under assumption (23) and follows by combining loc. cit. with a formula due to Hida for a special value of the L -function of the adjoint of a modular form and the precise formulae of Watson [Wa] and Woodbury [Wo1], [Wo2].

Proposition 2.1. **a)** For any test vector $(\check{g}_k, \check{f}_\ell, \check{h}_m)$,

$$|I(\check{g}_k, \check{f}_\ell, \check{h}_m)|^2 = \mathfrak{a}(k, \ell, m) \cdot \mathfrak{f}(k, \ell, m) \cdot \prod_{v|N\infty} C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot L(g_k \otimes f_\ell \otimes h_m, c) \quad (25)$$

where

- i) $\mathfrak{a}(k, \ell, m) = \left(\frac{k+\ell+m-4}{2}\right)! \cdot \left(\frac{k+\ell-m-2}{2}\right)! \cdot \left(\frac{k-\ell+m-2}{2}\right)! \cdot \left(\frac{k-\ell-m}{2}\right)! \cdot \pi^{-2k}$.
- ii) $\mathfrak{f}(k, \ell, m) = A \cdot 2^B$ where $A \in \mathbb{Q}^\times$ depends only on the levels of the eigenforms and of their characters, $B = B(k, \ell, m) \in \mathbb{Z}$ is a linear function on the weights k, ℓ, m .
- iii) for all $v \mid N\infty$, the local term $C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \in \mathbb{Q}(g_k, h_\ell, h_m)$ depends only on the admissible representations of $\mathbf{GL}_2(\mathbb{Q}_v)$ associated to the three cuspforms and on the local components at v of the choice of $(\check{g}_k, \check{f}_\ell, \check{h}_m)$.

b) There exists a test vector $(\check{g}_k, \check{f}_\ell, \check{h}_m)$ for which $C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \neq 0$ for all $v \mid N\infty$.

c) Assume N is square-free. Then there exists a particular test vector $(\check{g}_k, \check{f}_\ell, \check{h}_m)$ such that $W_v := C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \in \mathbb{Q}^\times$ is a rational constant for all $v \mid N$.

Proof. Let $\Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2)$ and $\Gamma_{\mathbb{C}}(s) := 2(2\pi)^{-s}\Gamma(s)$, where $\Gamma(s)$ is the usual gamma function. For any eigenform $\varphi \in S_w(N, \chi)$, let $L(\text{Ad}(\varphi), s) = L(\text{sym}^2(\varphi), \chi^{-1}, s)$ denote the L -function of the adjoint square of φ and let

$$L^*(\text{Ad}(\varphi), s) = L_\infty(\text{Ad}(\varphi), s)L(\text{Ad}(\varphi), s), \text{ with } L_\infty(\text{Ad}(\varphi), s) = \Gamma_{\mathbb{R}}(s - w + 2)\Gamma_{\mathbb{C}}(s)$$

denote the completed L -function. The Rankin-Selberg method relates the special value of $L(\text{Ad}(\varphi), s)$ at $s = w$ to the Petersson norm of φ . A completely explicit formula is given in [Hi1, Theorem 5.1], which, combined with the value of $L_\infty(\text{Ad}(\varphi), s)$ at $s = w$, yields

$$L^*(\text{Ad}(\varphi), w) = 2^w A_\varphi \times \langle \varphi, \varphi \rangle \quad (26)$$

where $A_\varphi \in \mathbb{Q}^\times$ is a constant which only depends on the level of φ and conductor of χ .

Likewise, let

$$L^*(g_k \otimes f_\ell \otimes h_m, s) := L_\infty(g_k \otimes f_\ell \otimes h_m, s)L(g_k \otimes f_\ell \otimes h_m, s)$$

denote the completed Garrett L -function, where

$$L_\infty(g_k \otimes f_\ell \otimes h_m, s) := \Gamma_{\mathbb{C}}(s)\Gamma_{\mathbb{C}}(s+1-m)\Gamma_{\mathbb{C}}(s+1-\ell)\Gamma_{\mathbb{C}}(s+2-\ell-m). \quad (27)$$

The main result of [Ich] (see also [Wo1, Theorem 2.1] for a reformulation of the same statement) asserts that for any test vector $(\check{g}_k, \check{f}_\ell, \check{h}_m)$,

$$\frac{|I(\check{g}_k, \check{f}_\ell, \check{h}_m)|^2}{\langle g_k, g_k \rangle \cdot \langle f_\ell, f_\ell \rangle \cdot \langle h_m, h_m \rangle} = C \cdot \frac{\prod_{v|N\infty} C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot L^*(g_k \otimes f_\ell \otimes h_m, c)}{L^*(\text{Ad}(g_k), k)L^*(\text{Ad}(f_\ell), \ell)L^*(\text{Ad}(h_m), m)}, \quad (28)$$

where

- $C \in \mathbb{Q}^\times$ is an absolute constant which only depends on the normalisation taken for the Petersson scalar product;
- for any $v \mid N\infty$, $C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \in \mathbb{Q}(g_k, f_\ell, h_m)$ is as described in iii) above.

Combining (28) with (26), and evaluating (27) at $s = c$ it follows that

$$|I(\check{g}_k, \check{f}_\ell, \check{h}_m)|^2 = \mathfrak{a}(k, \ell, m) \cdot \mathfrak{f}(k, \ell, m) \cdot \prod_{v|N\infty} C_v \cdot L(g_k \otimes f_\ell \otimes h_m, c),$$

where we have set $\mathfrak{f}(k, \ell, m) = 2^{-3k-\ell-m+4} \cdot C \cdot (A_{g_k} A_{f_\ell} A_{h_m})^{-1}$. This proves **a)**. Statement **b)** was originally known as Jacquet's conjecture, and was proved by Harris and Kudla in [HaKu]. As for **c)**, we refer to [Wo1, Theorem 5.1]. $\square$

Remark 2.2. The choice of test vector in c) is a completely explicit one, given by Watson [Wa] in the case $N_g = N_f = N_h$ and more recently by Woodbury [Wo1] for arbitrary square-free levels. We shall refer to it as the *Watson-Woodbury test vector*. The resulting value of $W_v \in \mathbb{Q}^\times$ for $v \mid N$ is also explicit (see [Wa, Theorem 3] and [Wo1, Theorem 5.1] for the recipes), and we shall refer to it as the *Watson-Woodbury local constant at v* . Note we say nothing about $C_\infty(\check{g}_k, \check{f}_\ell, \check{h}_m)$, although having a tight control of it would be desirable. See [Wo2] for the latest developments, which assert that in favorable scenarios (such as the one considered in Sec. 3.2 below) the Watson-Woodbury test vector yields $W_\infty = 1$.

Let $\mathbf{g}$, $\mathbf{f}$, and $\mathbf{h}$ be three Hida families of p -adic ordinary eigenforms of tame level N_g , N_f , and N_h , with tame nebentype characters χ_g , χ_f and χ_h , respectively, satisfying $\chi_g \chi_f \chi_h = 1$ as above. Assume the three Hida families are *cuspidal*, meaning that all their specialisations at classical weights $k > 1$ (though not necessarily at $k = 1$) are cuspidal modular forms. Let

$$\Omega := \mathrm{Hom}(\mathbb{Z}_p^\times, \mathbb{Z}_p^\times) \simeq \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p \quad (29)$$

denote the weight space, which contains the integers as a dense subset. Hida families are families of overconvergent modular forms parametrized by rigid analytic spaces equipped with a finite *weight map* to Ω . Fix classical specialisations g_1 , f_2 and h_1 of weights $(1, 2, 1)$ of $\mathbf{g}$, $\mathbf{f}$ and $\mathbf{h}$ respectively. When the weight map is étale at these points, one can choose suitably small neighborhoods of these points, denoted $\mathcal{U}_g$, $\mathcal{U}_f$ and $\mathcal{U}_h$ respectively, which can be identified with subsets of Ω (and are even contained in a single residue class modulo $p-1$). This identification will be made from now on in order to lighten the discussion.

Define $\mathcal{U}_g^{\mathrm{cl}} \subseteq \mathcal{U}_g \cap \mathbb{Z}_{\geq 1}$ to be the set of weights in $\mathcal{U}_g$ at which $\mathbf{g}$ specializes to a *classical* eigenform. By the work of Hida, $\mathcal{U}_g^{\mathrm{cl}}$ contains $\mathcal{U}_g \cap \mathbb{Z}_{\geq 2}$, while $k = 1$ belongs to $\mathcal{U}_g^{\mathrm{cl}}$ by assumption. For any $k \in \mathcal{U}_g^{\mathrm{cl}}$, let $g_k \in S_k(N_g, \chi_g)$ denote the classical eigenform whose ordinary p -stabilisation is the specialisation of $\mathbf{g}$ at k , written as

$$g_{k,\alpha} = \nu_k(\mathbf{g}) \in S_k(N_g p, \chi_g).$$

Similar notations are adopted for f_ℓ and h_m . Note that the subfield $\mathbb{Q}(g_k, f_\ell, h_m)$ of $\mathbb{C}_p$ is a finite extension of $\mathbb{Q}$ (whose degree varies in general as a function of the classical weights (k, ℓ, m)). Define a dense subset $\mathcal{U}_{gh}^{\mathrm{cl}} \subset \mathcal{U}_g \times \mathcal{U}_f \times \mathcal{U}_h$ by

$$\mathcal{U}_{gh}^{\mathrm{cl}} = \{(k, \ell, m) \in \mathcal{U}_g^{\mathrm{cl}} \times \mathcal{U}_f^{\mathrm{cl}} \times \mathcal{U}_h^{\mathrm{cl}}, \text{ such that } k \geq \ell + m\}.$$

Let $\check{\mathbf{g}}$, $\check{\mathbf{f}}$, $\check{\mathbf{h}}$ be Λ -adic test vectors of tame level N associated to $\mathbf{g}$, $\mathbf{f}$ and $\mathbf{h}$ as described in [DR1, Definition 2.16]. Such test vectors may be written down explicitly as $\check{\mathbf{g}} = \sum_{d \mid N/N_g} \lambda_d \mathbf{g}(q^d)$, where d ranges over the positive divisors of N/N_g and $\lambda_d \in \Lambda_g = \mathcal{O}(\mathcal{U}_g)$ are Iwasawa functions on the domain $\mathcal{U}_g$ (and similarly for $\check{\mathbf{f}}$ and $\check{\mathbf{h}}$).

Under the running assumption on local signs, a three-variable p -adic L -function

$$\mathcal{L}_p^g(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}}) : \mathcal{U}_g \times \mathcal{U}_f \times \mathcal{U}_h \longrightarrow \mathbb{C}_p \quad (30)$$

was constructed in [DR1, § 4.2], which is characterized by the following interpolation property at triplets $(k, \ell, m) \in \mathcal{U}_{gh}^{\mathrm{cl}}$:

$$\mathcal{L}_p^g(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}})(k, \ell, m) = \mathfrak{e}(k, \ell, m) \cdot J(\check{g}_k, \check{f}_\ell, \check{h}_m), \quad (31)$$

where

$$\mathfrak{e}(k, \ell, m) = \mathcal{E}(g_k, f_\ell, h_m) / \mathcal{E}_0(g_k) \mathcal{E}_1(g_k) = \mathcal{E}(g_k^*, f_\ell^*, h_m^*) / \mathcal{E}_0(g_k^*) \mathcal{E}_1(g_k^*)$$

denotes the Euler factor introduced in [DR1, Theorem 1.3].

The next proposition spells out the interpolation property tying this p -adic L -function to central critical L -values.

Proposition 2.3. *For any Λ -adic test vectors $\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}}$ and every $(k, \ell, m) \in U_{fgh}^{\text{cl}}$, the following formula is valid:*

$$\begin{aligned} \mathcal{L}_p^g(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}}) \cdot \mathcal{L}_p^g(\check{\mathbf{g}}^*, \check{\mathbf{f}}^*, \check{\mathbf{h}}^*)(k, \ell, m) &= \mathfrak{e}(k, \ell, m)^2 \cdot \mathfrak{a}(k, \ell, m) \cdot \mathfrak{f}(k, \ell, m) \\ &\times \prod_{v|N\infty} C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot \frac{L(g_k \otimes f_\ell \otimes h_m, \frac{k+\ell+m-2}{2})}{\langle g_k^*, g_k^* \rangle^2} \end{aligned} \quad (32)$$

Proof. Note that

$$|I(\check{g}_k, \check{f}_\ell, \check{h}_m)|^2 = I(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot \overline{I(\check{g}_k, \check{f}_\ell, \check{h}_m)} = I(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot I(\check{g}_k^*, \check{f}_\ell^*, \check{h}_m^*),$$

and that

$$\langle g_k, g_k \rangle = \langle g_k^*, g_k^* \rangle.$$

It thus follows from (31) and (25) that

$$\begin{aligned} \mathcal{L}_p^g(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}}) \cdot \mathcal{L}_p^g(\check{\mathbf{g}}^*, \check{\mathbf{f}}^*, \check{\mathbf{h}}^*)(k, \ell, m) &= \mathfrak{e}(k, \ell, m)^2 \frac{|I(\check{g}_k, \check{f}_\ell, \check{h}_m)|^2}{\langle g_k^*, g_k^* \rangle^2} \\ &= \mathfrak{e}(k, \ell, m)^2 \cdot \mathfrak{a}(k, \ell, m) \cdot \mathfrak{f}(k, \ell, m) \\ &\times \prod_{v|N\infty} C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \cdot \frac{L(g_k \otimes f_\ell \otimes h_m, \frac{k+\ell+m-2}{2})}{\langle g_k^*, g_k^* \rangle^2}. \end{aligned}$$

□

Remark 2.4. An inspection of the definition of the Watson-Woodbury test vector shows that, when we let it vary along the Hida families $\mathbf{g}, \mathbf{f}$ and $\mathbf{h}$, it gives rise to a triplet $(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}})$ of Λ -adic test vectors of tame level N , whose specialization at any $(k, \ell, m) \in \mathcal{U}_g^{\text{cl}} \times \mathcal{U}_f^{\text{cl}} \times \mathcal{U}_h^{\text{cl}}$ is an eigenform with coefficients in $\mathbb{Q}(g_k, f_\ell, h_m)$, a finite extension of $\mathbb{Q}$; and moreover, the constants $W_v = C_v(\check{g}_k, \check{f}_\ell, \check{h}_m) \in \mathbb{Q}^\times$ in Proposition 2.1 c) depend on v and N but not on the triplet (k, ℓ, m) of weights.

Denote by $k \mapsto \mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}_2, \check{h}_1)(k)$ the restriction of $\mathcal{L}_p^g(\check{\mathbf{g}}, \check{\mathbf{f}}, \check{\mathbf{h}})$ to $\mathcal{U}_g^{\text{cl}} \times \{f_2\} \times \{h_1\}$. Because it does not involve the variation of f_2 in a Hida family, this restriction has the virtue of being defined a bit more generally, without any ordinariness assumptions on f_2 . The next proposition relates this restriction of the Garrett-Hida p -adic L -function to p -adic iterated integrals:

Proposition 2.5. *Assume that $\nu_1(\mathbf{g}) = g_{1,\alpha} \in M_1(N_g p, \chi_g)$ and $\nu_1(\mathbf{h}) = h_{1,\alpha} \in M_1(N_h p, \chi_h)$ are classical, and that $g_{1,\alpha}^*$ satisfies the classicality property (Hypothesis C). Then there exists a linear form*

$$\check{\gamma}_\alpha : M_1(Np, \chi_g^{-1})_L[g_{1,\alpha}^*] \longrightarrow L$$

with $L = \mathbb{Q}(g_1, h_1)$ such that

$$\mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}_2, \check{h}_1)(1) = \int_{\check{\gamma}_\alpha} \check{f}_2 \cdot \check{h}_1.$$

Proof. Fix a finite flat extension $\Lambda^\dagger$ of Λ , sufficiently large so that it contains the coefficients of all the Λ -adic modular forms considered below, and write $\mathbf{S}^{\text{ord}}(N; \Lambda^\dagger)$ for the space of Λ -adic modular forms with coefficients in $\Lambda^\dagger$. The Hida family $\mathbf{g}$ gives rise to a subspace

$$\mathbf{S}^{\text{ord}}(N; \Lambda^\dagger)[\mathbf{g}] := \left\{ \check{\mathbf{g}} \in \mathbf{S}^{\text{ord}}(N; \Lambda^\dagger) \quad \text{such that } T_n \check{\mathbf{g}} = a_n(\mathbf{g}) \check{\mathbf{g}}, \quad \text{for all } (n, N) = 1 \right\}.$$

Letting $\mathcal{L}^\dagger$ denote the fraction field of $\Lambda^\dagger$, the vector space $\mathbf{S}^{\text{ord}}(N; \mathcal{L}^\dagger)[\mathbf{g}]$ is finite-dimensional over $\mathcal{L}^\dagger$ and has for basis the set

$$\{\mathbf{g}(q^d)\}_{d|(N/N_g)} \quad (33)$$

of Λ -adic forms. By [DR1, Lemma 2.19], there exists a linear operator

$$J(\check{\mathbf{g}}^*) : S^{\text{ord}}(N, \Lambda^\dagger) \longrightarrow \mathcal{L}^\dagger \otimes_\Lambda \Lambda^\dagger, \quad \phi \mapsto J(\check{\mathbf{g}}^*, \phi)$$

characterized uniquely by the interpolation property that for all integers $k \in \mathbb{Z}_{\geq 2} \cap \mathcal{U}_g \cap \mathcal{U}_\phi$, its specialisation at weight k is the linear operator

$$\nu_k(J(\check{\mathbf{g}}^*)) : S_k^{\text{ord}}(N, \chi_g^{-1})_{\mathbb{Q}_p(g_k)} \longrightarrow \mathbb{Q}_p(g_k), \quad \phi \mapsto \frac{\langle \check{g}_k^*, \phi \rangle}{\langle g_k^*, g_k^* \rangle}.$$

For any $d|(N/N_g)$ let $c_d : S^{\text{ord}}(N, \mathcal{L}^\dagger) \longrightarrow \mathcal{L}^\dagger$ denote the functional which associates to ϕ its coefficient in $\mathbf{g}(q^d)$ with respect to the basis in (33). It follows from the proofs of [DR1, Lemmas 2.12 and 2.19] that

$$J(\check{\mathbf{g}}^*) = \sum_{d|(N/N_g)} \lambda_d \cdot c_d$$

where $\lambda_d \in \mathcal{L}^\dagger$ are elements which, viewed as functions of $\mathbb{Z}_{\geq 2}$ by means of the rule $k \mapsto \nu_k(\lambda_d)$, can be expressed as polynomials in the expressions q^k , $a_q(g_k)$, $\frac{1}{q}$ and $\frac{1}{q+1}$ as q ranges over the divisors of N . Define

$$\check{\gamma}_\alpha = \nu_1(J(\check{\mathbf{g}}^*)) : S_1^{\text{oc,ord}}(N, \chi_g^{-1})_{L_p} \longrightarrow L_p$$

to be the specialisation of $J(\check{\mathbf{g}}^*)$ in weight 1. It factors through $S_1^{\text{oc,ord}}(N, \chi_g^{-1})_{L_p}[[g_{1,\alpha}^*]]$, which by the classicality hypothesis imposed on $g_{1,\alpha}^*$ is isomorphic to the corresponding space $M_1(Np, \chi_g^{-1})_{L_p}[g_{1,\alpha}^*]$ of classical modular forms.

As explained in the introduction, Hypothesis C equips $S_1^{\text{oc,ord}}(N, \chi_g^{-1})_{L_p}[[g_{1,\alpha}^*]]$ with an L -rational structure, denoted $S_1^{\text{oc,ord}}(N, \chi_g^{-1})_L[[g_{1,\alpha}^*]]$. In view of the description given above for λ_d as functions of k , it follows that $\nu_1(\lambda_d) \in L$ and hence γ_α is L -rational, that is to say, it belongs to $M_1(Np, \chi_g^{-1})_L[g_{1,\alpha}^*]^\vee = S_1^{\text{oc,ord}}(N, \chi_g^{-1})_L[[g_{1,\alpha}^*]]^\vee$.

Let $e_{\text{ord}}(d^\bullet \check{f}_2^{[p]} \times \check{h}_1)$ denote the Λ -adic modular form whose specialisation in weight k is $e_{\text{ord}}(d^{\frac{k-3}{2}} \check{f}_2^{[p]} \times \check{h}_1)$ for all $k \in \mathbb{Z}_{\geq 2}$. By [DR1, Definition 4.4], $\mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}_2, \check{h}_1) = J(\check{\mathbf{g}}, e_{\text{ord}}(d^\bullet \check{f}_2^{[p]} \times \check{h}_1))$ and, by construction,

$$\mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}_2, \check{h}_1)(1) = \check{\gamma}_\alpha(e_{\text{ord}}(d^{-1} \check{f}_2^{[p]} \times \check{h}_1)) = \int_{\check{\gamma}_\alpha} \check{f}_2 \cdot \check{h}_1,$$

as was to be shown. $\square$

Proposition 2.5 recasts the p -adic iterated integral of Conjecture ES with $(\check{\gamma}, \check{f}, \check{h}) = (\check{\gamma}_\alpha, \check{f}_2, \check{h}_1)$ as a p -adic avatar of the classical special value $L(E, \varrho_{gh}, 1)$. This suggests a possible strategy for proving Conjecture ES in the the rank $(1, 1)$ setting where ϱ_{gh} admits a decomposition of the form $\varrho_{gh} = \varrho_1 \oplus \varrho_2 \oplus \varrho_3$ as in (10). Assuming the regulator $R_{g_\alpha}(E, \varrho_{gh})$ is factorisable, Conjecture ES might then be proved by constructing appropriate “ p -adic Artin L -series” $\mathcal{L}_p(\text{Ad}_{g_\alpha})$ associated to (the irreducible constituent containing Ad_g^+ in) Ad_g and “ p -adic Hasse-Weil-Artin L -series” $\mathcal{L}_p(E, \varrho_i)$ attached to the irreducible constituents of ϱ_{gh} , and proving that

$$\mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}_2, \check{h}_1) \cdot \mathcal{L}_p(\text{Ad}_{g_\alpha}) \stackrel{?}{=} \mathcal{L}_p(E, \varrho_1) \cdot \mathcal{L}_p(E, \varrho_2) \cdot \mathcal{L}_p(E, \varrho_3) \quad (34)$$

and

$$\begin{aligned} \mathcal{L}_p(\text{Ad}_{g_\alpha})(1) &\stackrel{?}{=} \log_p(u_{g_\alpha}), & \mathcal{L}_p(E, \varrho_1)(1) &\stackrel{?}{=} \log_{E,p}(P_1), \\ \mathcal{L}_p(E, \varrho_3)(1) &\in L^\times, & \mathcal{L}_p(E, \varrho_2)(1) &\stackrel{?}{=} \log_{E,p}(P_2). \end{aligned} \quad (35)$$

This is the strategy carried out in §3.2: see Theorem 3.8 for the variant of (34) and (35) in the special setting considered therein. In general, identities like (34) appear to lie quite deep, and we are not even able to provide an independent definition of the p -adic L -values $\mathcal{L}_p(E, \varrho_i)$

that figure in its putative right-hand side; they are perhaps somewhat akin to the “genuine” p -adic L -functions sought for in [Hi3].

3. HEEGNER POINTS AND THETA SERIES OF IMAGINARY QUADRATIC FIELDS

Let K be an imaginary quadratic field of discriminant $-D_K$. The aim of this chapter is to study Conjecture ES in the case where both g and h are theta series associated to K . After setting up the basic notations and assumptions, the cases in which the prime p splits or remains inert in K are treated separately in Sections 3.2 and 3.3.

Just as in the introduction, let E be an elliptic curve over $\mathbb{Q}$ and $f \in S_2(N_f)$ denote its associated newform. Given a finite order character

$$\psi : G_K \longrightarrow \mathbb{C}^\times$$

of conductor $\mathfrak{c}_\psi \subseteq \mathcal{O}_K$ and central character χ (viewed as a Dirichlet character), let

$$\theta_\psi \in M_1(D_K \cdot N_{K/\mathbb{Q}}(\mathfrak{c}_\psi), \chi)$$

denote the theta series associated to ψ and let $V_\psi := \text{Ind}_K^{\mathbb{Q}} \psi$ denote the two-dimensional induced representation of ψ from G_K to $G_{\mathbb{Q}}$. It is known that θ_ψ is Eisenstein if and only if V_ψ is reducible, which occurs precisely when $\psi = \psi'$, where ψ' denotes the character of G_K defined by the rule $\psi'(\sigma) = \psi(\sigma_0 \sigma \sigma_0^{-1})$. Here σ_0 is any choice of element in $G_{\mathbb{Q}} \setminus G_K$.

Fix two finite order characters ψ_g and ψ_h of K of conductors $\mathfrak{c}_g$ and $\mathfrak{c}_h$. Assume that the central character χ of ψ_h is inverse to that of ψ_g , and set

$$g := \theta_{\psi_g} \in M_1(N_g, \bar{\chi}), \quad h := \theta_{\psi_h} \in M_1(N_h, \chi).$$

Define $\psi_1 = \psi_g \psi_h$ and $\psi_2 = \psi_g \psi'_h$. The characters ψ_1 and ψ_2 are ring class characters of K , associated to orders $\mathcal{O}_{c_1}$ and $\mathcal{O}_{c_2}$ in $\mathcal{O}_K$ of conductors c_1 and c_2 respectively. Such characters are sometimes also called, synonymously, dihedral or anticyclotomic, characters of K . Note that ψ_g and ψ_h were not assumed to be necessarily anticyclotomic.

One checks that there is a decomposition of Artin representations and, in parallel to that, a factorisation of L -series as follows:

$$V_{gh} = V_{\psi_1} \oplus V_{\psi_2}, \quad L(E, \varrho_{gh}, s) = L(E/K, \psi_1, s) \cdot L(E/K, \psi_2, s). \quad (36)$$

Assume also for simplicity that $\gcd(N_f, \mathfrak{c}_g \mathfrak{c}_h) = 1$. This implies that for all places of K above any place $v \mid N_\infty$ of $\mathbb{Q}$, the local signs of $L(E/K, \psi_1, s)$ and of $L(E/K, \psi_2, s)$ are all equal, and hence $\varepsilon_v(E, \varrho_{gh}) = (\pm 1)^2 = 1$. This automatically implies that Hypothesis A holds.

Let H denote the ring class field of conductor $c := \text{lcm}(c_1, c_2)$ of K , which contains the fields cut out by ψ_1 and ψ_2 . The field L of coefficients that is implicit in (36) can be taken to be any finite extension of the field generated by the traces of V_{ψ_1} and V_{ψ_2} .

3.1. Heegner points and elliptic units. In the rank $(1, 1)$ setting of Conjecture ES in which $r(E, V_{\psi_1}) = r(E, V_{\psi_2}) = 1$, the global points and units that arise on the right hand side are expected to be expressible in terms of Heegner points and elliptic units. Not surprisingly, these global constructions are the key to the theoretical evidence that has been amassed in support of Conjecture ES in the rank $(1, 1)$ setting when p is split in K .

The Heegner point construction is available under the following

Heegner hypothesis: *The level of f may be factorised as $N_f = N_f^+ N_f^-$ where*

- *there exists an integral ideal $\mathfrak{N}$ of K such that $\mathcal{O}_K/\mathfrak{N} \simeq \mathbb{Z}/N_f^+ \mathbb{Z}$,*
- *N_f^- is the square-free product of an even number of primes which are all inert in K .*

This hypothesis implies that the signs in the functional equations for $L(E, V_{\psi_1}, s)$ and $L(E, V_{\psi_2}, s)$ are -1 , and hence that the analytic ranks $r_{\text{an}}(E, V_{\psi_1})$ and $r_{\text{an}}(E, V_{\psi_2})$ are both

odd. In particular, Hypothesis B of the introduction is automatically satisfied. It is further expected that, in “generic” instances,

$$r_{\text{an}}(E, V_{\psi_1}) = r_{\text{an}}(E, V_{\psi_2}) = 1, \quad (37)$$

and that these equalities certainly hold when the p -adic iterated integrals attached to the triple (f, g, h) are not identically zero.

Recall from the previous chapter the running assumption that g_α satisfies the classicality Hypothesis C. If g is cuspidal (that is, if $\psi_g \neq \psi'_g$), we in fact assume $\alpha_g \neq \beta_g$ and (12), which ensures Hypothesis C by Proposition 1.2. If $g = E_1(\chi_1, \chi_2)$ is Eisenstein (that is, if $\psi_g = \psi'_g$), recall that Proposition 1.3 asserts that Hypothesis C implies $\chi_1(p) = \chi_2(p)$.

The decomposition in (36) can be further refined to

$$\varrho_{gh} = \varrho_1 \oplus \varrho_2 \oplus \varrho_3, \quad (38)$$

where

- for $i = 1, 2$, ϱ_i is the unique irreducible constituent of V_{ψ_i} for which $r_{\text{an}}(E, \varrho_i)$ is odd;
- ϱ_3 is the $G_{\mathbb{Q}}$ -stable complement of $\varrho_1 \oplus \varrho_2$ in ϱ_{gh} .

This decomposition is uniquely determined by (f, g, h) , up to possibly switching ϱ_1 and ϱ_2 . When (37) is satisfied, the decomposition (38) satisfies the conditions

$$r_{\text{an}}(E, \varrho_1) = r_{\text{an}}(E, \varrho_2) = 1, \quad r_{\text{an}}(E, \varrho_3) = 0. \quad (39)$$

As will be explained shortly, the work of Gross, Zagier, Kolyvagin and Zhang further shows that the same equations as in (39) hold for the corresponding algebraic ranks:

$$r(E, \varrho_1) = r(E, \varrho_2) = 1, \quad r(E, \varrho_3) = 0, \quad (40)$$

consistent with the predictions made by the Birch and Swinnerton-Dyer conjecture, and so we are in the rank $(1, 1)$ setting of Conjecture ES. It also leads to the construction of explicit non-trivial elements in $E(H)_L^{\varrho_1}$ and $E(H)_L^{\varrho_2}$ from *Heegner points*, which we now describe.

Let X be the Shimura curve associated to an Eichler order of level N_f^+ in the (indefinite, thanks to the Heegner hypothesis) quaternion algebra of discriminant N_f^- over $\mathbb{Q}$. Let

$$\pi_E : \text{Jac}(X) \rightarrow E$$

denote a modular parametrisation of E arising from the fact that E is an abelian variety quotient of the jacobian of X .

Let $X(H)_{\mathcal{O}_c} \subset X(H)$ denote the set of Heegner points attached to the order of conductor c in K . It is a finite set, and any two points in it are related by the action of an Atkin-Lehner involution and an element of $\text{Gal}(H/K)$. A *Heegner divisor* of conductor c on X is any degree 0 divisor supported on $X(H)_{\mathcal{O}_c}$, and a *Heegner point* on E of conductor c is the image of such a divisor.

After fixing a choice of $P \in X(H)_{\mathcal{O}_c}$, any ring class character ψ of conductor c gives rise to a degree zero divisor on X by setting

$$D_\psi := \begin{cases} \sum_{\sigma \in G} \psi^{-1}(\sigma) P^\sigma, & \text{if } \psi \neq 1, \\ \sum_{\sigma \in G} P^\sigma - \#G \cdot \infty & \text{if } \psi = 1, \end{cases} \quad (41)$$

where ∞ is the degree one divisor in $\text{Div}(X) \otimes \mathbb{Q}$ defined by $\infty = \frac{1}{2g-2} \mathcal{K}_X$, and $\mathcal{K}_X$ denotes the canonical class on X . The *Heegner point attached to ψ* is then defined to be

$$P_\psi := \pi_E(D_\psi) \in E(H)_L.$$

Note that the point P_ψ actually belongs to $E(H)_L^{V_\psi}$, and that the same is true of $P_{\bar{\psi}}$.

Lemma 3.1. *Assume that (39) is satisfied. Then (40) holds as well, and, for $i = 1$ and 2 , the Mordell-Weil group $E(H)_L^{\varrho_i}$ is generated by the vectors P_{ψ_i} and $P_{\bar{\psi}_i}$. If $\psi_i \neq \bar{\psi}_i$, these points are linearly independent and form a basis for $E(H)_L^{\varrho_i}$. If $\psi_i = \bar{\psi}_i$ (i.e., if ψ_i is a genus character) then $P_{\psi_i} = P_{\bar{\psi}_i}$ and this vector is a basis for $E(H)_L^{\varrho_i}$.*

Proof. Assumption (39) asserts that the central critical derivative $L'(E, \varrho_i, 1)$ is non-zero. The Gross-Zagier formula and its extension to ring class characters given in [Zh] shows that the vectors P_{ψ_i} and $P_{\bar{\psi}_i}$ have non-zero canonical height and hence are non-zero in $E(H)_L^{V_{\psi_i}}$. The main theorem of [BD0] (and a suitable extension to Heegner points arising from Shimura curves) implies that $r(E, V_{\psi_i}) = 1$ and that $E(H)_L^{\varrho_i}$ is generated by the Heegner vectors P_{ψ_i} and $P_{\bar{\psi}_i}$. The result follows. $\square$

Note that the points P_{ψ_i} and $P_{\bar{\psi}_i}$ depend on the choice of a point $P \in X(H)_{\mathcal{O}_c}$ that was made in (41), but only up to multiplication by a root of unity. More precisely, replacing P by wP^{σ_0} for $\sigma_0 \in G$ and w a Fricke involution on X has the effect of replacing P_{ψ_i} by $\pm\psi_i(\sigma_0)P_{\psi_i}$ and $P_{\bar{\psi}_i}$ by $\pm\bar{\psi}_i(\sigma_0)P_{\bar{\psi}_i}$. In particular, the L -vector space in $E(H)_L$ spanned by P_{ψ_i} is completely canonical and does not depend on the choice of P .

It will also be convenient to introduce a related basis of $E(H)_L^{\varrho_i}$ attached to a choice of reflection τ in the dihedral Galois group $\text{Gal}(H/\mathbb{Q})$, by setting

$$P_{\psi_i}^+ = P_{\psi_i} + P_{\psi_i}^\tau, \quad P_{\psi_i}^- = P_{\psi_i} - P_{\psi_i}^\tau. \quad (42)$$

Sometimes, the basic Heegner point P used to define the points $P_{\psi_i}^\pm$ can be chosen to be fixed (up to sign) by the reflection τ , i.e., in such a way that $\tau P = \pm P$. When this is the case,

$$P_{\psi_i}^+ = P_{\psi_i} \pm P_{\bar{\psi}_i}, \quad P_{\psi_i}^- = P_{\psi_i} \mp P_{\bar{\psi}_i}.$$

But this simpler expression for the basis $(P_{\psi_i}^+, P_{\psi_i}^-)$ need not be invoked in general.

Let

$$\psi_0 := \psi_g / \psi'_g$$

be the non-trivial ring class character attached to the adjoint representation of g . In addition to the Heegner points P_{ψ_i} and $P_{\bar{\psi}_i}$, Conjecture ES also involves the elliptic unit $u \in \mathcal{O}_{H_g}^\times$ and its ψ_0 -component

$$u_{\psi_0} = \sum_{\sigma \in G} \psi_0^{-1}(\sigma) u^\sigma \in (\mathcal{O}_{H_g}^\times)_L^{V_{\psi_0}},$$

where the group law in the unit group $\mathcal{O}_{H_g}^\times$ is written additively, and the units $u_{\psi_0}^\pm$ relative to any choice of a reflection $\tau \in \text{Gal}(H/\mathbb{Q})$ are defined just as in (42):

$$u_{\psi_0}^+ = u_{\psi_0} + u_{\psi_0}^\tau, \quad u_{\psi_0}^- = u_{\psi_0} - u_{\psi_0}^\tau. \quad (43)$$

Note that, just as for the points P_{ψ_i} and $P_{\bar{\psi}_i}$, the group $\text{Gal}(H/K)$ acts on the element u_{ψ_0} through the character ψ_0 but does not preserve the line spanned by either $u_{\psi_0}^+$ or $u_{\psi_0}^-$.

3.2. Conjecture ES when p splits in K . When p is split in K , the Frobenius element σ_p belongs to $\text{Gal}(H/K)$, and the choice of a prime $\wp$ of K above p is sufficient to determine σ_p as an element of this group. The choices of $\alpha_g, \beta_g, \alpha_h$ and β_h can then be made so that

$$\alpha_g = \psi_g(\wp), \quad \beta_g = \psi_g(\bar{\wp}), \quad \alpha_h = \psi_h(\wp), \quad \beta_h = \psi_h(\bar{\wp}).$$

With the above choices,

$$\alpha_g \alpha_h = \psi_g \psi_h(\wp) = \psi_1(\sigma_p), \quad \alpha_g \beta_h = \psi_g \psi'_h(\wp) = \psi_2(\sigma_p).$$

The following lemma describes the p -adic regulator $\det R_{g_\alpha}(E, \varrho_{gh})$ that enters into Conjecture ES.

Lemma 3.2. *The regulator $\det(R_{g_\alpha}(E, \varrho_{gh}))$ is given by the following expressions:*

(1) If $r(E, V_{\psi_1}) = r(E, V_{\psi_2}) = 1$,

$$\det R_{g_\alpha}(E, \varrho_{gh}) = \log_{E,p}(P_{\psi_1}) \cdot \log_{E,p}(Q_{\psi_2}),$$

where P and Q are generators of $\text{hom}_{G_{\mathbb{Q}}}(V_{\psi_1}, E(H)_L)$ and $\text{hom}_{G_{\mathbb{Q}}}(V_{\psi_2}, E(H)_L)$ respectively;

(2) If $r(E, V_{\psi_1}) \geq 2$ or $r(E, V_{\psi_2}) \geq 2$,

$$\det R_{g_\alpha}(E, \varrho_{gh}) = 0;$$

Proof. The first part follows from the fact that, after choosing a σ_p -eigenbasis for $V_{gh}^{g_\alpha}$, the matrix $R_{g_\alpha}(E, \varrho_{gh})$ takes the form

$$R_{g_\alpha}(E, \varrho_{gh}) = \begin{pmatrix} \log_{E,p}(P_{\psi_1}) & 0 \\ 0 & \log_{E,p}(Q_{\psi_2}) \end{pmatrix}.$$

In the rank 2 setting, assume without loss of generality that $r(E, V_{\psi_1}) = 2$ and that $r(E, V_{\psi_2}) = 0$. The two dimensional subspace $V_{gh}^{g_\alpha}$ then intersects V_{ψ_1} and V_{ψ_2} in one-dimensional subspaces which determine a decomposition of $V_{gh}^{g_\alpha}$. Choosing a basis (v_1, v_2) for $V_{gh}^{g_\alpha}$ which is compatible with this decomposition, and a basis Φ_1, Φ_2 of $\text{Hom}_{G_{\mathbb{Q}}}(V_{\psi_1}, E(H)_L)$, produces a matrix $R_{g_\alpha}(E, \varrho_{gh})$ whose second column is zero. $\square$

The lemma above motivates focussing our attention on the rank $(1, 1)$ setting of Conjecture ES when p is split, which is what shall be done for the remainder of Section 3.2. Let us assume then our Heegner hypothesis is satisfied and let

$$P_1 := P_{\psi_1}, \quad P_2 := P_{\psi_2}$$

be the Heegner points constructed in the previous section. The main result of this section is:

Theorem 3.3. *Assume $N = \text{lcm}(N_f, N_g, N_h)$ is square-free and $p \nmid 2N$. If $r_{\text{an}}(E, V_{\psi_1}) > 1$ or $r_{\text{an}}(E, V_{\psi_2}) > 1$, the iterated integrals attached to the triple (f, g, h) vanish identically. Otherwise, there exists a finite extension L of $\mathbb{Q}(\psi_g, \psi_h)$, a scalar $\lambda \in L^\times$ and test vectors*

$$(\check{f}, \check{\gamma}_{g_\alpha}, \check{h}) \in S_2(N)_L[f] \times M_1(Np, \chi)_L^\vee[g_\alpha] \times M_1(N, \chi)_L[h]$$

for which

$$\int_{\check{\gamma}_{g_\alpha}} \check{f} \cdot \check{h} = \lambda \cdot \frac{\log_{E,p}(P_1) \log_{E,p}(P_2)}{\log_p(u_{g_\alpha})}.$$

Remark 3.4. In the setting in which h is an Eisenstein series an alternative and simpler proof is available which leads to a more precise version of the theorem. Namely, in the proof one interpolates h through an Eisenstein family (rather than a CM cuspidal family) and uses the more explicit Rankin-Selberg formula in place of the Garrett formula. The field L can be shown to be the field $\mathbb{Q}(\psi_g)$ generated by the values of ψ_g in that case. The authors believe that in fact L can always be taken to be $\mathbb{Q}(\psi_g, \psi_h)$, but fall short of proving this in general.

For example, let χ_K denote a quadratic Dirichlet character of prime conductor D_K associated to an imaginary quadratic field K . Assume $N = D_K$ and that $\psi_g = \psi_h = 1$, so that

$$g = h = \theta_1 \in M_1(D_K, \chi_K),$$

is the Eisenstein theta series associated to the trivial character of K . Then one proves the striking formula

$$\int_{\gamma_{g_\alpha}} f \cdot h = \frac{|E(\mathbb{F}_p)|^2 \log_{E,p}(P_K)^2}{p(p-1)h_K \log_p(u_\varphi)} \quad (44)$$

where $|E(\mathbb{F}_p)|$ is the number of points on E over the finite field $\mathbb{F}_p$, h_K is the class number of K , P_K is the Heegner point in $E(K)$, and $u_\varphi \in K^\times$ is any p -unit satisfying $(u_\varphi) = \wp^{h_K}$.

Formula (44) was first discovered experimentally by the authors (on Christmas day), after a long theoretical and experimental investigation trying to uncover the arithmetic meaning of these p -adic iterated integrals. It is the prototype for all the formulae in this paper.

The remainder of this section is devoted to proving Theorem 3.3. The proof rests on a comparison between the Garrett-Hida p -adic L -function of the previous section and two different types of “Katz-style” p -adic L -functions whose definitions are recalled below.

1. Katz’s two-variable p -adic L -function associated to the imaginary quadratic field K .

Let $\mathfrak{c} \subseteq \mathcal{O}_K$ be an integral ideal of K and denote Σ the set of Hecke characters of K of conductor dividing $\mathfrak{c}$. Define $\Sigma_K = \Sigma_K^{(1)} \cup \Sigma_K^{(2)} \subset \Sigma$ to be the disjoint union of the sets

$$\Sigma_K^{(1)} = \{\psi \in \Sigma \text{ of infinity type } (\kappa_1, \kappa_2), \text{ with } \kappa_1 \leq 0, \kappa_2 \geq 1\},$$

$$\Sigma_K^{(2)} = \{\psi \in \Sigma \text{ of infinity type } (\kappa_1, \kappa_2), \text{ with } \kappa_1 \geq 1, \kappa_2 \leq 0\}.$$

For all $\psi \in \Sigma_K$, the complex argument $s = 0$ is a critical point for the Hecke L -function $L(\psi^{-1}, s)$, and Katz’s p -adic L -function is constructed by interpolating the (suitably normalized) values $L(\psi^{-1}, 0)$ as ψ ranges over $\Sigma_K^{(2)}$.

More precisely, let $\hat{\Sigma}_K$ denote the completion of $\Sigma_K^{(2)}$ with respect to the compact open topology on the space of functions on a certain subset of $\mathbf{A}_K^\times$, as described in [BDP1, §5.2]. By the work of Katz [Katz1], there exists a p -adic analytic function

$$L_p(K) : \hat{\Sigma}_K \longrightarrow \mathbb{C}_p$$

which is uniquely characterized by the following interpolation property: for all $\psi \in \Sigma_K^{(2)}$ of infinity type (κ_1, κ_2) ,

$$L_p(K)(\psi) = \mathfrak{a}(\psi) \times \mathfrak{c}(\psi) \times \mathfrak{f}(\psi) \times \frac{\Omega_p^{\kappa_1 - \kappa_2}}{\Omega^{\kappa_1 - \kappa_2}} \times L_c(\psi^{-1}, 0), \quad (45)$$

where

- $\mathfrak{a}(\psi) = (\kappa_1 - 1)! \pi^{-\kappa_2}$, $\mathfrak{c}(\psi) = (1 - \psi(\wp)p^{-1})(1 - \psi^{-1}(\bar{\wp}))$, and $\mathfrak{f}(\psi) = D_K^{\kappa_2/2} 2^{-\kappa_2}$;
- $\Omega_p \in \mathbb{C}_p^\times$ is a p -adic period attached to K , as defined in [BDP1, (140)], [BDP2, (2-17)];
- $\Omega \in \mathbb{C}^\times$ is the complex period associated to K defined in [BDP1, (137)];
- $L_c(\psi^{-1}, s)$ is Hecke’s L -function associated to ψ^{-1} with the Euler factors at primes dividing c removed.

Formula (45) is [BDP2, Proposition 3.1], which in turn follows from [Katz2, §5.3.0] (cf. also [Gro, Theorem 2.3] and [deS, II, Theorem 4.14]; the reader is cautioned that the latter reference needs to be slightly modified, for the power of 2π is not correct there.)

Letting $\mathbf{N}_K$ denote the norm character of K , Katz’s p -adic L -function satisfies the functional equation $L_p(K)(\psi) = L_p(K)((\psi')^{-1} \mathbf{N}_K)$ (cf. [Gro, p. 90-91]).

The values of p -adic L -functions at classical points lying outside their region of interpolation are frequently of great arithmetic interest, since they may encode further invariants of the associated motives. An instance of this philosophy is Katz’s p -adic analogue of the Kronecker limit formula (cf. [Katz1, §10.4, 10.5], [Gro, p. 90], [deS, Ch. II, §5.2]), expressing the value of $L_p(K)$ at a *finite order* character ψ of G_K in terms of a suitable linear combination of p -adic logarithms of elliptic units:

$$L_p(K)(\psi) = \begin{cases} \frac{1}{2}(\frac{1}{p} - 1) \cdot \log_p(u_\wp) & \text{if } \psi = 1 \\ \frac{-1}{24c}(1 - \psi^{-1}(\bar{\wp}))(1 - \psi(\wp)p^{-1}) \cdot \sum_{\sigma \in G} \psi^{-1}(\sigma) \log_p(\sigma(u)) & \text{if } \psi \neq 1. \end{cases} \quad (46)$$

Here, as above, $u_\wp \in K^\times$ is any element satisfying $(u_\wp) = \wp^{h_K}$ and $u \in \mathcal{U}_{\text{CM}}$ is an elliptic unit; the integer $c > 0$ is the smallest positive integer in the conductor ideal of ψ .

2. *The p -adic Rankin L -function attached to a cusp form and an imaginary quadratic field.* For any Hecke character ψ of K of infinity type (κ_1, κ_2) , let

$$L(f, \psi, s) := L\left(\pi_f \times \pi_\psi, s - \frac{\kappa_1 + \kappa_2 + 1}{2}\right)$$

denote the L -series associated with the product of the global automorphic representations attached to the weight two cuspform f and the Hecke character ψ , normalised conveniently. As usual, $L(f, \psi, s) = \prod_q L_q(q^{-s})$ is defined as a product of Euler factors ranging over the set of prime numbers. The Euler factors at primes $q \nmid N$ are exactly the same as that of the Rankin L -series $L(\theta_\psi \otimes f, s)$ introduced above, but may differ at primes $q \mid N$: see [Ja, §15] for more details.

Fix a positive integer $c \geq 1$ which is relatively prime to pN_f . Let $\Sigma_{f,c}$ be the set of Hecke characters $\psi \in \Sigma$ of conductor c and trivial central character for which $L(f, \psi^{-1}, s)$ is self-dual and has $s = 0$ as its central critical point. This set can be expressed as the disjoint union of the three subsets

$$\Sigma_{f,c}^{(1)} = \{\psi \in \Sigma_{f,c} \text{ of infinity type } (1, 1)\},$$

$$\Sigma_{f,c}^{(2)} = \{\psi \in \Sigma_{f,c} \text{ of infinity type } (\kappa + 2, -\kappa), \kappa \geq 0\},$$

and

$$\Sigma_{f,c}^{(2')} = \{\psi \in \Sigma_{f,c} \text{ of infinity type } (-\kappa, \kappa + 2), \kappa \geq 0\}.$$

The three sets $\Sigma_{f,c}^{(1)}$, $\Sigma_{f,c}^{(2)}$ and $\Sigma_{f,c}^{(2')}$ are each dense in the completion $\hat{\Sigma}_{f,c}$ of $\Sigma_{f,c}$ with respect to the p -adic compact open topology as explained in [BDP1, §5.2]. As shown in loc. cit., there exists a unique p -adic analytic function

$$L_p(f, K) : \hat{\Sigma}_{f,c} \rightarrow \mathbb{C}_p$$

interpolating the critical values $L(f, \psi^{-1}, 0)$ for $\psi \in \Sigma_{f,c}^{(2)}$, suitably normalized. The function $L_p(f, K)$ is referred to as the p -adic Rankin L -function attached to the pair (f, K) . By [BDP2, Proposition 3.11], the interpolation formula reads precisely as follows: for any character $\psi \in \Sigma_{f,c}^{(2)}$ of infinity type $(\kappa + 2, -\kappa)$,

$$L_p(f, K)(\psi) = \mathfrak{a}(f, \psi) \times \mathfrak{e}^2(f, \psi) \times \mathfrak{f}(f, \psi) \times \frac{\Omega_p^{4\kappa+4}}{\Omega^{4\kappa+4}} \times L(f, \psi^{-1}, 0) \quad (47)$$

where

$$\begin{aligned} \bullet \quad \mathfrak{a}(f, \psi) &= \kappa!(\kappa + 1)! \pi^{2\kappa+1}, \quad \mathfrak{e}(f, \psi) = 1 - a_p(f) \psi^{-1}(\bar{\rho}) + \psi^{-2}(\bar{\rho})p \quad \text{and} \\ \bullet \quad \mathfrak{f}(f, \psi) &= (2/c\sqrt{D_K})^{2\kappa+1} \cdot \prod_{q|c} \frac{q - \chi_K(q)}{q-1} \cdot \omega(f, \psi)^{-1}. \end{aligned}$$

Here $\omega(f, \psi)$ denotes the complex number defined in [BDP1, (5.1.11)]. It is an algebraic scalar of absolute value 1 relative to any complex embedding of $\bar{\mathbb{Q}}$.

Consider now a Hecke character $\psi \mathbf{N}_K \in \Sigma_{f,c}^{(1)}$, where ψ has finite order. The character $\psi \mathbf{N}_K$ lies outside the region of interpolation of $L_p(f, K)$ and the main theorem of [BDP1]—cf. [Br] for a proof of it under the more relaxed Heegner hypothesis imposed here—asserts that

$$L_p(f, K)(\psi \mathbf{N}_K) = \left(1 - \frac{a_p(f)}{\psi(\bar{\rho})p} + \frac{1}{\psi^2(\bar{\rho})p}\right)^2 \times \log_{\omega_E}(P_\psi)^2. \quad (48)$$

We now turn to the comparison of the Rankin p -adic L -function with the Garrett-Hida p -adic L -function of the previous section: the Katz p -adic L -function then arises indirectly from the ratio of periods (50) used in constructing these two p -adic L -functions.

Recall the theta series g of weight 1, level $N_g = D_K \mathbf{N}_{K/\mathbb{Q}}(\mathfrak{c}(\psi_g))$ and character χ^{-1} appearing in the statement of Theorem 3.3.

Let $\mathbf{g}$ be the primitive cuspidal Hida family of theta series of tame level N_g and tame character $\chi_g = \chi^{-1}$ constructed in e.g. [Hi2, p. 235-236] and [Gh, §5], specialising in weight 1 to the p -stabilisation g_α of g satisfying $U_p(g_\alpha) = \alpha_g \cdot g_\alpha$. It can be described explicitly by fixing a Hecke character λ of infinity type $(1, 0)$ and conductor $\wp$. Let H denote the number field generated by the values of λ and let H_p be the completion of H at the prime determined by an embedding $H \hookrightarrow \bar{\mathbb{Q}}_p$ which is fixed at the outset. Writing $\mathcal{O}_{H_p}$ for its ring of integers, decompose its group of units as $\mathcal{O}_{H_p}^\times = \mu \times W$, where μ is finite and W is free over $\mathbb{Z}_p$. Let $\langle \cdot \rangle : \mathcal{O}_{H_p}^\times \rightarrow W$ denote the natural projection.

For every integer $k \geq 1$ define

$$\psi_{g,k-1}^{(p)} = \psi_g \langle \lambda \rangle^{k-1}.$$

This definition does not depend on the choice of λ , for any other such choice λ' differs from λ by a finite character and hence $\langle \lambda' \rangle = \langle \lambda \rangle$. Let now $\psi_{g,k-1}$ be the Hecke character defined by

$$\psi_{g,k-1}(\mathfrak{q}) = \begin{cases} \psi_{g,k-1}^{(p)}(\mathfrak{q}) & \text{if } \mathfrak{q} \neq \wp \\ \chi^{-1}(p)p^{k-1}/\psi_{g,k-1}^{(p)}(\bar{\wp}) & \text{if } \mathfrak{q} = \wp. \end{cases}$$

The Hecke character $\psi_{g,k-1}$ has conductor $\mathfrak{c}(\psi_g)$ and infinity type $(k-1, 0)$. For any $k \in \mathbb{Z}_{\geq 1} \cap \mathcal{U}_g$, the weight k specialisation of $\mathbf{g}$ is the ordinary p -stabilisation of the theta series $g_k = \theta_{\psi_{g,k-1}} = \theta_{\psi'_{g,k-1}}$ associated with $\psi_{g,k-1}$ (or with its $\text{Gal}(K/\mathbb{Q})$ -conjugate $\psi'_{g,k-1}$). In particular $\psi_{g,0} = \psi_g$ is the character we considered at the outset and $g_1 = g$.

Together with g , recall also the eigenforms $f \in S_2(N_f)$ and $h \in M_1(N_h, \chi)$ of the statement. Let $(\check{g}, \check{f}, \check{h})$ be Watson-Woodbury's test vector of level N associated to these three modular forms as described in Remark 2.2, and let also $\check{\mathbf{g}}$ denote Watson-Woodbury's Λ -adic test vector discussed in Remark 2.4. Recall that, as discussed in Chapter 2, $W_v = C_v(\check{g}_k, \check{f}, \check{h}) \in \mathbb{Q}^\times$ are constants independent of k for all $v \mid N$. Moreover, the results of [Wo2] guarantee that in this setting $C_\infty(\check{g}_k, \check{f}, \check{h}) = 1$ for all k .

Proposition 2.3 implies, for all $k \in \mathbb{Z}_{\geq 1} \cap \mathcal{U}_g$,

$$\begin{aligned} \mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}, \check{h})(k) \cdot \mathcal{L}_p^g(\check{\mathbf{g}}^*, \check{f}, \check{h}^*)(k) &= \mathfrak{c}(k, 2, 1)^2 \cdot \mathfrak{f}(k, 2, 1) \cdot \mathfrak{a}(k, 2, 1) \cdot \prod_{v \mid N} W_v \\ &\quad \times \frac{L(g_k \otimes f \otimes h, \frac{k+1}{2})}{\langle g_k^*, g_k^* \rangle^2}. \end{aligned}$$

Here the factors on the right are described in Proposition 2.3. In particular,

$$\mathfrak{f}(k, 2, 1) = A \cdot 2^B \quad \text{and} \quad \mathfrak{a}(k, 2, 1) = \left(\left(\frac{k-3}{2} \right)! \left(\frac{k-1}{2} \right)! \right)^2 \cdot \pi^{-2k}.$$

For every integer $k \geq 1$ define the Hecke characters

$$\Psi_{gh}(k) = (\psi'_{g,k-1})^{-1} \cdot \psi_h^{-1} \cdot \mathbf{N}^{\frac{k+1}{2}} \quad \text{and} \quad \Psi_{gh'}(k) = (\psi'_{g,k-1})^{-1} \cdot (\psi'_h)^{-1} \cdot \mathbf{N}^{\frac{k+1}{2}}.$$

While for $k = 1$ the characters $\Psi_{gh}(1) = \mathbf{N} \cdot (\psi'_g \psi_h)^{-1}$ and $\Psi_{gh'}(1) = \mathbf{N} \cdot (\psi'_g \psi'_h)^{-1}$ lie in $\Sigma_{f,c}^{(1)}$, for $k \in U_g := \mathbb{Z}_{\geq 2} \cap \mathcal{U}_g$ the characters $\Psi_{gh}(k), \Psi_{gh'}(k)$ belong to $\Sigma_{f,c}^{(2)}$ with $c = \text{lcm}(c_1, c_2)$, because their infinity type is $(\kappa + 2, -\kappa)$ with $\kappa = \frac{k-3}{2} \geq 0$.

Hence for $k \in U_g$ the values $L_p(f, K)(\Psi_{gh}(k)), L_p(f, K)(\Psi_{gh'}(k))$ and $\mathcal{L}_p^g(\check{\mathbf{g}}, \check{f}, \check{h})(k)$ all lie in the region of interpolation. An inspection of the Euler factors defining their classical counterparts yields the following relationship between critical values: for every $k \in U_g$ there is a fudge factor $\mathfrak{f}_0(k) \neq 0$ such that

$$L\left(g_k \otimes f \otimes h, \frac{k+1}{2}\right) = \mathfrak{f}_0(k) \cdot L(f, \Psi_{gh}^{-1}(k), 0) \cdot L(f, \Psi_{gh'}^{-1}(k), 0), \quad (49)$$

and we shall prove the theorem by letting $k \in U_g$ in (49) tend p -adically to 1 within $\mathcal{U}_g$, so that the triplets $(k, 2, 1)$ tend to $(1, 2, 1)$.

Several of the fudge factors that arise in comparing different L -values, viewed as functions of weight space, satisfy the property described in the following definition:

Definition 3.5. Let F be a number field. A function

$$\mathfrak{f} : U_g \longrightarrow \bar{\mathbb{Q}}$$

is F -admissible if it extends to an element of the fraction field $\mathcal{L}_g$ of the algebra $\Lambda_g = \mathcal{O}(\mathcal{U}_g)$ of Iwasawa functions on $\mathcal{U}_g$, and if $\mathfrak{f}(1)$ is well-defined and belongs to $F^\times$.

As in the statement of the definition, an F -admissible function $\mathfrak{f}$ will customarily be identified with the function on $\mathcal{U}_g \subseteq \mathbb{Z}_p$ which it induces. In order to make the calculations described below easier to follow, vowels will generally be used to denote non-admissible functions (such as rational functions of p^k , π^k or $k!$), and consonants to denote admissible ones (such as q^k or $a_q(g_k)$ for some prime number $q \neq p$).

Instances of admissible functions were already encountered in the proof of Proposition 2.5: using the notation employed there, the functions λ_d were shown in loc.cit. to be $\mathbb{Q}(g_1, h_1)$ -admissible. Recall also that the fudge factor $\mathfrak{f}_0(k)$ appearing in (49) accounts for the discrepancy of the Euler factors at the primes $q \mid cD_K N_f$ in the definitions of the above L -series. It may be computed explicitly, but we content to note here that it is $\mathbb{Q}(\psi_g, \psi_h)$ -admissible because $\mathfrak{f}_0(k)$ is a rational function on q^k for $q \mid cD_K N_f$ with coefficients in $\mathbb{Q}(\psi_g, \psi_h)$, with $\mathfrak{f}_0(k) \neq 0$ for all $k \in \mathbb{Z}$.

For every $k \in U_g$, define

$$\sigma(k) := \langle g_k^*, g_k^* \rangle \times \left(\frac{\pi \Omega_p}{\Omega} \right)^{2k-2} \quad (50)$$

and

$$\mathfrak{e}_1(k) = \frac{\mathfrak{e}(k, 2, 1)^2}{\mathfrak{e}^2(f, \Psi_{gh}(k)) \mathfrak{e}^2(f, \Psi_{gh'}(k))}, \quad \mathfrak{f}_1(k) = \frac{\prod_{v \mid N} W_v \cdot \mathfrak{f}_0(k) \cdot \mathfrak{f}(k, 2, 1)}{\mathfrak{f}(f, \Psi_{gh}(k)) \mathfrak{f}(f, \Psi_{gh'}(k))}.$$

Lemma 3.6. *The function $\mathfrak{f}_1(k)$ is $K(\psi_g, \psi_h)$ -admissible and*

$$\mathcal{L}_p^g(\check{\mathfrak{g}}, \check{f}, \check{h})(k) \cdot \mathcal{L}_p^g(\check{\mathfrak{g}}^*, \check{f}, \check{h}^*)(k) \cdot \sigma(k)^2 = \mathfrak{e}_1(k) \mathfrak{f}_1(k) \cdot L_p(f, K)(\Psi_{gh}(k)) \cdot L_p(f, K)(\Psi_{gh'}(k)).$$

Proof. We already saw that $\mathfrak{f}_0(k)$ is $\mathbb{Q}(\psi_g, \psi_h)$ -admissible, and Proposition 2.1 (b) (ii) implies that $\mathfrak{f}(k, 2, 1) = A \cdot 2^{B(k)}$ is $\mathbb{Q}$ -admissible because $p \neq 2$ and $B(k)$ is a linear function of k . It remains to prove that the denominator of $\mathfrak{f}_1(k)$ is $K(\sqrt{N_g}, \sqrt{N_h}, \psi_g, \psi_h)$ -admissible and not zero at $k = 1$. Recall from (47) that $\mathfrak{f}(f, \psi) = (2/c\sqrt{D_K})^{k-2} \cdot \prod_{q \mid c} \frac{q - \chi_K(q)}{q-1} \cdot \omega(f, \psi)^{-1}$ for any Hecke character $\psi \in \Sigma_{f,c}^{(2)}$ of infinity type $(\frac{k-3}{2} + 2, -\frac{k-3}{2})$. The first two factors are K -admissible, because p is assumed not to divide $2N$. Hence, in order to conclude, we need to show that the functions $\omega(f, \Psi_{gh}(k))$, $\omega(f, \Psi_{gh'}(k))$ and $\omega(k)$ (or at least their products) are admissible. By [BDP1, (5.1.11)]:

$$\omega(f, \Psi_{gh}(k)) \cdot \omega(f, \Psi_{gh'}(k)) = \frac{\mathbf{N}^{2k-2}(\mathfrak{b}) N^{k-1}}{(\psi'_{g,k-1})^2 \psi_h \psi'_h(\mathfrak{b}) \cdot b^{2k-2}}, \quad (\mathfrak{b}, pNc) = 1, \mathfrak{b} \cdot \mathfrak{N} = (b). \quad (51)$$

This shows that the function in (51) is $K(\psi_g, \psi_h)$ -admissible, because $p \nmid \mathfrak{b}\mathfrak{N}$ and its value at $k = 1$ is the non-zero algebraic number $((\psi'_g)^2 \psi_h \psi'_h(\mathfrak{b}))^{-1} = (\psi'_g)^2 \psi_h \psi'_h(\mathfrak{N})$, which does not depend on the choice of $\mathfrak{b}$.

Finally, the factorisation formula of Lemma 3.6 is a direct consequence of the interpolating formulae (32) and (47), combined with (49), once one observes that the factorial terms and powers of π on both sides cancel out. $\square$

At this point it is convenient to invoke the following classical formula:

Lemma 3.7. *There is a K -admissible function $\mathfrak{f}_2(k)$ such that*

$$\langle g_k^*, g_k^* \rangle = \frac{(k-1)!}{\pi^k} \cdot \mathfrak{f}_2(k) \cdot L(\psi_{g,k-1}^2 \cdot \chi, k).$$

Proof. Since the L -series of the adjoint of a CM-form factors as

$$L(\mathrm{Ad}(g_k^*), s) = L(\mathrm{Ad}(g_k), s) = L(\chi_K, s - k + 1) L(\psi_{g,k-1}^2 \cdot \chi_g^{-1}, s)$$

and $\chi_g = \chi^{-1}$, Lemma 3.7 follows directly from (26) after evaluating at $s = k$ and applying Dirichlet's class number formula $L(\chi_K, 1) = 2\pi h_K(|\mathcal{O}_K^\times| \sqrt{D})^{-1}$. $\square$

It now becomes clear how Katz's p -adic L -function enters in the calculation: note that

$$L(\psi_{g,k-1}^2 \cdot \chi, k) = \mathfrak{f}_3(k) \cdot L_c(\psi_{g,k-1}^2 \cdot \chi \cdot \mathbf{N}_K^{-k}, 0),$$

where again $\mathfrak{f}_3(k)$ accounts for the discrepancy between the Euler factors at primes $q \mid cD_K$ defining both L -series above, and hence is $\mathbb{Q}(\psi_g, \psi_h)$ -admissible.

Define $\Psi_g(k) := (\psi'_{g,k-1})^{-2} \chi^{-1} \mathbf{N}_K^k$. In light of Katz's interpolation formula (45), since the infinity type of $\Psi_g(k)$ is $(\kappa_1, \kappa_2) = (k, -k + 2)$, it follows that

$$L(\psi_{g,k-1}^2 \cdot \chi, k) = L((\psi'_{g,k-1})^2 \cdot \chi, k) = \frac{\pi^{-k+2}}{(k-1)!} \cdot \frac{1}{\mathfrak{e}(\Psi_g(k))} \cdot \frac{\mathfrak{f}_3(k)}{\mathfrak{f}(\Psi_g(k))} \cdot \frac{\Omega^{2k-2}}{\Omega_p^{2k-2}} \cdot L_p(K)(\Psi_g(k)).$$

Comparing Lemma 3.7 and the above formula with the definition of $\sigma(k)$, one finds that the powers of π and the factorials (which, as functions of k , are not admissible) cancel out and

$$\sigma(k) = \frac{1}{\mathfrak{e}(\Psi_g(k))} \cdot \frac{\mathfrak{f}_2(k)\mathfrak{f}_3(k)}{\mathfrak{f}(\Psi_g(k))} \cdot L_p(K)(\Psi_g(k)). \quad (52)$$

For all integers $k \in U_g$ define

$$\omega(k) = I(\check{g}_k^*, \check{f}, \check{h}^*) / I(\check{g}_k, \check{f}, \check{h})$$

as in (24) and

$$\mathfrak{f}(k) = \frac{\mathfrak{f}_1(k)}{\omega(k)} \left(\frac{\mathfrak{f}(\Psi_g(k))}{\mathfrak{f}_2(k)\mathfrak{f}_3(k)} \right)^2.$$

Let $\mathfrak{g}(\chi) \in \bar{\mathbb{Q}} \subset \mathbb{C}_p$ denote the Gauss sum associated to the character χ and define the number field $L_0 = K(\sqrt{N_g}, \sqrt{N_h}, \mathfrak{g}(\chi), \psi_g, \psi_h)$. Theorem 3.3 will follow from the next result:

Theorem 3.8. *The function $\mathfrak{f}(k)$ is L_0 -admissible and the following factorisation of p -adic L -series is valid*

$$\mathcal{L}_p^g(\check{\mathfrak{g}}, \check{f}, \check{h})(k)^2 \cdot L_p(K)(\Psi_g(k))^2 = \mathfrak{f}(k) \cdot L_p(f, K)(\Psi_{gh}(k)) \cdot L_p(f, K)(\Psi_{gh'}(k)). \quad (53)$$

Moreover, there exist explicit elements $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{Q}(\psi_g, \psi_h)^\times$ such that

$$L_p(f, K)(\Psi_{gh'}(1)) = \lambda_1^2 \cdot \log_{E,p}(P_1)^2, \quad L_p(f, K)(\Psi_{gh}(1)) = \lambda_2^2 \cdot \log_{E,p}(P_2)^2, \quad (54)$$

$$\text{and} \quad L_p(K)(\Psi_g(1)) = \lambda_3 \cdot \log_p(u_{g_\alpha}).$$

Proof. We first prove (54): note that $\Psi_g(1) = (\psi'_g)^{-2} \cdot \chi^{-1} \cdot \mathbf{N}_K = \frac{\psi_g}{\psi'_g} \mathbf{N}_K$. By the functional equation of Katz's p -adic L -function quoted above, since $\frac{\psi_g}{\psi'_g}$ is anticyclotomic we have $L_p(K)(\frac{\psi_g}{\psi'_g} \mathbf{N}_K) = L_p(K)(\frac{\psi_g}{\psi'_g})$ and hence (46) applies, yielding $L_p(K)(\Psi_g(k))(1) = \lambda_3^2 \log_p(u_{g_\alpha})$ for some explicit value of $\lambda_3 \in \mathbb{Q}(\psi_g, \psi_h)^\times$. Moreover,

$$\Psi_{gh}(1) = (\psi'_g)^{-1} \cdot \psi_h^{-1} \cdot \mathbf{N} = \psi_2 \cdot \mathbf{N}, \quad \Psi_{gh'}(1) = (\psi'_g)^{-1} \cdot (\psi'_h)^{-1} \cdot \mathbf{N} = \psi_1 \cdot \mathbf{N},$$

and hence (48) can be applied, thereby proving (54).

In order to prove the admissibility of $\mathfrak{f}$ and the factorisation formula, recall we already saw that $\mathfrak{f}_1, \mathfrak{f}_2$ and $\mathfrak{f}_3$ are $K(\psi_g, \psi_h)$ -admissible, and an inspection to the fudge factor $\mathfrak{f}(f, \Psi_g(k))$

shows that it also is. Hence it remains to prove that $\omega(k)$ is L_0 -admissible. Let us first show that ω interpolates to a meromorphic Iwasawa function on $\mathcal{U}_g$ such that (53) holds and $\omega(1) \in \mathbb{C}_p^\times$ is a non-zero p -adic number, leaving for later that $\omega(1)$ actually lies in $L_0^\times$.

Note that Lemma 3.6 combined with (52) gives rise to the following identity for all integers $k \geq 0$:

$$\begin{aligned} & \mathcal{L}_p^g(\check{g}, \check{f}, \check{h})(k) \cdot \mathcal{L}_p^g(\check{g}^*, \check{f}, \check{h}^*)(k) \cdot L_p(K)(\Psi_g(k))^2 = \\ & = \mathfrak{e}(k) \cdot \mathfrak{f}_1(k) \left(\frac{\mathfrak{f}(\Psi_g(k))}{\mathfrak{f}_2(k)\mathfrak{f}_3(k)} \right)^2 \cdot L_p(f, K)(\Psi_{gh}(k)) \cdot L_p(f, K)(\Psi_{gh'}(k)) \end{aligned} \quad (55)$$

where

$$\mathfrak{e}(k) := \mathfrak{e}(\Psi_g(k))^2 \mathfrak{e}_1(k) = \left(\frac{\mathfrak{e}(\Psi_g(k)) \cdot \mathfrak{e}(k, 2, 1)}{\mathfrak{e}(f, \Psi_{gh}(k)) \mathfrak{e}(f, \Psi_{gh'}(k))} \right)^2.$$

An inspection of the factors appearing above shows that

$$\mathfrak{e}(k) = 1 \quad \text{for all } k \geq 0. \quad (56)$$

More precisely, one checks that the four factors in the numerator $\mathcal{E}(g_k, f, h)$ of $\mathfrak{e}(k, 2, 1)$ are equal to the four factors appearing in $\mathfrak{e}(f, \Psi_{gh}(k)) \mathfrak{e}(f, \Psi_{gh'}(k))$, and hence cancel out. In addition, the two factors showing up in $\mathfrak{e}(\Psi_g(k))$ are identical to those in the denominator of $\mathfrak{e}(k, 2, 1)$, and thus also cancel out.

Since the unit u_{g_α} and the points P_1 and P_2 are non-torsion, it also follows from (54) and (55) that neither $\mathcal{L}_p^g(\check{g}, \check{f}, \check{h})(k)$ nor $\mathcal{L}_p^g(\check{g}^*, \check{f}, \check{h}^*)(k)$ vanish at $k = 1$. Note that (31) implies that

$$\omega(k) = \mathcal{L}_p^g(\check{g}^*, \check{f}, \check{h}^*)(k) / \mathcal{L}_p^g(\check{g}, \check{f}, \check{h})(k), \quad (57)$$

from which it follows that $\omega \in \mathcal{L}_g$, with no poles at $k = 1$ and satisfying $\omega(1) \neq 0$. The factorisation formula claimed in the theorem is also a consequence of (55), (56) and (57).

To finish the proof, we need to show that $\omega(1) \in L_0$. To do that, let W_N denote the Atkin-Lehner operator induced by the matrix $\begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$. As shown in [AL78] and [Hi2, Ch. X], it commutes with the Shimura-Maass operator, and its adjoint on $S_{2k+3}(N, \chi)$ with respect to the Petersson scalar product is $-W_N$. This implies that

$$I(\check{g}_k, \check{f}, \check{h}) = I(W_N(\check{g}_k), W_N(\check{f}), W_N(\check{h})).$$

By the formulae in [AL78], $W_N(\check{h}(q))$ is a linear combination with coefficients in $\mathbb{Q}(h)$ of the modular forms $\lambda_h \cdot h^*(q^d)$ where d ranges over the positive divisors of N/N_h and λ_h is the pseudo-eigenvalue of the newform h of level N_h in the sense of [AL78], characterized by the equation $W_{N_h}(h) = \lambda_h h^*$. The analogous claim, with similar notations, holds for g_k and f . It thus follows from e.g. [DR1, Lemma 2.12] that the ratio

$$\omega(k) = I(\check{g}_k^*, \check{f}, \check{h}^*) / I(W_N(\check{g}_k), W_N(\check{f}), W_N(\check{h}))$$

lies in $\mathbb{Q}(g_k, f, h, \lambda_f, \lambda_g, \lambda_h)$.

By [AL78, Theorem 2.1], since $a_{N_g}(g_k)$, $a_{N_f}(f)$ and $a_{N_h}(h)$ are all non-zero, the pseudo-eigenvalues of f , g_k and h are

$$\lambda_f = -a_{N_f}(f), \quad \lambda_g = N_g^{\frac{k-2}{2}} \mathfrak{g}(\chi^{-1}) / a_{N_g}(g_k), \quad \lambda_h = \mathfrak{g}(\chi) / (\sqrt{N_h} \cdot a_{N_h}(h)).$$

This implies that $\omega(1)$ lies in L_0 , as claimed. Note that, since $p \nmid N$ and the coefficients $a_{N_g}(g_k)$ are Iwasawa functions on $\mathcal{U}_g$, the above formulae give rise to a rather explicit expression of $\omega(k)$ as a p -adic continuous function. In any case, Theorem 3.8 follows. $\square$

To conclude the proof of Theorem 3.3, recall that Proposition 2.5 recasts $\mathcal{L}_p^g(\check{g}, \check{f}, \check{h})(1)$ as the p -adic iterated integral. Combined with the above result we deduce that

$$\left(\int_{\check{\gamma}_\alpha} \check{f} \cdot \check{h} \right)^2 \cdot \log_p(u_{g_\alpha})^2 = \check{f}(1) \log_{E,p}(P_1)^2 \log_{E,p}(P_2)^2 \quad (58)$$

up to *squares* in $\mathbb{Q}(\psi_g, \psi_h)^\times$. Set $L = L_0(\sqrt{\check{f}(1)})$. Taking square-roots in (58) shows that

$$\int_{\check{\gamma}_\alpha} \check{f} \cdot \check{h} = \lambda \cdot \frac{\log_{E,p}(P_1) \log_{E,p}(P_2)}{\log_p(u_{g_\alpha})}$$

for some $\lambda \in L^\times$, thereby proving Theorem 3.3.

3.3. Numerical evidence for inert primes. The proof of Conjecture ES based on Katz-style p -adic L -series and their factorisations breaks down when the prime p is *inert* in K . Yet the units and points that arise can still be expressed in terms of elliptic units and Heegner points, at least in the rank $(1, 1)$ setting, making this a natural first testing ground for Conjecture ES.

3.3.1. General remarks. Retain all the notations and assumptions of Section 3.2, but assume now that the prime p is *inert* in K .

Let K_p denote the completion of K at p , a quadratic unramified extension of $\mathbb{Q}_p$. Since the Frobenius element σ_p is a reflection in the dihedral group $\text{Gal}(H/\mathbb{Q})$, the prime p splits completely in H/K and the choice of σ_p amounts to choosing an embedding $H \hookrightarrow K_p$, which shall be fixed from now on.

Because ϱ_g and ϱ_h have dihedral projective image, it follows that σ_p acts on V_g (resp. on V_h) with eigenvalues λ and $-\lambda$ (resp. λ^{-1} and $-\lambda^{-1}$), for a suitable root of unity λ . Label the four values

$$\alpha_g = \lambda, \quad \beta_g = -\lambda, \quad \alpha_h = \lambda^{-1}, \quad \beta_h = -\lambda^{-1},$$

so that

$$\alpha_g \alpha_h = 1, \quad \alpha_g \beta_h = -1,$$

as shall be assumed from now on. The decomposition

$$V_{gh} \simeq V_{\psi_1} \oplus V_{\psi_2}$$

of $\text{Gal}(H/\mathbb{Q})$ -modules continues to hold, but now σ_p acts on both V_{ψ_1} and V_{ψ_2} as reflections, with eigenvalues 1 and -1 .

The asymmetry in the roles of g and h in the definition of the p -adic iterated integral continues to determine a pair (ψ_1, ψ_2) of global characters of G_K by requiring that ψ_1 and ψ_2 “each be divisible by ψ_g ”. This pair is uniquely defined up to the action of $\text{Gal}(K/\mathbb{Q})$, since replacing ψ_g by ψ'_g has the effect of replacing (ψ_1, ψ_2) by (ψ'_2, ψ'_1) . Fix $v_{\psi_j} \in V_{\psi_j}$ to be a non-zero vector in the one-dimensional ψ_j -eigenspace for the action of G_K on V_{ψ_j} . With this choice in hand, set $v_{\psi'_j} := \sigma_p v_{\psi_j}$, as in the previous section. Given any $P \in \text{Hom}(V_{\psi_j}, E(H)_L)$, we shall write

$$P_{\psi_j} := P(v_{\psi_j}), \quad P_{\psi'_j} := P(v_{\psi'_j}).$$

Lemma 3.9. *The regulator $\det(R_{g_\alpha}(E, \varrho_{gh}))$ is given by*

(1) *If $r(E, V_{\psi_1}) = r(E, V_{\psi_2}) = 1$,*

$$\det R_{g_\alpha}(E, \varrho_{gh}) = \log_{E,p}(P_{\psi_1}) \cdot \log_{E,p}(Q_{\psi_2}) - \log_{E,p}(P_{\psi'_1}) \cdot \log_{E,p}(Q_{\psi'_2}).$$

In particular, if ψ_1 (say) is a quadratic (genus) character, so that the point P_{ψ_1} is a σ_p -eigenvector with eigenvalue $\iota = \pm 1$, then

$$\det R_{g_\alpha}(E, \varrho_{gh}) = \log_{E,p}(P_{\psi_1}) \cdot \log_{E,p}(Q_{\psi_2} - \iota Q_{\psi'_2}).$$

(2) If $r(E, V_{\psi_1}) = 2$ and $r(E, V_{\psi_2}) = 0$,

$$\det R_{g\alpha}(E, \varrho_{gh}) = \log_{E,p}(P_{\psi_1}) \cdot \log_{E,p}(Q_{\psi'_1}) - \log_{E,p}(P_{\psi'_1}) \cdot \log_{E,p}(Q_{\psi_1}),$$

where (P, Q) is a basis for $\text{hom}_{G_{\mathbb{Q}}}(V_{\psi_1}, E(H)_L)$.

Observe in particular that the regulator vanishes in the special case where there are no global points $(P_1, P_2) \in E(H)_L^{\varrho_1} \times E(H)_L^{\varrho_2}$ belonging to distinct eigenspaces for the action of σ_p ; this exceptional scenario in which the regulator is degenerate occurs precisely when

$$\dim(\varrho_1) = \dim(\varrho_2) = 1 \text{ (so that } \psi_1^2 = \psi_2^2 = 1 \text{) and } \varrho_1(p) = \varrho_2(p). \quad (59)$$

If (59) holds, Conjecture ES predicts that the iterated integrals attached to (f, g, h) vanish, in spite of the running hypothesis $r_{\text{an}}(E, \varrho_1) = r_{\text{an}}(E, \varrho_2) = 1$. See Example 4.3 below for a numerical verification of this interesting phenomenon.

Suppose for the rest of this section that (59) is not satisfied, so that, switching ϱ_1 and ϱ_2 if necessary, it may be assumed that 1 and -1 are eigenvalues of $\varrho_1(\sigma_p)$ and $\varrho_2(\sigma_p)$ respectively. Recall furthermore the points $P_{\psi_i}^{\pm}$ defined from Heegner points in Lemma 3.1 and and (42), and the unit $u_{\psi_0}^-$ defined in (43) from elliptic units. Assuming (37) or (39), Conjecture ES predicts that there is some choice of test vectors $\check{f}$, $\check{\gamma}$ and $\check{h}$ for which

$$\int_{\check{\gamma}} \check{f} \cdot \check{h} = \frac{1}{\log_p(u_{\psi_0}^-)} \cdot \begin{cases} \log_{E,p}(P_{\psi_1}^+) \log_{E,p}(P_{\psi_2}^-) - \log_{E,p}(P_{\psi_1}^-) \log_{E,p}(P_{\psi_2}^+) & \text{if } \psi_1^2, \psi_2^2 \neq 1, \\ & \text{and } V_{\psi_1} \neq V_{\psi_2}, \\ \log_{E,p}(P_{\psi_1}^+) \log_{E,p}(P_{\psi_2}^-) & \text{otherwise.} \end{cases} \quad (60)$$

3.3.2. Numerical evidence. The following experiment in support of (60) gave the authors their first evidence that the p -adic iterated integrals continue to have geometric meaning beyond the setting treated in Section 3.2.

Example 3.10. Consider the elliptic curve

$$E : y^2 + xy + y = x^3 + x^2 + x$$

of conductor 83 labelled 83a in Cremona's tables and write f for the associated weight 2 newform. Let χ be the odd quadratic character of conductor 83. The space $S_1(83, \chi)$ has dimension 1 and is spanned by the theta series $g := \theta_{\psi_g}$ attached to a cubic character ψ_g of $G = \text{Gal}(H/K)$, where H is the Hilbert class field of $K = \mathbb{Q}(\sqrt{-83})$, which has class number 3. The representation ρ_g is equal to the induced representation V_{ψ_g} . Let $h = E(1, \chi) \in M_1(83, \chi)$ be the Eisenstein series attached to the pair $(1, \chi)$, whose Galois representation ϱ_h is induced from the trivial character $\psi_h = 1$ of G_K . Since g and h have rational Fourier coefficients it will be enough to take $L = \mathbb{Q}$ (although the representations are in fact defined over $\mathbb{Q}(\zeta_3)$). The representations V_{gg} and V_{gh} decompose into irreducible constituents as follows:

$$\begin{aligned} V_{gg} &= V_{\psi_g^2} \oplus V_{\psi_g \cdot \psi_g^{-1}} &= V_{\psi_g} \oplus \mathbb{Q} \oplus \mathbb{Q}(\chi) &=: \varrho_1 \oplus \varrho_2 \oplus \varrho_3, \\ V_{gh} &= V_{\psi_g \cdot \psi_h} \oplus V_{\psi_g \cdot \psi_h^{-1}} &= V_{\psi_g} \oplus V_{\psi_g} &= \varrho_1 \oplus \varrho_1. \end{aligned}$$

A direct calculation shows that

$$r(E, \varrho_1) = r(E, \varrho_2) = 1, \quad r(E, \varrho_3) = 0,$$

and similarly for the analytic ranks. More precisely, the Mordell-Weil group

$$E(H)_{\mathbb{Q}} = E(H)^{V_{\psi_g}} \oplus E(\mathbb{Q})_{\mathbb{Q}}$$

is of rank 3, and is generated by the Heegner point

$$P = (t, -t^2 - 2) \in E(H), \quad \text{where } t^3 - t^2 + t - 2 = 0, \quad (61)$$

along with its Galois translates P^σ and P^{σ^2} , where σ is a generator of $\text{Gal}(H/K)$. The vector spaces $E(\mathbb{Q})_{\mathbb{Q}}$ and $E(H)_{\mathbb{Q}}^{V_{\psi_g}}$ are generated by

$$P_{\mathbb{Q}} := P + P^\sigma + P^{\sigma^2} = (1, -3), \quad \text{and} \quad \begin{cases} P_{\psi_g}^+ := 2P - P^\sigma - P^{\sigma^2}, \\ P_{\psi_g}^- := P^\sigma - P^{\sigma^2}. \end{cases}$$

Let $p = 5$, which is inert in K . The cubic polynomial in (61) has exactly one root over $\mathbb{Q}_5$. There is therefore a unique embedding of H into $\mathbb{C}_5$ for which the Heegner point P maps to $E(\mathbb{Q}_5)$; the images of the points P^σ and P^{σ^2} are then interchanged by the Frobenius element σ_5 .

The Stark unit that enters into the denominator of the conjectural expression for the p -adic iterated integral is the elliptic unit defined by $u_{\psi_0}^- = u^\sigma / u^{\sigma^2}$, so that

$$\log_p(u_{\psi_0}^-) = \log_p(u^\sigma) - \log_p(u^{\sigma^2}),$$

where $u \in H$ is the unique root of the polynomial

$$x^3 - 2x^2 - 2x - 1$$

whose image under the previously chosen embedding of H belongs to $\mathbb{Q}_5$.

Let $\gamma_{\pm 1} \in S_1^{\text{oc,ord}}(83, \chi)^\vee$ be the dual basis to $g_{\pm 1} = g(q) \mp g(q^5)$ in the space of 5-adic overconvergent modular forms, which maps $g_{\pm 1}$ to 1 and all other generalised eigenspaces in $S_1^{\text{oc,ord}}(83, \chi)$ to zero. A numerical calculation of 5-adic iterated integrals shows that

$$\begin{aligned} \int_{\gamma_{\pm 1}} f \cdot g &= -1688341751390720654446425615337716320464561776401 \bmod 5^{70} \\ \int_{\gamma_{\pm 1}} f \cdot h &= -2981481189608571355501040629889815717824676581453 \bmod 5^{70}. \end{aligned}$$

Let $\mathcal{E}(g_{\pm 1}, f, g)$ denote the algebraic factor defined at the start of Section 5.2. In all the examples where p is inert, this factor is always equal to

$$\mathcal{E}(g_{\pm 1}, f, g) := \left(1 - \frac{a_p(f)}{p} + \frac{1}{p}\right) \left(1 + \frac{a_p(f)}{p} + \frac{1}{p}\right) \in \mathbb{Q}.$$

The following identities were checked to 70 digits of 5-adic precision:

$$\int_{\gamma_{\pm 1}} f \cdot g = \frac{5 \cdot \mathcal{E}(g_{\pm 1}, f, g) \cdot \log_{E,5}(P_{\mathbb{Q}}) \log_{E,5}(P_{\psi_g}^-)}{24 \cdot \log_5(u_{\psi_0}^-)} \quad (62)$$

$$\int_{\gamma_{\pm 1}} f \cdot h = -\frac{5 \cdot \mathcal{E}(g_{\pm 1}, f, h) \cdot \log_{E,5}(P_{\psi_g}^+) \log_{E,5}(P_{\psi_g}^-)}{24 \cdot \log_5(u_{\psi_0}^-)} \quad (63)$$

in perfect agreement with the prediction (60) for both γ_{+1} and γ_{-1} . Note that $u_{\psi_0}^-$ is a unit in the (-1) -eigenspace for σ_5 while the numerators involve points in both eigenspaces, so that the resulting ratio belongs to $\mathbb{Q}_5$.

Although the Heegner point P is only defined up to sign, it is worth noting that the right-hand side of (62) and (63) are unchanged if P is replaced by $-P$.

Formulae similar to (62) and (63) (with the terms 5 and 24 in the numerator and denominator replaced by p and $4(p+1)$ respectively) were verified to 15 digits of 13-adic precision (respectively, 10 digits of 19-adic precision) replacing $p = 5$ by the inert primes $p = 13$ and $p = 19$. Similar experiments were also performed with the class number 3 field $K = \mathbb{Q}(\sqrt{-331})$ and the inert prime $p = 7$, where the same formulae (mutatis mutandis) were verified to 10 significant 7-adic digits.

The next example also involves Heegner points but is one of the most illuminating in this paper, insofar as it concerns a rank $(1, 1)$ setting of Conjecture ES in which nonetheless the regulator $R_{g_\alpha}(E, \varrho_{gh})$ is not factorisable. It illustrates a feature of Conjecture ES which

initially surprised the authors, namely, that the numerator on the right hand side of Conjecture ES needs not, in contrast with Theorem 3.3, be the product of logarithms of two points but may involve a linear combination of such products. This ran counter to the intuition the authors had first gained in their proof of Conjecture ES in the setting of Section 3.2.

Example 3.11. Consider the curve

$$E : y^2 + xy + y = x^3 + x^2 - 2x$$

of conductor 79 labelled 79a in Cremona's tables and write f for the associated weight 2 newform.

Let χ be the odd Dirichlet character of order 2 and conductor 79 associated to the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-79})$, of class number 5. As in the previous example, denote by H the Hilbert class field of K and set $G = \text{Gal}(H/K)$. The space $S_1(79, \chi)$ has dimension 2 and is spanned by a pair of dihedral eigenforms with coefficients in the field $L = \mathbb{Q}(\sqrt{5})$ generated by the golden ratio $\omega = (1 + \sqrt{5})/2$, namely

$$g = q + (-1 + \omega)q^2 + (1 - \omega)q^4 - \omega q^5 - q^8 + q^9 - q^{10} + \dots$$

and its Galois conjugate $g^\dagger$. Let $h = E(1, \chi) \in M_1(79, \chi)$ be the Eisenstein series attached to the pair $(1, \chi)$. After fixing a quintic character ψ of G , it can be assumed that

$$V_g = V_{\psi^2} = V_{\psi^3}, \quad V_{g^\dagger} = V_\psi = V_{\psi^4}, \quad V_h = V_{\psi_h} = L \oplus L(\chi_K),$$

where ψ_h is the trivial character of G_K . The representations $V_{g^\dagger g}$ and $V_{g^\dagger h}$ decompose into irreducible summands as follows:

$$\begin{aligned} V_{g^\dagger g} &= V_{\psi \cdot \psi^{-2}} \oplus V_{\psi \cdot \psi^2} = V_\psi \oplus V_{\psi^2} =: \varrho_1 \oplus \varrho_2, \\ V_{g^\dagger h} &= V_{\psi \cdot \psi_h} \oplus V_{\psi \cdot \psi_h^{-1}} = V_\psi \oplus V_\psi = \varrho_1 \oplus \varrho_1. \end{aligned}$$

A Heegner point calculation reveals that the Mordell-Weil group $E(H)$ has rank 5, and is generated by the Heegner point

$$P = \left(t, \frac{-t^4 - 3t^3 + 2t^2 + 7t - 6}{3} \right), \quad \text{where} \quad t^5 + t^4 - 5t^3 - 3t^2 + 8t - 3 = 0,$$

and its Galois translates $P^\sigma, P^{\sigma^2}, P^{\sigma^3}$ and P^{σ^4} , where σ is a generator of $\text{Gal}(H/K)$, chosen in such a way that $\psi(\sigma) = \zeta_5$ with ζ_5 a primitive fifth root of unity satisfying $\zeta_5 + \zeta_5^4 = -\omega$. Let $\omega' = (1 - \sqrt{5})/2$ denote the conjugate of the golden ratio. The L -vector space $E(H)_L^{\varrho_1}$ is then spanned by the points

$$\begin{aligned} P_1^+ &= 2P - \omega P^\sigma - \omega' P^{\sigma^2} - \omega' P^{\sigma^3} - \omega P^{\sigma^4}, \\ P_1^- &= (\zeta_5 - \zeta_5^4) \left(P^\sigma - \omega P^{\sigma^2} + \omega P^{\sigma^3} - P^{\sigma^4} \right), \end{aligned}$$

and $E(H)_L^{\varrho_2}$ has a similar basis (P_2^+, P_2^-) obtained by replacing ζ_5 by ζ_5^3 , and thus ω by its conjugate ω' , in the above expressions for P_1^+ and P_1^- .

Let $p = 29$ be the smallest prime which is inert in K but split in $L/\mathbb{Q}$. (The latter property is not essential to the conjectures, but makes it easier to calculate the p -adic iterated integrals.) Fix the unique embedding of H into $\mathbb{C}_p$ for which the image of P belongs to $E(\mathbb{Q}_p)$, and for which σ_p interchanges P^σ with P^{σ^4} and P^{σ^2} with P^{σ^3} .

For the elliptic unit, one may start with u as the unique root of $x^5 - 1014x^4 + 299243x^3 - 21035x^2 + 195x - 1$ in $H \cap \mathbb{Q}_{29}$ and define $u_{g_\pm^\dagger}$ in exactly the manner as P_2^- was defined above (only replacing P by u and using additive notation for $(\mathcal{O}_H^\times)_L$), i.e.,

$$u_{g_+^\dagger} := u_{g_-^\dagger} := (\zeta_5^3 - \zeta_5^2) \left(u^\sigma - \omega u^{\sigma^2} + \omega u^{\sigma^3} - u^{\sigma^4} \right).$$

Let $\gamma_{\pm 1}^\dagger \in S_1^{\text{oc,ord}}(79 \cdot 23, \chi)_{L_p}^\vee$ be the dual element to $g_{\pm 1}^\dagger$, defined as before. A numerical calculation of p -adic iterated integrals reveals that

$$\begin{aligned} \int_{\gamma_{\pm 1}^\dagger} f \cdot h &= 371806430046362 \bmod 29^{10} \\ \int_{\gamma_{\pm 1}^\dagger} f \cdot g &= 259114157339642 \bmod 29^{10}. \end{aligned}$$

The following identity was then verified to 10 digits of 29-adic precision:

$$\int_{\gamma_{\pm 1}^\dagger} f \cdot g = \frac{p \cdot \mathcal{E}(g_{\pm 1}^\dagger, f, g) \cdot (\log_{E,29}(P_1^+) \log_{E,29}(P_2^-) - \log_{E,29}(P_1^-) \log_{E,29}(P_2^+))}{4 \cdot 5 \cdot \sqrt{5} \log_{29}(u_{g^\dagger})}. \quad (64)$$

where here $\sqrt{5}$ denotes the unique 29-adic square root of 5 which is congruent to 18 modulo 29.

Turning to $\int_{\gamma_{\pm 1}^\dagger} f \cdot h$, notice that $\mathcal{E}(g_{\pm 1}^\dagger, f, h) = \mathcal{E}(g_{\pm 1}^\dagger, f, g) (= 864/841)$ and the transcendental factor $\log_{29}(u_{g^\dagger})$ expected on the righthand side of the formula is the same as for $\int_{\gamma_{\pm 1}^\dagger} f \cdot g$. Thus these factors should cancel upon taking a ratio and indeed one observes (modulo 29^{10}) the striking identity

$$\begin{aligned} \frac{\int_{\gamma_{\pm 1}^\dagger} f \cdot g}{\int_{\gamma_{\pm 1}^\dagger} f \cdot h} &= \frac{1}{2} \cdot \frac{\log_{E,29}(P_1^+) \log_{E,29}(P_2^-) - \log_{E,29}(P_1^-) \log_{E,29}(P_2^+)}{\log_{E,29}(P_1^+) \log_{E,29}(P_1^-)} \\ &= \frac{1}{2} \left(\frac{\log_{E,29}(P_2^-)}{\log_{E,29}(P_1^-)} - \frac{\log_{E,29}(P_2^+)}{\log_{E,29}(P_1^+)} \right), \end{aligned} \quad (65)$$

which suggests that

$$\int_{\gamma_{\pm 1}^\dagger} f \cdot h = \frac{p \cdot \mathcal{E}(g_{\pm 1}^\dagger, f, h) \cdot \log_{E,29}(P_1^+) \log_{E,29}(P_1^-)}{2 \cdot 5 \cdot \sqrt{5} \log_{29}(u_{g^\dagger})}. \quad (66)$$

Both equations (64) and (66) are in perfect agreement with Conjecture ES.

Remark 3.12. It was a ratio identity like the one in (65) (for a simpler ratio in Example 3.10) which first convinced the authors that the p -adic iterated integral had geometric meaning in the setting where p is inert, even prior to their full understanding of the transcendental factor occurring in the denominator of the identity of Conjecture ES.

We now turn to our first numerical example in the rank 2 setting, illustrating a notable feature of Conjecture ES, namely that its right hand side can be a non-trivial p -adic combination of logarithms of global points in a “genuine rank two” setting involving a primitive L -function with a double zero at the center.

Example 3.13. We take the curve

$$E : y^2 + xy + y = x^3 - 7x + 5$$

of conductor 57 labelled 57b in Cremona’s tables and write f for the attached newform of weight 2.

Let χ be the odd Dirichlet character of order 2 and conductor 23 associated to the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-23})$, of class number 3. The space $S_1(23, \chi)$ has dimension 1 and is spanned by the theta series $g := \theta_{\psi_g}$ attached to a cubic character ψ_g of $G = \text{Gal}(H/K)$, where H is the class field of K . Let $h = E(1, \chi) \in M_1(23, \chi)$ be the Eisenstein series. As in Example 3.10 we find

$$V_{gg} = V_{\psi_g} \oplus L \oplus L(\chi) \text{ with } L = \mathbb{Q}(\zeta_3)$$

but a calculation reveals that here

$$r(E, V_{\psi_g}) = 2, \quad r(E, L) = r(E, L(\chi)) = 0$$

and likewise for the analytic ranks. Here the Heegner hypothesis is not satisfied, but the descent code in MAGMA can be used to construct the required global points in $E(H)_L^{V_{\psi_g}}$. Over the cubic subfield

$$\mathbb{Q}(a), \quad a^3 - a^2 + 2a - 1 = 0$$

of H , the curve E has rank 2 with generators for the Mordell-Weil lattice

$$P := (9a^2 - 4a + 17, -45a^2 + 20a - 80), \quad Q := (a^2 + 3, 2a^2 + 2).$$

We embed $\mathbb{Q}(a)$ into $H = \mathbb{Q}(b)$, where $b^6 - 6b^4 + 9b^2 + 23 = 0$, by taking $a = (b^4 - 2b^2 + 1)/9$. Taking σ as a generator for G we compute four points

$$\begin{aligned} P_{\psi_g}^+ &:= 2P - P^\sigma - P^{\sigma^2}, & P_{\psi_g}^- &:= P^\sigma - P^{\sigma^2} \\ Q_{\psi_g}^+ &:= 2Q - Q^\sigma - Q^{\sigma^2}, & Q_{\psi_g}^- &:= Q^\sigma - Q^{\sigma^2} \end{aligned}$$

and similarly a unit $u_{\psi_g}^-$ starting from the root $(b^4 - 5b^2 + 4)/9$ of $x^3 - x - 1$ in H . We take $p = 19$ and find we may embed H into $\mathbb{C}_{19}$ so that P, Q and u are defined over $\mathbb{Q}_{19}$.

To compute the relevant 19-adic iterated integral we must work in 19-adic level $3 \cdot 23$ and so choose a 3-stabilisation $\check{g}$ of g . We find $x^2 - a_3(g)x + \chi(3) = (x - \zeta_3)(x - \zeta_3^2)$ and take $\check{g}$ to be the 3-stabilisation with U_3 -eigenvalue ζ_3 . (With the other choice the formula is exactly the same, except the algebraic factor is replaced by its complex conjugate.) We embed $\mathbb{Q}(\zeta_3)$ into $\mathbb{Q}_{19}$ so that $\zeta_3 \equiv 7 \pmod{19}$, and find

$$\int_{\check{\gamma}_{\pm 1}} f \cdot g = -25103076413984358720047537708218 \pmod{19^{25}}.$$

One checks to 25 digits of 19-adic precision that

$$\int_{\check{\gamma}_{\pm 1}} f \cdot g = \frac{63\zeta_3 + 18}{26 \cdot 19} \cdot \frac{\log_{E,19}(P_{\psi_g}^+) \log_{E,19}(Q_{\psi_g}^-) - \log_{E,19}(P_{\psi_g}^-) \log_{E,19}(Q_{\psi_g}^+)}{\log_p u_{\psi_g}^-},$$

as predicted by Conjecture ES.

In the next example, the relevant Heegner point arises from a Shimura curve rather than a classical modular curve parametrisation. Note that Conjecture ES in principle allows the efficient computation of these Heegner points in terms of the Fourier expansions of classical modular forms rather than working with their Jacquet-Langlands lifts. For the purpose of verifying Conjecture ES, the authors did not actually calculate the Heegner points arising from the Shimura curve parametrisation, which raises significant computational challenges, but rather found appropriate global points by general descent methods.

Example 3.14. Consider the curve

$$E : y^2 + y = x^3 + x^2 + 9x + 1$$

of conductor 35 labelled 35a in Cremona's tables and write f for the associated weight 2 newform. Let $K = \mathbb{Q}(\sqrt{-23})$ be the smallest imaginary quadratic field of class number 3, and note that both primes dividing 35 are inert in K . This is the simplest setting where there is no Heegner point on E arising from a modular curve, and in which one must rather work with the Shimura curve attached to the indefinite quaternion algebra ramified at 5 and 7.

Let χ be the quadratic Dirichlet character of conductor 23. Then $S_1(23, \chi)$ is one-dimensional and is spanned by the single dihedral form

$$g = q - q^2 - q^3 + q^6 + q^8 - q^{13} - q^{16} + \dots$$

Let H be the Hilbert class field of K , set $G = \text{Gal}(H/K)$, and let ϱ_g be the representation induced from a cubic character ψ_g of G with image contained in $L = \mathbb{Q}(\zeta_3)$.

The representation V_{gg} decomposes as

$$V_{gg} = V_{\psi_g \cdot \psi_g} \oplus V_{\psi_g \cdot \psi_g^{-1}} = V_{\psi_g} \oplus L(\chi) \oplus L.$$

A direct calculation reveals that

$$r(E, V_{\psi_g}) = r(E, L(\chi)) = 1, \quad r(E, L) = \text{rank}(E(\mathbb{Q})) = 0,$$

and the same is true of the analytic ranks. Let P_K be a generator of $E(K)$ up to torsion, which lies in the -1 eigenspace for the action of the involution in $\text{Gal}(K/\mathbb{Q})$, and let P be a $\mathbb{Q}[\text{Gal}(H/K)]$ -module generator of $E(H) \otimes \mathbb{Q}$. The points P_K and P were not constructed using Lemma 3.1, since the Heegner point evoked in this lemma is difficult to compute directly in practice. Instead, one just takes

$$P_K := \left(-5, \frac{-5\sqrt{-23}-1}{2}\right), \quad P := \left(\frac{-7t^4 + 14t^2 + 29}{36}, \frac{7t^4 - 14t^2 - 68}{24}\right) \in E(H),$$

where t is a normal basis generator for H over $\mathbb{Q}$ satisfying the equation

$$t^6 - 6t^4 + 9t^2 + 23 = 0.$$

The point P just turned out to be the first point in a basis for the rank 3 Mordell-Weil group $E(H)$ computed by MAGMA.

Let $p = 5$ be the smallest prime which is inert in K . After a suitable choice of embedding of H into K_5 for which P maps to $E(\mathbb{Q}_5)^-$, the Frobenius element σ_5 interchanges P^σ and P^{σ^2} . With these conventions, we may set

$$P_{\psi_g}^+ := 2P - (P^\sigma + P^{\sigma^2}).$$

Both f and g are 5-adic overconvergent modular forms of level 7 and 23 respectively. Stabilise g to an eigenform $\check{g} = g(q) - g(q^7)$ of level $161 = 7 \cdot 23$ with U_7 eigenvalue 1. Then $\check{g}$ has two stabilisations

$$\check{g}_{\pm 1} = \check{g}(q) \mp \check{g}(q^5)$$

at 5, which can be viewed either as classical modular forms of level $161 \cdot 5$ or as 5-adic overconvergent forms of level 161. Write $\check{\gamma}_{\pm 1} \in S_1^{\text{oc,ord}}(7 \cdot 23, \chi)^\vee$ for the usual dual element to $\check{g}_{\pm 1}$. A direct evaluation of the 5-adic iterated integral reveals that

$$\int_{\check{\gamma}_{+1}} f \cdot g = - \int_{\check{\gamma}_{-1}} f \cdot g = 376357248757242028241 \pmod{5^{30}}.$$

Starting with the elliptic unit u which is the unique root of $x^3 + 6x^2 + 29x - 1$ in $H \cap \mathbb{Q}_5$, the unit u^- is defined by $u^- = (u^\tau / u^{\tau^2})$. The following identity was then checked to 30 digits of 5-adic precision:

$$\int_{\check{\gamma}_{+1}} f \cdot g = \frac{3 \cdot 7 \cdot \log_{E,5}(P_K) \cdot \log_{E,5}(P_{\psi_g}^+)}{2 \cdot 5 \cdot \log_5(u^-)}$$

as predicted by (60). Note that once the point P_K is known, this identity leads to an explicit analytic formula for a global point on E over the Hilbert class field H , namely $P_{\psi_g}^+$, without passing through the calculation of the Shimura curve parametrisation of E , as would be necessary if one were to resort to Heegner points instead of p -adic iterated integrals.

4. STARK-HEEGNER POINTS AND THETA SERIES OF REAL QUADRATIC FIELDS

4.1 Theoretical preliminaries. This chapter focusses on the case where g and h are theta series attached to characters—denoted ψ_g and ψ_h respectively—of mixed signature $(+, -)$ of the *same* real quadratic field K . The self-duality assumption on V_{gh} implies that

$$\psi_1 := \psi_g \psi_h, \quad \psi_2 := \psi_g \psi'_h$$

are dihedral characters of K , which are totally even and totally odd respectively and that

$$V_g = V_{\psi_g}, \quad V_h = V_{\psi_h}, \quad V_{gh} = V_{\psi_1} \oplus V_{\psi_2}.$$

As in the previous chapter, the Hasse-Weil-Artin L -function factors as $L(E, \varrho_{gh}, s) = L(E/K, \psi_1, s) \cdot L(E/K, \psi_2, s)$, and Hypothesis A is fulfilled as soon as the conductors of ψ_g and ψ_h are coprime to N_f . In all examples where Conjecture ES is tested below, Hypothesis B holds because we have there $L(E/K, \psi_1, 1) = L(E/K, \psi_2, 1) = 0$, and Hypothesis C holds too by Proposition 1.2.

Let p be a rational prime, at which the representations V_g and V_h are unramified, and let α_g and β_g (resp. α_h and β_h) denote as before the frobenius eigenvalues for V_g (resp. V_h) attached to this prime. In order for g to satisfy the classicality condition of Hypothesis C, it is necessary to assume that

$$p \text{ is inert in } K. \quad (67)$$

In that case, by the remarks surrounding (21), the Stark unit $u_{g_\alpha} =: u_K$ of Conjecture ES can be taken to be equal to the fundamental unit of K , and in particular it depends only on K and not on ψ_g .

Assumption (67) also implies that the frobenius element σ_p corresponds to reflections in the dihedral groups cut out by the representations V_{ψ_1} and V_{ψ_2} . As in Section 3.3, it is assumed that

$$\alpha_g \alpha_h = 1, \quad \alpha_g \beta_h = -1, \quad \beta_g \alpha_h = -1, \quad \beta_g \beta_h = 1.$$

The eigenvalues of σ_p on V_{ψ_1} are $(1, -1)$, and likewise for V_{ψ_2} . Note in particular that in this setting, the representation V_{gh} is never regular at p , even though the representations V_g , V_h , V_{ψ_1} and V_{ψ_2} are all regular.

Consider first the “generic” setting where neither of the characters ψ_1 or ψ_2 is a quadratic (genus) character, so that the representations V_{ψ_1} and V_{ψ_2} are both *irreducible*. Conjecture ES then predicts the existence of points $P_1 \in E(H)_L^{V_{\psi_1}}$ and $P_2 \in E(H)_L^{V_{\psi_2}}$ whose projections onto the different σ_p -eigenspaces

$$\begin{aligned} P_1^+ &\in (E(H)_L^{V_{\psi_1}})^{\sigma_p=+1}, & P_2^- &\in (E(H)_L^{V_{\psi_2}})^{\sigma_p=-1}, \\ P_1^- &\in (E(H)_L^{V_{\psi_1}})^{\sigma_p=-1}, & P_2^+ &\in (E(H)_L^{V_{\psi_2}})^{\sigma_p=+1}, \end{aligned}$$

are related to a suitable p -adic iterated integral by

$$\int_{\gamma_\alpha} \check{f} \cdot \check{h} \stackrel{?}{=} \frac{\log_{E,p}(P_1^+) \log_{E,p}(P_2^-) - \log_{E,p}(P_1^-) \log_{E,p}(P_2^+)}{\log_p(u_K)}. \quad (68)$$

This conjecture becomes slightly easier to verify (if not to state!) when one of the characters ψ_1 or ψ_2 is a quadratic (genus) character. Assume without loss of generality that this is the case for ψ_1 , so that

$$V_{\psi_1} = L(\chi_1) \oplus L(\chi_2)$$

becomes reducible as a $G_{\mathbb{Q}}$ -module, breaking up into two one-dimensional representations attached to quadratic Dirichlet characters χ_1 and χ_2 satisfying $\chi_1 \chi_2 = \epsilon_K$, so that in particular $\chi_1(p) = -\chi_2(p)$. When the iterated integral of the left-hand side of (68) is non-zero, Conjecture ES predicts that precisely one of the characters χ_1 or χ_2 occurs in the Mordell-Weil group of E . Assume that it is χ_1 which arises, and set $\iota := \chi_1(p) = \pm 1$.

One then has

$$P_1 \in E(H)_L^{\chi_1}.$$

Conjecture ES predicts that, for a suitable choice of test vectors,

$$\int_{\gamma_\alpha} \check{f} \cdot \check{h} \stackrel{?}{=} \frac{\log_{E,p}(P_1) \log_{E,p}(P_2^{-\iota})}{\log_p(u_K)}. \quad (69)$$

Finally consider the case in which $\psi_1^2 = \psi_2^2 = 1$. The representation V_{gh} then decomposes as a sum of one dimensional summands, and Conjecture ES takes the form

$$\int_{\gamma_\alpha} \check{f} \cdot \check{h} \stackrel{?}{=} \frac{\log_{E,p}(P_1) \log_{E,p}(P_2)}{\log_p(u_K)}, \quad (70)$$

provided the points P_1 and P_2 , which both belong to quadratic extensions of $\mathbb{Q}$, belong to *distinct* eigenspaces for σ_p ; in the case where they are in the same eigenspace, Conjecture ES predicts a subtle vanishing of the p -adic iterated integral arising from a degeneracy in the regulator term. This case is illustrated in Example 4.3 below.

4.2. Numerical evidence. Weight one modular forms whose Artin representations can be induced *only* from characters of class groups of real quadratic fields are rather rare in small level. The first such occurrence is in level 145, where one finds such a form coming from a quartic character of $\mathbb{Q}(\sqrt{5})$ with conductor a prime of norm 29. The next example is concerned with this modular form.

Example 4.1. Let $K = \mathbb{Q}(\sqrt{5})$ and let

$$\psi_g = \psi_h^{-1}$$

be the quartic character of K which is ramified at the prime $\lambda = \left(29, \frac{11+\sqrt{5}}{2}\right)$ of norm 29. Let χ_5 and χ_{29} be the quadratic and quartic Dirichlet characters of conductors 5 and 29 respectively, and let $\chi := \chi_5 \chi_{29}$ be the odd quartic character of conductor $5 \cdot 29$. Both ψ_g and χ take values in the field $L := \mathbb{Q}(i)$ and are unique up to conjugation by $\text{Gal}(L/\mathbb{Q})$. Fix a choice of these characters in such a way that the theta series

$$g \in S_1(5 \cdot 29, \chi^{-1}), \quad h \in S_1(5 \cdot 29, \chi)$$

associated to ψ_g and ψ_h belong to spaces of weight 1 modular forms with those characters (and Fourier coefficients in L). The Artin representation V_{gh} is given by

$$V_{gh} = L \oplus L(\chi_5) \oplus V_\psi,$$

where ψ is the unique (up to conjugation) quartic ring class character of K of conductor 29. Let H be the abelian quartic extension of K cut out by ψ , and write $\text{Gal}(H/K) =: \langle \sigma \rangle$. The field H is Galois over $\mathbb{Q}$, with Galois group $G = D_8$, the dihedral group of order 8. The subfield of H fixed by the center $\langle \sigma^2 \rangle$ of G is the totally real biquadratic field $M := \mathbb{Q}(\sqrt{5}, \sqrt{29})$, and H is the quadratic extension of M given by

$$H = M(\sqrt{\delta}) = M(\sqrt{\delta'}), \quad \delta = \left(\frac{-29 + 3\sqrt{29}}{2} \right), \quad \delta' = \left(\frac{-29 - 3\sqrt{29}}{2} \right).$$

Note that δ and δ' are conjugate totally negative elements of M and that $\delta\delta' = 5 \cdot 29$.

Let f be the weight two cusp form of level 17 attached to the unique elliptic curve

$$E: y^2 + xy + y = x^3 - x^2 - x - 14$$

of conductor 17 denoted $17a$ in the tables of Cremona. The conductor of E is inert in the quadratic field K . An explicit calculation shows that

$$E(\mathbb{Q})_L = 0, \quad E(\mathbb{Q})_L^{\chi_5} = L \cdot P_K, \quad \text{where} \quad P_K = \left(\frac{379}{20}, \frac{-1995 + 7218\sqrt{5}}{200} \right).$$

The group G admits four distinct one-dimensional representations, on which σ^2 acts trivially, and a unique irreducible representation of dimension two with rational traces, which is isomorphic to V_ψ and on which σ^2 acts as multiplication by -1 . Hence

$$E(H)_{\mathbb{Q}}^{V_\psi} = E(H)_{\mathbb{Q}}^{\sigma^2=-1}.$$

Considerations involving signs in the functional equations of L -series suggest that the representation V_ψ occurs with *odd multiplicity* in $E(H)$ (cf. for example the introduction of [Dar]). Note that the field H , a totally odd cyclic quartic extension of K , contains no imaginary quadratic subfields. The classical Heegner point construction is therefore unavailable to produce the global points in $E(H)_{\mathbb{Q}}^{V_\psi}$ whose existence is predicted by the Birch and Swinnerton-Dyer conjecture. The Stark-Heegner point algorithms described in [DP] can be used as a substitute for the method based on Heegner points, and yield four non-trivial points in $E(H)$ given by

$$\begin{aligned} P &= \left(\frac{-220777 - 17703\sqrt{145}}{5800}, \frac{214977 + 17703\sqrt{145}}{11600} + \frac{28584525 + 3803103\sqrt{5} + 1645605\sqrt{29} + 2364771\sqrt{145}}{290000} \sqrt{\delta} \right) \\ P^\sigma &= \left(\frac{-220777 + 17703\sqrt{145}}{5800}, \frac{214977 - 17703\sqrt{145}}{11600} + \frac{28584525 + 3803103\sqrt{5} - 1645605\sqrt{29} - 2364771\sqrt{145}}{290000} \sqrt{\delta'} \right) \\ P^{\sigma^2} &= \left(\frac{-220777 - 17703\sqrt{145}}{5800}, \frac{214977 + 17703\sqrt{145}}{11600} - \frac{28584525 + 3803103\sqrt{5} + 1645605\sqrt{29} + 2364771\sqrt{145}}{290000} \sqrt{\delta} \right) \\ P^{\sigma^3} &= \left(\frac{-220777 + 17703\sqrt{145}}{5800}, \frac{214977 - 17703\sqrt{145}}{11600} - \frac{28584525 + 3803103\sqrt{5} - 1645605\sqrt{29} - 2364771\sqrt{145}}{290000} \sqrt{\delta'} \right). \end{aligned}$$

These four Galois conjugate points are constructed from periods arising from a p -adic integration on $\mathcal{H}_{17} \times \mathcal{H}$ based on the *minus* modular symbol of f , as described in [Dar]. The conjectures of loc.cit. predict that these points should lie in the minus eigenspace for complex conjugation, which coincides in this case with the central involution σ^2 of $\text{Gal}(H/\mathbb{Q})$. It can indeed be observed that $\sigma^2 P = -P$ and that $\sigma^3 P = -\sigma P$. In particular, the Stark-Heegner point P and its conjugates generate a rank two subgroup of $E(H)$ which is isomorphic to V_ψ . This shows that $r(E, V_\psi) \geq 1$, and the belief that Stark-Heegner points ought to behave like their quadratic imaginary counterparts strongly suggests that in fact $r_{\text{an}}(E, V_\psi) = 1$. If such is the case, Conjecture ES predicts the non-vanishing of some p -adic iterated integral attached to (f, g, h) .

The global point P and its conjugates were obtained from a 17-adic calculation. This is an essential limitation of [Dar], where one is forced to work p -adically at a prime p dividing the conductor of E . On the other hand, the p -adic iterated integral appearing on the left of (69) can be calculated using any prime p which is inert in K . It turns out to be expedient (albeit, not essential, as in the calculations relying on the algorithms of [DP]) to work at $p = 17$. This is because, with this choice of p , the classical modular form f of level 17 is p -adically of level 1, and the p -adic iterated integral can therefore be calculated by applying the ordinary projection algorithms of [La2] to a space of ordinary overconvergent 17-adic modular forms of weight one and relatively modest level $5 \cdot 29$.

It can be checked that, for a suitable embedding of H into the quadratic extension $\mathbb{Q}_{17^2}$ of $\mathbb{Q}_{17}$, the points P and $\sigma^2 P$ are defined over $\mathbb{Q}_{17}$ while the points σP and $\sigma^3 P$ are defined over $\mathbb{Q}_{17^2}$ and are interchanged by the Frobenius element σ_{17} . Fix such an embedding from now on, and let $\gamma_\pm \in S_1(\Gamma_1(5 \cdot 29), \chi)_L^\vee[g_\pm]$ be the unique element satisfying

$$\gamma_\pm(g_\pm^*) = 1.$$

Here $g_{\pm} = g_{\pm\zeta_8^3}$ denotes the 17-stabilisations of g . A numerical evaluation of the 17-adic iterated integrals, to 16 significant digits shows that

$$\int_{\gamma_{\pm}} f \cdot h = 1259389260500681328 \times 17 \bmod 17^{16},$$

and that

$$\int_{\gamma_{\pm}} f \cdot h = \frac{\log_{E,17}(P_K) \log_{E,17}(P)}{3 \cdot 17 \cdot \log_{17}(u_K)} \quad (71)$$

to an accuracy of 16 significant 17-adic digits. Since $\sigma_{17}P_K = -P_K$, while P generates $E(H)_L^{V_{\psi}} \cap E(\mathbb{Q}_{17})$, this is in perfect agreement with the prediction of (69).

Remark 4.2. It is worth emphasising that the Stark-Heegner point P is *a priori* quite mysterious and not known to correspond to a global point, while the 17-adic iterated integral appearing on the left-hand side of (71) can be related to the 17-adic logarithm of the *global* cohomology class $\kappa(\check{f}, \check{g}, \check{h}) \in H^1(H, V_{17}(E))$ constructed in [DR2] as a limit of 17-adic étale Abel-Jacobi images of suitable “generalised Gross-Schoen diagonal cycles”. Since both the 17-adic iterated integral appearing on the left hand side of (71) and the “integrals on $\mathcal{H}_{17} \times \mathcal{H}$ ” used to compute the Stark-Heegner point P are given by explicit 17-adic analytic expressions, a proof of formula (71) does not seem entirely out of reach. Such a proof would represent substantial progress in understanding the global properties of Stark-Heegner points, making it possible, for instance, to use them in Euler system arguments of the kind exploited in [DR2] in the setting of “analytic rank 0”.

The next example involves forms g and h with representation having projective image D_4 . Such odd representations are induced from class characters of *both* real and imaginary quadratic fields, so this example would be equally at home in Chapter 3. Its principal novelty is that it illustrates situations where the p -adic iterated integrals vanish in rank $(1, 1)$ settings because the regulator term of Conjecture ES is 0.

Example 4.3. Consider the curves

$$\begin{aligned} E_a : y^2 + y &= x^3 - x^2 + 10x + 6 \\ E_b : y^2 + xy + y &= x^3 + x^2 - x - 2 \\ E_c : y^2 + y &= x^3 - x^2 - x + 1 \end{aligned}$$

of conductor 155 labelled 155a, 155b, and 155c in Cremona’s tables and write f_a , f_b and f_c for the associated weight two newforms.

Let χ_1 denote the trivial character, let χ_5 and χ_{31} denote the (even and odd, respectively) characters of conductor 5 and 31, and set $\chi := \chi_{155} := \chi_5 \cdot \chi_{31}$, which takes values in $L = \mathbb{Q}$. The space $S_1(155, \chi)$ has dimension 3 and contains three dihedral forms: one form g with integer Fourier coefficients

$$g = q + q^4 - q^5 - q^9 + q^{16} - 2q^{19} - \dots$$

and a second pair of complex conjugate forms each defined over the cyclotomic field $\mathbb{Q}(\zeta_3)$. The form g has associated linear representation ρ_g with projective image D_4 , and ρ_g can be described in the following equivalent ways:

- (1) It is induced from a quartic character of the Hilbert class field of $\mathbb{Q}(\sqrt{-155})$;
- (2) it is induced from a quadratic character of the imaginary quadratic field $\mathbb{Q}(\sqrt{-31})$, whose conductor is a prime ideal of norm 5;
- (3) it is induced from a quadratic character of the real quadratic field $\mathbb{Q}(\sqrt{5})$, whose conductor is a prime ideal of norm 31.

(The remaining pair of complex conjugate forms are associated to representations which arise from quartic characters of $\mathbb{Q}(\sqrt{-31})$ with conductor a prime ideal of norm 5—they are not considered further in this example.)

The representation V_{gg} decomposes as

$$V_{gg} = L \oplus L(\chi_5) \oplus L(\chi_{31}) \oplus L(\chi_{155}),$$

and the corresponding ranks are given by

$$\begin{aligned} r(E_a/\mathbb{Q}) &= 1, & r(E_a, \chi_5) &= 0, & r(E_a, \chi_{31}) &= 1, & r(E_a, \chi_{155}) &= 0 \\ r(E_b/\mathbb{Q}) &= 0, & r(E_b, \chi_5) &= 0, & r(E_b, \chi_{31}) &= 0, & r(E_b, \chi_{155}) &= 0 \\ r(E_c/\mathbb{Q}) &= 1, & r(E_c, \chi_5) &= 1, & r(E_c, \chi_{31}) &= 0, & r(E_c, \chi_{155}) &= 0, \end{aligned}$$

and likewise for the analytic ranks. In particular, the central critical value $L(E_b, \varrho_{gg}, 1)$ is non-zero, and hence the curve E_b fails to satisfy Hypothesis B of the introduction. It will therefore be ignored from now on.

As predicted by the BSD conjecture, the points on E_a and E_c over $\mathbb{Q}$ and the fields $\mathbb{Q}(\sqrt{-31})$ and $\mathbb{Q}(\sqrt{5})$ are given by

$$\begin{aligned} P_a &= (2, 5) \in E_a(\mathbb{Q}), & P_{a,31} &= \left(\frac{-1630}{729}, \frac{40175\sqrt{-31} - 19683}{39366} \right) \in E_a(\mathbb{Q}(\sqrt{-31}))^{\chi_{31}}, \\ P_c &= (1, 0) \in E_c(\mathbb{Q}), & P_{c,5} &= \left(0, \frac{-1 + \sqrt{5}}{2} \right) \in E_c(\mathbb{Q}(\sqrt{5}))^{\chi_5}. \end{aligned}$$

Consider both the primes $p = 7$ and $p = 13$, which have different splitting behaviours in the three quadratic fields, as shown in the table below.

p	$\chi_5(p)$	$\chi_{31}(p)$	$\chi_{155}(p)$
7	-1	+1	-1
13	-1	-1	+1

For $p = 7$ the eigenvalues of σ_p on V_g are $\alpha_g = 1$ and $\beta_g = -1$, and so

$$(\alpha_g \cdot \alpha_g, \alpha_g \cdot \beta_g) = (1, -1), \quad (\beta_g \cdot \alpha_g, \beta_g \cdot \beta_g) = (-1, 1).$$

The 7-adic iterated integrals attached to $(f_a, \gamma_{\pm 1}, g)$ and $(f_c, \gamma_{\pm 1}, g)$ are

$$\begin{aligned} \int_{\gamma_1} f_a \cdot g &= \int_{\gamma_{-1}} f_a \cdot g = 0 \pmod{7^{30}} \\ \int_{\gamma_1} f_c \cdot g &= \int_{\gamma_{-1}} f_c \cdot g = 1416540068067896281523634 \times 7 \pmod{7^{30}}. \end{aligned}$$

For $p = 13$ the eigenvalues of σ_p on V_g are $\alpha_g = i$ and $\beta_g = -i$, and so

$$(\alpha_g \cdot \alpha_g, \alpha_g \cdot \beta_g) = (-1, 1), \quad (\beta_g \cdot \alpha_g, \beta_g \cdot \beta_g) = (1, -1).$$

The 13-adic iterated integrals attached to $(f_a, \gamma_{\pm i}, g)$ and $(f_c, \gamma_{\pm i}, g)$ are

$$\begin{aligned} \int_{\gamma_i} f_a \cdot g &= - \int_{\gamma_{-i}} f_a \cdot g = 10546307292618612 \pmod{13^{15}} \\ \int_{\gamma_i} f_c \cdot g &= \int_{\gamma_{-i}} f_c \cdot g = 21913581880269709 \pmod{13^{15}}. \end{aligned}$$

Following the notations in the rank (1, 1) setting of Conjecture ES, the pair (ϱ_1, ϱ_2) of irreducible subrepresentations of V_{gg} attached to the curves E_a and E_c can be taken to be (χ_1, χ_{31}) and (χ_1, χ_5) , respectively, where χ_1 is the trivial character. Observe that the eigenvalues of σ_{13} are $(1, -1)$ in both cases. One checks to 15-digits of 13-adic precision that

$$\int_{\gamma_i} f_a \cdot g = \frac{80 \cdot \log_{E_a, 13}(P_a) \log_{E_a, 13}(P_{a,31})}{i \cdot 63 \cdot 13 \cdot \log_{13}(u^-)}$$

where i is the unique square root of -1 in $\mathbb{Z}_{13}$ which is $\equiv 5 \pmod{13}$, and $u^- = (1 + \sqrt{5})/2$ is a fundamental unit for $\mathbb{Q}(\sqrt{5})$. Note that the same formula holds if one replaced i by $-i$ on both sides. Similarly one checks to the same amount of 13-adic precision that

$$\int_{\gamma_i} f_c \cdot g = -\frac{80 \cdot \log_{E_c,13}(P_c) \log_{E_c,13}(P_{c,5})}{21 \cdot 13 \cdot \log_{13}(u^-)}.$$

Turning now to $p = 7$ and the curve E_a , observe that $(\chi_1(7), \chi_{31}(7)) = (1, 1)$, and hence the regulator term in Conjecture ES vanishes. This explains why the iterated integrals attached to $(f_a, \gamma_{\pm 1}, g)$ were observed to vanish, at least numerically to 30 digits of 7-adic precision. The situation for the curve E_c is different, since $(\chi_1(7), \chi_5(7)) = (1, -1)$. As predicted by Conjecture ES, the authors were able to verify, to 30 digits of 7-adic precision, that

$$\int_{\gamma_{\pm 1}} f_c \cdot g = -\frac{8 \cdot \log_{E_c,7}(P_c) \log_{E_c,7}(P_{c,5})}{3 \cdot 7 \cdot \log_7(u^-)}.$$

5. STARK POINTS OVER A_4 , S_4 OR A_5 EXTENSIONS AND EXOTIC WEIGHT ONE FORMS

The remaining chapters summarize some of the numerical evidence that was obtained in support of Conjecture ES in settings where g and h are not both theta series. It is an apt moment to pause momentarily and thank again Steve Donnelly: most of the interesting global points in both chapters were found by him upon our request using his descent code. We might also stress here that although the overconvergent projection algorithms used to compute p -adic iterated integrals are efficient, such computations still remain something of a technological *tour de force*: many of our examples pushed the available hardware and software (the algorithms of [La2] in particular) to their current limits.

This chapter focusses on the case where V_{gh} is reducible and at least one of g or h is an exotic cusppform whose associated projective Galois representation cuts out a non-totally real extension of $\mathbb{Q}$ with Galois group A_4 , S_4 or A_5 .

Lemma 5.1. *Assume that one of g or h is an exotic weight one form and that g and h satisfy the self-duality condition. Then the representation V_{gh} is irreducible, unless g is the twist of h^* by a quadratic character ι , in which case*

$$V_{gh} = L(\iota) \oplus \text{Ad}_g(\iota).$$

Proof. Lemma 6.2. of [DR2] rules out the case where V_{gh} contains an irreducible subrepresentation of dimension two. Hence V_{gh} contains an irreducible one-dimensional subrepresentation, and the result follows from Lemma 6.1. of [DR2]. $\square$

So if V_{gh} is reducible, Lemma 5.1 implies that g is a quadratic twist of h^* . Since it is always possible to replace the form f by one of its quadratic twists, there is no real loss in generality in assuming that $g = h^*$ which is what shall be done in the rest of this chapter. In this setting, Conjecture ES suggests a completely new, efficiently computable (conjectural) analytic formula for the p -adic logarithm of global points defined over the A_4 , S_4 or A_5 -extension cut out by the adjoint Ad_g (or, equivalently, by the projective representation attached to g).

5.1. Theoretical preliminaries. Let h be an exotic weight one form, let $g = h^*$ be its conjugate form, and let L be as before the field generated by the Fourier coefficients of h , or of $g = h^*$. The representation $V_{gh} = L \oplus \text{Ad}_g$, where L refers to the trivial representation of $G_{\mathbb{Q}}$ with coefficients in L , factors through $\text{Gal}(H/\mathbb{Q})$, where H is the field cut out by the *projective* representation attached to g , so that

$$\text{Gal}(H/\mathbb{Q}) \simeq A_4, S_4 \text{ or } A_5.$$

Conjecture ES predicts that the trilinear form I'_p in (6) arising from p -adic iterated integrals may be non-identically zero only when

$$r(E, L) = r(E, \text{Ad}_g) = 1, \quad (72)$$

or

$$r(E, L) = 0, \quad r(E, \text{Ad}_g) = 2. \quad (73)$$

In the rank $(1, 1)$ setting, Conjecture ES asserts that, for a suitable choice of elements

$$P_{\mathbb{Q}} \in E(\mathbb{Q}), \quad P_{g_{\alpha}} \in (E(H)_L^{\text{Ad}_g})^{\sigma_p = \alpha_g / \beta_g}, \quad u_{g_{\alpha}} \in ((\mathcal{O}_H^{\times})_L^{\text{Ad}_g})^{\sigma_p = \alpha_g / \beta_g},$$

which, when $\alpha_g \neq \pm\beta_g$, are *unique* up to scaling by $L^{\times}$, and for a suitable choice $\check{f}$, $\check{\gamma}_{g_{\alpha}}$ and $\check{h}$ of test vectors attached to (f, g_{α}, h) ,

$$\int_{\check{\gamma}_{g_{\alpha}}} \check{f} \cdot \check{h} \stackrel{?}{=} \frac{\log_{E,p}(P_{\mathbb{Q}}) \log_{E,p}(P_{g_{\alpha}})}{\log(u_{g_{\alpha}})}.$$

In the rank 2 setting, Conjecture ES asserts that

$$\int_{\check{\gamma}_{g_{\alpha}}} \check{f} \cdot \check{h} \stackrel{?}{=} \frac{\log_{E,p}(P_1) \log_{E,p}(Q_{g_{\alpha}}) - \log_{E,p}(Q_1) \log_{E,p}(P_{g_{\alpha}})}{\log(u_{g_{\alpha}})},$$

where (P, Q) is a basis of $\text{Hom}_{G_{\mathbb{Q}}}(\text{Ad}_g, E(H)_L)$, and P_1, Q_1 (resp. $P_{g_{\alpha}}, Q_{g_{\alpha}}$) denote the images of a vector $v_1 \in \text{Ad}_g^{\sigma_p=1}$ (resp. $v_2 \in \text{Ad}_g^{\sigma_p=\alpha_g/\beta_g}$) under these homomorphisms.

In the rank $(1, 1)$ setting, the conjectural formula above gives an explicit, albeit conjectural, p -adic analytic expression for the logarithm of a global point—namely $P_{g_{\alpha}}$ —defined over the field H cut out by Ad_g . This point can thus be viewed as an elliptic curve analogue of the Stark unit $u_{g_{\alpha}}$. For the sake of numerical experiments, it is worth giving a more explicit description of $P_{g_{\alpha}}$, depending on the nature of $\text{Gal}(H/\mathbb{Q})$.

5.1.1. Cases where g is of tetrahedral type. The group $G = \text{Gal}(H/\mathbb{Q}) \simeq A_4$ has three representations of dimension one which factor through the quotient $A_4/V_4 \simeq A_3$, and a unique faithful irreducible representation of dimension three, which is therefore isomorphic to the adjoint representation Ad_g . Let $R(x)$ be any degree four polynomial whose splitting field is H , and let M be the quartic field which is generated by a single root of R . Then Ad_g is related to a permutation representation by the formula

$$\text{Ad}_g = \text{Ind}_{G_M}^{G_{\mathbb{Q}}} L - L.$$

It follows that

$$r(E, \text{Ad}_g) = r(E, \text{Ind}_M^{\mathbb{Q}} L) - r(E, L) = \text{rank}(E(M)) - \text{rank}(E(\mathbb{Q})).$$

(The Artin formalism for L -series implies similar equalities for the analytic ranks.) In particular when $r(E, \text{Ad}_g) = 1$, there is (up to scaling) a unique point $P \in E(M)$ whose trace to $E(\mathbb{Q})$ is zero, and $E(H)_L^{\text{Ad}_g}$ is generated, as an L -vector space, by the four Galois translates of P , whose sum is zero. Let $P =: P_1, P_2, P_3$ and P_4 denote these Galois translates, on which $\text{Gal}(H/\mathbb{Q}) = A_4$ acts via the natural permutation action. The point $P_{g_{\alpha}}$ can be expressed as a linear combination of the P_j , depending on the conjugacy class of σ_p .

Case 1: $\sigma_p = 1$. In the case $\alpha_g/\beta_g = \beta_g/\alpha_g = 1$, the representation Ad_g is not regular at p , and Hypothesis C need not be satisfied. Conjecture ES then only asserts, when the p -adic iterated integrals attached to g_{α} are defined, that $P_{g_{\alpha}}$ is some undetermined linear combination of P_1, P_2, P_3 and P_4 , which all belong to $E(\mathbb{Q}_p)$ after a p -adic embedding of H has been chosen.

Case 2: σ_p is of order 2. In that case it may be assumed that the points P_1 and P_2 are defined over $\mathbb{Q}_{p^2}$ and are interchanged by σ_p , and likewise for P_3 and P_4 . It follows that $\alpha_g/\beta_g = -1$ and that

$$\begin{aligned} (E(H)_L^{\text{Ad}_g})^{\sigma_p=1} &= L \cdot (P_1 + P_2) = L \cdot (P_3 + P_4) && \text{is one-dimensional,} \\ (E(H)_L^{\text{Ad}_g})^{\sigma_p=-1} &= L \cdot (P_1 - P_2) \oplus L \cdot (P_3 - P_4) && \text{is two-dimensional.} \end{aligned}$$

In this scenario, in which V_g is regular but Ad_g is not, it can only be asserted that P_{g_α} is some L -linear combination of the points $P_1 - P_2$ and $P_3 - P_4$.

Case 3: σ_p is of order 3. In that case, it may be assumed that σ_p fixes the point P_1 , which is thus defined over $\mathbb{Q}_p$, and cyclically permutes the points P_2, P_3 and P_4 . The eigenvalues of σ_p on Ad_g are then equal to 1, $\alpha_g/\beta_g = \zeta_3$ and $\beta_g/\alpha_g = \zeta_3^{-1}$, for some primitive cube root ζ_3 of 1, and P_{g_α} is determined, up to scaling, by the rule

$$P_{g_\alpha} = P_2 + \zeta_3^{-1}P_3 + \zeta_3P_4. \quad (74)$$

In particular,

$$(E(H)_L^{\text{Ad}_g})^{\sigma_p=\alpha_g/\beta_g} = L \cdot P_{g_\alpha} \quad \text{is one-dimensional.}$$

This is the only regular case, in which the point P_{g_α} is completely pinned down by its specified behaviour under the action of σ_p .

Remark 5.2. There are two distinct conjugacy classes of elements of order 3 in A_4 , which are rational over $\mathbb{Q}(\zeta_3)$ and interchanged by the action of $\text{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$. Note that 8 of the 12 elements in A_4 are of order 3, and hence, by the Chebotarev density theorem, the Frobenius element σ_p is regular for Ad_g for a set of primes p of Dirichlet density $2/3$.

5.1.2. Cases where g is of octahedral type. The dimensions of the irreducible representations of the group $G = \text{Gal}(H/\mathbb{Q}) \simeq S_4$ are 1, 1, 2, 3, and 3. The first two representations are the trivial and sign representations, and the two-dimensional representation factors through the quotient $S_4/V_4 = S_3$. In particular, the only faithful irreducible representations of G are three-dimensional, and there are two such, whose characters are given in the following table:

size	1	6	3	8	6
class	1	(12)	(12)(34)	(123)	(1234)
ϱ_1	3	1	-1	0	-1
ϱ_2	3	-1	-1	0	1

The representation ϱ_1 occurs in the natural permutation representation of G on 4 letters, while ϱ_2 is just the twist of ϱ_1 by the sign character. Among the two representations it is ϱ_2 which is the adjoint representation. This can be seen by noting that the eigenvalues of $\varrho_1((1234))$ are visibly $-1, i$ and $-i$, and that those of $\varrho_2((1234))$ are $1, i$, and $-i$, and that only the latter set of eigenvalues is consistent with the property of the adjoint representation that all elements in G admit 1 as eigenvalue.

Let $R(x)$ be any degree four polynomial whose splitting field is H , and let M be the quartic field which is generated by a single root of R . The field H is an S_3 -extension of M and hence there is a unique quadratic extension $\tilde{M}$ of M contained in H . The field H also contains a unique quadratic field $K/\mathbb{Q}$, which is cut out by the sign character of S_4 and satisfies $\tilde{M} = MK$. The representation Ad_g is described by the formula

$$\text{Ad}_g = \text{Ind}_{G_M}^{G_{\mathbb{Q}}} \text{sgn} - \text{sgn},$$

where sgn is viewed both as the quadratic character of G_M and $G_{\mathbb{Q}}$ induced by the sign characters on the finite quotients $\text{Gal}(H/M) \simeq S_3$ and $\text{Gal}(H/\mathbb{Q}) \simeq S_4$. It follows that

$$r(E, \text{Ad}_g) = r(E, \text{Ind}_M^H \text{sgn}) - r(E, \text{sgn}) = \dim_L E(\tilde{M})_L^- - \dim_L E(K)_L^-,$$

where the superscript $-$ refers to the -1 eigenspace for the involutions in $\text{Gal}(\tilde{M}/M)$ and $\text{Gal}(K/\mathbb{Q})$ respectively. When $r(E, \text{Ad}_g) = 1$, there is, by Frobenius reciprocity, a unique point $P \in E(\tilde{M})^-$, up to scaling. Let P_1, P_2, P_3 and P_4 denote the A_4 translates of $P = P_1$, and observe that $\sigma \in G = S_4$ acts on these points by the rule

$$\sigma P_j = \begin{cases} P_{\sigma j} & \text{if } \sigma \in A_4; \\ -P_{\sigma j} & \text{if } \sigma \in S_4 - A_4. \end{cases}$$

The resulting explicit description of the action of G on these four points, which together generate $E(H)_L^{\text{Ad}_g}$, make it possible to describe the point P_{g_α} explicitly, depending on the conjugacy class of σ_p in G .

Case 1: $\sigma_p = 1$. In that case, $\alpha_g/\beta_g = \beta_g/\alpha_g = 1$, and the representation Ad_g is not regular at p . Conjecture ES then only asserts that P_{g_α} is some undetermined linear combination of P_1, P_2, P_3 and P_4 .

Case 2: σ_p is of order 2 and odd, i.e., it is conjugate to the transposition (12). In that case, after relabelling the P_j , it can be assumed that

$$\sigma_p P_1 = -P_2, \quad \sigma_p P_2 = -P_1, \quad \sigma_p P_3 = -P_3, \quad \sigma_p P_4 = -P_4.$$

In particular, the x -coordinates of the points P_1 and P_2 lie in $\mathbb{Q}_{p^2}$ and are permuted by $\text{Gal}(\mathbb{Q}_{p^2}/\mathbb{Q}_p)$, while the x coordinates of P_3 and P_4 are defined over $\mathbb{Q}_p$, even though all four points are defined over $\mathbb{Q}_{p^2}$. The eigenvalues of σ_p are equal to 1 and $-1 = \alpha_g/\beta_g = \beta_g/\alpha_g$, and

$$\begin{aligned} (E(H)_L^{\text{Ad}_g})^{\sigma_p=1} &= L \cdot (P_1 - P_2) && \text{is one-dimensional,} \\ (E(H)_L^{\text{Ad}_g})^{\sigma_p=-1} &= L \cdot P_3 \oplus L \cdot P_4 && \text{is two-dimensional.} \end{aligned}$$

This is another irregular case, in which it can only be asserted that P_{g_α} is some L -linear combination of P_3 and P_4 .

Case 3: σ_p is of order 2 and even, i.e., it is conjugate to the permutation (12)(34). After eventually relabelling the P_j , it can be assumed that

$$\sigma_p P_1 = P_2, \quad \sigma_p P_2 = P_1, \quad \sigma_p P_3 = P_4, \quad \sigma_p P_4 = P_3,$$

so that

$$\begin{aligned} (E(H)_L^{\text{Ad}_g})^{\sigma_p=1} &= L \cdot (P_1 + P_2) = L \cdot (P_3 + P_4) && \text{is one-dimensional,} \\ (E(H)_L^{\text{Ad}_g})^{\sigma_p=-1} &= L \cdot (P_1 - P_2) \oplus L \cdot (P_3 - P_4) && \text{is two-dimensional.} \end{aligned}$$

This is yet another irregular case, and P_{g_α} is some unspecified L -linear combination of $P_1 - P_2$ and $P_3 - P_4$.

Case 4: σ_p is of order 3. In that case, it can be assumed that σ_p fixes the point P_1 , which is defined over $\mathbb{Q}_p$, and permutes the points P_2, P_3 and P_4 cyclically. The eigenvalues of σ_p on Ad_g are then equal to 1, $\alpha_g/\beta_g = \zeta_3$ and $\beta_g/\alpha_g = \zeta_3^{-1}$, and P_{g_α} is determined, up to scaling, by the rule

$$P_{g_\alpha} = P_2 + \zeta_3^{-1} P_3 + \zeta_3 P_4.$$

In particular,

$$(E(H)_L^{\text{Ad}_g})^{\sigma_p=\alpha_g/\beta_g} = L \cdot P_{g_\alpha} \quad \text{is one-dimensional.}$$

Case 5: σ_p is of order 4, and hence is conjugate to (1234). In that case, it can be assumed that

$$\sigma_p P_1 = -P_2, \quad \sigma_p P_2 = -P_3, \quad \sigma_p P_3 = -P_4, \quad \sigma_p P_4 = -P_1.$$

The eigenvalues of σ_p are equal to 1, i and $-i$, and

$$\begin{aligned} (E(H)_L^{\text{Ad}_g})^{\sigma_p=1} &= L \cdot (P_1 - P_2 + P_3 - P_4), \\ (E(H)_L^{\text{Ad}_g})^{\sigma_p=i} &= L \cdot (P_1 + iP_2 - P_3 - iP_4), \end{aligned} \quad (75)$$

$$(E(H)_L^{\text{Ad}_g})^{\sigma_p=-i} = L \cdot (P_1 - iP_2 - P_3 + iP_4). \quad (76)$$

In particular, the α_g/β_g -eigenspace for σ_p is one-dimensional if and only if σ_p has order > 2 , which occurs for 14 of the 24 permutations in S_4 . This regularity condition is therefore satisfied for a set of primes p of density $7/12$.

5.1.3. Cases where g is of icosahedral type. The dimensions of the irreducible representations of the group $G = \text{Gal}(H/\mathbb{Q}) \simeq A_5$ are 1, 3, 3, 4 and 5, and the characters of the two distinct three-dimensional representations are described in the following table:

size	1	15	20	12	12
class	1	(12)(34)	(123)	(12345)	(12354)
ϱ_1	3	-1	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
ϱ_2	3	-1	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$

In this case, both the representations ϱ_1 and ϱ_2 arise as the adjoint of the Artin representation ϱ_g and ϱ'_g respectively, where ϱ'_g denotes the Artin representation of the conjugate form of g , whose Fourier coefficients lie in the field $\mathbb{Q}(\sqrt{5})$.

It turns out to be easier to describe the direct sum $\varrho_1 \oplus \varrho_2$ in Galois-theoretic terms rather than each ϱ_i individually. Let D_{10} be the dihedral group of order 10, embedded naturally in A_5 , and let M be the non-Galois sextic field fixed by D_{10} . The field M is generated by a root of a sextic polynomial $R(x)$, the so-called *Lagrange sextic resolvent* attached to the A_5 extension. Since $\text{Gal}(H/M) = D_{10}$, there is a unique quadratic extension $\tilde{M}$ of M contained in H , which is cut out by the unique non-trivial quadratic character χ^- of D_{10} . It is not hard to check the identity of virtual representations:

$$\varrho_1 \oplus \varrho_2 = \text{Ind}_{G_M}^{G_{\mathbb{Q}}} \chi^-,$$

which implies that the six-dimensional representation $E(H)_L^{\text{Ad}_g} \oplus E(H)_L^{W_{g'}}$ is generated as an L -vector space, by the unique point $P \in E(\tilde{M})^-$ together with its 12 Galois translates

$$P =: P_1, P_2, P_3, P_4, P_5, P_6, \quad -P_1, -P_2, -P_3, -P_4, -P_5, -P_6.$$

The Stark point P_{g_α} is a $\mathbb{Q}(\sqrt{5})$ -linear combination of the 6 linearly independent points P_j with $1 \leq j \leq 6$. The coefficients that enter in this linear combination are completely determined by the requirement that $\sigma_p P_{g_\alpha} = \alpha_g/\beta_g P_{g_\alpha}$, provided the Frobenius element σ_p is not of order ≤ 2 . Since A_5 has precisely 44 elements of order > 2 , this condition holds for a set of primes of Dirichlet density $11/15$.

5.2. Numerical evidence. In contrast to the situation in Sections 4.2 and 7.2, numerical evidence is somewhat more bountiful in the setting of exotic modular forms. In particular, despite the appellation “exotic” such forms seem to be more plentiful in small level than those studied in Section 4.2. (Note that in all our exotic examples we checked that hypotheses A, B and C hold.)

As in Example 3.10 it is helpful to consider algebraic factors similar to those defined in [DR1, Theorem 3.1], given by

$$\begin{aligned} \mathcal{E}(g_\alpha, f, h) &:= (1 - \alpha_g \alpha_f \alpha_h p^{-1}) \times (1 - \alpha_g \alpha_f \beta_h p^{-1}) \\ &\quad \times (1 - \alpha_g \beta_f \alpha_h p^{-1}) \times (1 - \alpha_g \beta_f \beta_h p^{-1}) \\ \mathcal{E}_0(g_\alpha) &:= (1 - \alpha_g^2 \chi^{-1}(p)) \\ \mathcal{E}_1(g_\alpha) &:= (1 - \alpha_g^2 \chi^{-1}(p) p^{-1}). \end{aligned}$$

Here when p divides the level of f we set $\alpha_f = a_p(f)$ and $\beta_f = 0$, and otherwise define these so that $x^2 - a_p(f)x + p = (x - \alpha_f)(x - \beta_f)$. Economising on notation, we define $\mathcal{E}_\alpha$ as

$$\mathcal{E}_\alpha := \frac{\mathcal{E}(g_\alpha, f, h)}{\mathcal{E}_0(g_\alpha)\mathcal{E}_1(g_\alpha)} \in L. \quad (77)$$

We define $\mathcal{E}_\beta$ similarly, switching the roles of α_g and β_g in the above.

Our first exotic example involves the tetrahedral form of level 133 first constructed by Tate.

Example 5.3. Let χ be a Dirichlet character of level $133 = 7 \cdot 19$ of order 2 at 7 and order 3 at 19. (There are two such characters which are Galois conjugate to each other.) Denote by ζ a primitive 12-th root of unity and define $\zeta_3 := \zeta^{-4}$, a primitive cube root of unity. Then $S_1(133, \chi)$ is defined over $\mathbb{Q}(\zeta_3)$ and has dimension two. It contains two eigenforms defined over $\mathbb{Q}(\zeta)$, namely

$$h = q + (\zeta^2 - 1)q^2 - \zeta q^3 + \zeta q^5 + (-\zeta^3 + \zeta)q^6 + \zeta^3 q^7 - q^8 + (\zeta^3 - \zeta)q^{10} + (-\zeta^3 + \zeta)q^{13} - \dots$$

and its Galois conjugate obtained by replacing ζ by $-\zeta$. Let $g = h^*$, the twist of h by χ^{-1} (this has Fourier coefficients complex conjugate to those of h). The A_4 -extension H cut out by the projective representation attached to g is the normal closure of the quartic field

$$M := \mathbb{Q}(w), \quad w^4 + 3w^2 - 7w + 4 = 0,$$

which has discriminant 133^2 .

Consider the elliptic curves

$$\begin{aligned} E_a : y^2 + y &= x^3 + x \\ E_b : y^2 + y &= x^3 + x^2 - 7x + 5 \end{aligned}$$

of conductor $91 = 7 \cdot 13$ labelled $91a$ and $91b$ in Cremona's tables and write f_a and f_b for the associated weight 2 newforms. These curves both satisfy the conditions in the rank $(1, 1)$ setting of Conjecture ES, namely

$$r(E_a, 1) = 1, \quad r(E_a, \text{Ad}_g) = 1, \quad r(E_b, 1) = 1, \quad r(E_b, \text{Ad}_g) = 1.$$

Let $P_{\mathbb{Q}}^{(a)}$ and $P_{\mathbb{Q}}^{(b)}$ denote Mordell-Weil generators of $E_a(\mathbb{Q})$ and $E_b(\mathbb{Q})$ respectively (modulo torsion), and let $P^{(a)}$ and $P^{(b)}$ denote generators of the one-dimensional subspaces of $E_a(M)_{\mathbb{Q}}$ and $E_b(M)_{\mathbb{Q}}$ respectively, consisting of points with trace 0. A combination of naive search and two-descent shows that one can take

$$\begin{aligned} P_{\mathbb{Q}}^{(a)} &= (0, 0), \quad P^{(a)} = \left(\frac{-156w^3 - 64w^2 - 598w + 654}{289}, \frac{823w^3 + 1464w^2 + 5515w - 7663}{4913} \right), \\ P_{\mathbb{Q}}^{(b)} &= (-1, 3), \quad P^{(b)} = (9w^3 + 5w^2 + 31w - 45, -w^3 + 16w^2 + 16w + 83). \end{aligned}$$

Letting $p = 13$, the Hecke polynomial $t^2 - a_{13}(h)t + \chi(13)$ has roots $\beta_h = \zeta$ and $\alpha_h = -\zeta^3$. Hence the two 13-stabilisations g_α and g_β of g with U_p -eigenvalue $\alpha_g = \overline{\beta_h} = \zeta^{-1}$ and $\beta_g = \overline{\alpha_h} = \zeta^3$ are distinct. (Note we are twisting by χ^{-1} here and $\chi^{-1}(p)\beta_h = (\alpha_h\beta_h)^{-1}\beta_h = \overline{\alpha_h}$, and likewise $\chi^{-1}(p)\alpha_h = \overline{\beta_h}$.) Let γ_{g_α} and γ_{g_β} denote the corresponding dual elements.

Choose the embedding of $\mathbb{Q}(\zeta)$ into $\mathbb{Q}_{13}$ which sends ζ to the unique 12th root of unity in $\mathbb{Z}_{13}$ which is congruent to 6 modulo 13. The values of the p -adic iterated integrals, evaluated to 20 digits of 13-adic accuracy, are then given by the four quantities

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_a \cdot h &= -6603156733219298602528 \bmod 13^{20} \\ \int_{\gamma_{g_\beta}} f_a \cdot h &= 2705941994573919260213 \bmod 13^{20} \\ \int_{\gamma_{g_\alpha}} f_b \cdot h &= -679336918469999467881 \times 13 \bmod 13^{20} \\ \int_{\gamma_{g_\beta}} f_b \cdot h &= -2983964554989928769557 \bmod 13^{20}. \end{aligned}$$

The ratio $\alpha_g/\beta_g = \zeta^{-4} = \zeta_3$ is a primitive cube root of unity, and hence the eigenvalues of σ_{13} on Ad_g are 1, ζ_3 and ζ_3^2 . The fact that σ_{13} has order 3 puts us in *Case 3* in the analysis in Section 5.1.1. Let

$$P_{g_\alpha}^a, P_{g_\beta}^{(a)} \in E_a(H)_L^{\text{Ad}_g}, \quad P_{g_\alpha}^b, P_{g_\beta}^{(b)} \in E_b(H)_L^{\text{Ad}_g}$$

be the points constructed from $P^{(a)}$ and $P^{(b)}$ using (74). The units u_{g_α} and u_{g_β} are constructed using the same formula with the point P replaced by the unit $u_1 := -w + 1$ which is the root of $x^4 - 4x^3 + 9x^2 - 3x + 1 = 0$ in M and its Galois conjugates u_2, u_3 and u_4 .

It then turns out that to 20 digits of 13-adic accuracy,

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_a \cdot g &= -\zeta_3^2 \cdot \mathcal{E}_\alpha \cdot \frac{\log_{E_a,13}(P_{\mathbb{Q}}^{(a)}) \log_{E_a,13}(P_{g_\alpha}^{(a)})}{\log_{13}(u_{g_\alpha})}, \\ \int_{\gamma_{g_\beta}} f_a \cdot g &= -\zeta_3 \cdot \mathcal{E}_\beta \cdot \frac{\log_{E_a,13}(P_{\mathbb{Q}}^{(a)}) \log_{E_a,13}(P_{g_\beta}^{(a)})}{\log_{13}(u_{g_\beta})}, \end{aligned}$$

where $\mathcal{E}_\alpha$ is defined as in (77), and likewise for $\mathcal{E}_\beta$.

A similar pattern recurs with the elliptic curve E_b , for which it was checked to 20 digits of 13-adic precision that

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_b \cdot h &= \frac{\zeta}{3} \cdot \mathcal{E}_\alpha \cdot \frac{\log_{E_b,13}(P_{\mathbb{Q}}^{(b)}) \log_{E_b,13}(P_{g_\alpha}^{(b)})}{\log_{13}(u_{g_\alpha})}, \\ \int_{\gamma_{g_\beta}} f_b \cdot h &= -\frac{\zeta^{-1}}{3} \cdot \mathcal{E}_\beta \cdot \frac{\log_{E_b,13}(P_{\mathbb{Q}}^{(b)}) \log_{E_b,13}(P_{g_\beta}^{(b)})}{\log_{13}(u_{g_\beta})}, \end{aligned}$$

with $\mathcal{E}_\alpha$ as in (77), and likewise for $\mathcal{E}_\beta$. (Here $\zeta \cdot \mathcal{E}_\alpha/3 = \zeta^{-1} \cdot \mathcal{E}_\beta/3 = 4/(13\sqrt{3})$.)

The choice of the prime $p = 13$ in the above calculations was dictated largely by convenience. This prime has the virtue of splitting completely in $L = \mathbb{Q}(\zeta)$, so that L embeds into $\mathbb{Q}_{13}$, and it divides the level of f_a and f_b , which can thus be viewed as overconvergent 13-adic modular forms of tame level 133.

The weight one tetrahedral form g of level 133 was considered by Chinburg [Ch] in connection with Stark's original conjectures, motivated by the final sentence in [St2],

A meaningful verification for $N = 133$ would be of some interest.

Stark's concluding comment follows the verification of his conjecture for the weight one modular form considered in Example 4.1.

The next example involves the tetrahedral modular form of minimal level $124 = 4 \cdot 31$ [Bu, § 2.2] which is also considered in [Ch]. It illustrates our conjecture in both rank (1, 1) and rank 2 exotic settings.

Example 5.4. Let χ be a sextic Dirichlet character of conductor $124 = 4 \cdot 31$, having order 2 at 4 and order 3 at 31. Denote by ζ a primitive 12-th root of unity and define $\zeta_3 := \zeta^2 - 1$, a primitive cube root of unity. The space $S_1(124, \chi)$ is defined over $\mathbb{Q}(\zeta_3)$ and has dimension two. Just as in the previous example, it contains two eigenforms, which are defined over $\mathbb{Q}(\zeta)$ and conjugate over $\mathbb{Q}(\zeta_3)$, namely

$$h = q + \zeta^3 q^2 + \zeta q^3 - q^4 - \zeta^2 q^5 + (\zeta^2 - 1)q^6 - \zeta q^7 - \zeta^3 q^8 + (-\zeta^3 + \zeta)q^{10} + (\zeta^3 - \zeta)q^{11} - \zeta q^{12} + \zeta^2 q^{13} + \dots$$

and its conjugate obtained by replacing ζ by $-\zeta$. Let $g = h^*$. The field cut out by the projective representation attached to g is the normal closure H of the quartic field

$$M = \mathbb{Q}(w), \quad w^4 + 7w^2 - 2w + 14 = 0,$$

which has discriminant 124^2 and Galois group isomorphic to A_4 .

Consider the elliptic curve

$$E : y^2 = x^3 + x^2 - 2x + 1$$

of conductor 124 labelled 124a in Cremona's tables and write f for the associated weight 2 newform. The curve E has rank one over $\mathbb{Q}$ (with a torsion subgroup of order 3) and rank two over M —this places us in the rank (1, 1) setting. Following the same notations of the previous example, generators for $E(\mathbb{Q})$ and for the subgroup of points in $E(M)$ of trace zero are given by

$$P_{\mathbb{Q}} = (-2, 1), \quad P = \left(\frac{w^3 - w^2 + 3w - 7}{2}, \frac{w^3 + 4w^2 + 9w + 10}{4} \right).$$

These points were found using a naive search.

Choosing the prime $p = 13$ as in the previous example, the Hecke polynomial $t^2 - a_{13}(h)t + \chi(13)$ has roots $\beta_h = 1$ and $\alpha_h = \zeta_3$. Hence $\alpha_g = 1$ and $\beta_g = \zeta_3^2$ and therefore the 13-stabilisations g_{α} and g_{β} of g are distinct.

Choose the embedding of $\mathbb{Q}(\zeta)$ into $\mathbb{Q}_{13}$ which sends ζ to the unique 12-th root of unity in $\mathbb{Z}_{13}$ which is congruent to 7 modulo 13. The values of the 13-adic iterated integrals were computed to be

$$\begin{aligned} \int_{\gamma_{g_{\alpha}}} f \cdot h &= 684158979097768450057 \times 13 \bmod 13^{20} \\ \int_{\gamma_{g_{\beta}}} f \cdot h &= 5953034774844704381281 \bmod 13^{20}. \end{aligned}$$

Note that α_g/β_g and β_g/α_g are both primitive cube roots of unity, and hence, just as in the previous example, the setting is the one discussed in *Case 3* in the analysis in Section 5.1.1. Just as before, the points $P_{g_{\alpha}}$ and $P_{g_{\beta}}$ in $E(H)_L^{\text{Ad}_g}$ can be defined as suitable linear combinations of the point P and its Galois conjugates, following the formula in (74). The units $u_{g_{\alpha}}$ and $u_{g_{\beta}}$ are likewise constructed using the same formula with the point P replaced by the unit $u_1 := w^2 - w + 3$, the unique root of $x^4 + 2x^3 + 18x^2 - 6x + 1$ in M .

With these conventions in place, it has been checked to 20 digits of 13-adic precision that

$$\begin{aligned} \int_{\gamma_{g_{\alpha}}} f \cdot h &= \frac{2}{\sqrt{3}} \cdot \mathcal{E}_{\alpha} \cdot \frac{\log_{E,13}(P_{\mathbb{Q}}) \log_{E,13}(P_{g_{\alpha}})}{\log_{13}(u_{g_{\alpha}})}, \\ \int_{\gamma_{g_{\beta}}} f \cdot h &= \frac{2}{\sqrt{3}} \cdot \mathcal{E}_{\beta} \cdot \frac{\log_{E,13}(P_{\mathbb{Q}}) \log_{E,13}(P_{g_{\beta}})}{\log_{13}(u_{g_{\beta}})}, \end{aligned}$$

where $\mathcal{E}_{\alpha}$ is as in (77), and likewise for $\mathcal{E}_{\beta}$, with $\sqrt{3} = \zeta^3 - 2\zeta$.

We now turn to the curve

$$E : y^2 + xy + y = x^3 - x^2 - 3x + 3$$

of conductor 26 labelled 26b. One checks

$$r(E) = 0, \quad r(E, \text{Ad}_g) = 2, \tag{78}$$

and likewise for analytic ranks, and so here we are in the rank 2 setting. The required points in $E(H)^{\text{Ad}_g}$ may be constructed from the generators

$$\begin{aligned} P &= \left(\frac{283w^3 + 1295w^2 + 1442w + 9528}{4802}, \frac{29371w^3 - 35875w^2 + 146552w - 319814}{235298} \right) \\ Q &= \left(\frac{-25w^3 + 98w^2 + 41w + 775}{121}, \frac{67w^3 + 3832w^2 + 2465w + 17404}{1331} \right) \end{aligned}$$

of the rank 2 Mordell-Weil lattice in $E(M)$. We define P_1 and $P_{g_{\alpha}}$ as in the Case 3 analysis for tetrahedral forms (and similarly $P_{g_{\beta}}$ switching ζ_3 for ζ_3^{-1}). Likewise we define the points Q_1 , $Q_{g_{\alpha}}$ and $Q_{g_{\beta}}$.

Writing now f for the form attached to E we find, with the same embedding as before, that

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f \cdot h &= 6721481652857547558786 \bmod 13^{20} \\ \int_{\gamma_{g_\beta}} f \cdot h &= -2549577176522983012568 \bmod 13^{20}. \end{aligned}$$

Then to 20-digits of 13-adic precision

$$\int_{\gamma_{g_\alpha}} f \cdot h = \frac{\zeta_3 \cdot \mathcal{E}_\alpha}{\sqrt{3}} \cdot \frac{\log_{E,13}(P_1) \log_{E,13}(Q_{g_\alpha}) - \log_{E,13}(Q_1) \log_{E,13}(P_{g_\alpha})}{\log_p u_{g_\alpha}} \quad (79)$$

$$\int_{\gamma_{g_\beta}} f \cdot h = -\frac{\zeta_3^{-1} \cdot \mathcal{E}_\beta}{\sqrt{3}} \cdot \frac{\log_{E,13}(P_1) \log_{E,13}(Q_{g_\beta}) - \log_{E,13}(Q_1) \log_{E,13}(P_{g_\beta})}{\log_p u_{g_\beta}} \quad (80)$$

where $\mathcal{E}_\alpha$ is as in (77), and likewise for $\mathcal{E}_\beta$, exactly as predicted by Conjecture ES.

Finally we consider the curve

$$E : y^2 = x^3 + x - 10$$

of conductor 52 labelled 52b for which again (78) holds and so we are in the rank 2 setting. Here we take

$$\begin{aligned} P &= \left(\frac{-225w^3 - 1533w^2 - 1458w + 1078}{2809}, \frac{-247041w^3 + 68295w^2 - 1265094w + 317530}{297754} \right) \\ Q &= \left(\frac{-63391w^3 - 51546w^2 - 698338w - 942346}{390625}, \frac{84293978w^3 - 33352782w^2 + 603648754w - 1348508882}{244140625} \right) \end{aligned}$$

as generators for the Mordell-Weil lattice in $E(M)$ and construct the points $P_1, P_{g_\alpha}, P_{g_\beta}$ and $Q_1, Q_{g_\alpha}, Q_{g_\beta}$ from them. Letting f now be the newform attached to E we have

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f \cdot h &= -1602512412162131808818 \bmod 13^{20} \\ \int_{\gamma_{g_\beta}} f \cdot h &= 4989460414989865668627 \bmod 13^{20} \end{aligned}$$

and one checks to 20-digits of 13-adic precision that again (79) and (80) hold. (The algebraic constants which occur in equation (79) for 26b and 52a are in fact exactly the same, and likewise for (80): both curves have non-split multiplicative reduction at 13.)

The following example illustrates the vanishing of p -adic iterated integrals predicted by Conjecture ES in a rank 2 setting but where the regulator term vanishes for trivial reasons.

Example 5.5. Consider the curve

$$E : y^2 + xy = x^3 - 18x + 31$$

of conductor $4811 = 17 \cdot 283$ labelled 4811a in Cremona's table and write f for the associated weight 2 newform. Letting h be the octahedral weight one form in level 283 with quadratic character and $g = h^*$, it is readily checked that

$$r(E/\mathbb{Q}) = r_{\text{an}}(E/\mathbb{Q}) = 2, \quad r_{\text{an}}(E, \text{Ad}_g) = 0.$$

(The third equality was obtained by calculating $L(E, \text{Ad}_g, 1)$ numerically to a modest precision sufficient to verify its non-vanishing.) Taking p a regular prime for ρ_g , it is clear that hypotheses A,B,C are satisfied but that the matrix $R_{g_\alpha}(E, \varrho_{gh})$ of Conjecture ES has determinant 0. The prediction tht all associated p -adic iterated integrals should vanish was verified for $p = 17$ to 5 digits of 17-adic accuracy.

We note also here that we tested the vanishing prediction of Conjecture ES in several examples when $r(E, \varrho_{gh}) = 4$ and 6 (involving CM rather than exotic forms).

Our final exotic example involves the exotic octahedral modular form of minimal level $148 = 4 \cdot 37$ [Bu, § 2.3] and rank (1, 1) settings of Conjecture ES.

Example 5.6. Let χ be the Dirichlet character of level $148 = 2^2 \cdot 37$ which is trivial at 2 and has order 4 at 37 (there are two such characters and they are Galois conjugate). The space $S_1(148, \chi)$ is defined over $L = \mathbb{Q}(i)$ and has dimension 1. It contains the single eigenform

$$h = q - iq^3 - q^7 + iq^{11} + (i-1)q^{17} + (-i+1)q^{19} + \dots$$

The field H cut out by the projective representation attached to $g = h^*$ is the normal closure of the quartic field

$$M := \mathbb{Q}(a), \quad a^4 - a^3 + 5a^2 - 7a + 12$$

of discriminant $2^2 \cdot 37^3$ and having Galois group S_4 . The unique quadratic subfield of H is $K = \mathbb{Q}(\sqrt{37})$, and hence $\tilde{M} = MK$ is the unique quadratic extension of M contained in H .

Testing Conjecture ES in the rank (1, 1) setting requires elliptic curves E of rank 1 over $\mathbb{Q}$ for which in addition $r(E, \text{Ad}_g) = 1$. We choose $p = 17$ and consider elliptic curves E of rank one and conductor dividing $148p$ which satisfy the necessary hypotheses. (Restricting the conductor of E in this way makes it possible to confine the 17-adic iterated integral computations to spaces of overconvergent p -adic modular forms tame level 148.) Analytic rank computations suggest these are the curves $37a$, $629a$, $629c$ and $629d$ in Cremona's tables.

Following the general notations introduced in Section 5.1.2, Conjecture ES was verified for each of $E = 37a$, $629a$ and $629d$ using points $P \in E(\tilde{M})^-$ found by Steve Donnelly. (For $629c$ the authors were unable to find the point P . This leaves open possibility of constructing it using Conjecture ES, a challenge that has not yet been addressed.)

The elliptic curves of interest are

$$\begin{aligned} E_{37a} : \quad & y^2 + y = x^3 - x \\ E_{629a} : \quad & y^2 + xy = x^3 - x^2 + 11x - 18 \\ E_{629d} : \quad & y^2 + xy + y = x^3 - x^2 - 171x + 1904 \end{aligned}$$

with associated modular forms f_{37a} , f_{629a} and f_{629d} .

With $p = 17$ the Hecke eigenvalues for the octahedral form h are $\beta_h = i$ and $\alpha_h = -1$, and therefore $\alpha_g = -i$ and $\beta_g = -1$. It follows that the Frobenius eigenvalues at 17 for the adjoint representation Ad_g are $1, \alpha/\beta = i$ and $\beta/\alpha = -i$, which places this experiment in *Case 5* of the analysis in Section 5.1.2.

Choose the embedding of $L = \mathbb{Q}(i)$ into $\mathbb{Q}_{17}$ which sends i to the unique square root of -1 which is congruent to 4 modulo 17. A calculation of iterated integrals to 15-digits of 17-adic precision reveals that

$$\begin{aligned} \int_{\gamma_{g\alpha}} f_{37a} \cdot h &= 795182232610993468 \bmod 17^{15} \\ \int_{\gamma_{g\beta}} f_{37a} \cdot h &= 591309544450892129 \bmod 17^{15} \\ \int_{\gamma_{g\alpha}} f_{629a} \cdot h &= 231921888434186915 \bmod 17^{15} \\ \int_{\gamma_{g\beta}} f_{629a} \cdot h &= 776162921415439615 \bmod 17^{15} \\ \int_{\gamma_{g\alpha}} f_{629d} \cdot h &= -612176557243786922 \times 17 \bmod 17^{15} \\ \int_{\gamma_{g\beta}} f_{629d} \cdot h &= 329220710071055453 \bmod 17^{15}. \end{aligned}$$

Consider first the elliptic curve $E = E_{37a}$. The generator of $E(\mathbb{Q})$ is chosen to be $P_{\mathbb{Q}} = (0, 0)$. The point

$$\left(\frac{-152551a^3 + 327117a^2 - 384541a + 1037221}{21316}, \frac{-110958819a^3 + 69702635a^2 + 95157821a + 314525497}{3112136} \right)$$

on the quadratic twist $y^2 + y = x^3 - 1369x + 12663$ of E by 37 gives rise to the point

$$\begin{aligned} P = \left(\frac{-10817b^7 - 15535b^6 - 11412b^5 - 43863b^4 - 97162b^3 + 30857b^2 + 39063b - 42}{21316}, \right. \\ \left. \frac{2553946b^7 - 379967b^6 - 1189613b^5 + 10482748b^4 + 12706439b^3 - 35039416b^2 + 19176309b + 6012477}{15560680} \right) \end{aligned}$$

in $E(\tilde{M})^-$, where

$$\tilde{M} = \mathbb{Q}(b), \quad b^8 + b^7 + b^6 + 4b^5 + 8b^4 - 4b^3 + b^2 - b + 1 = 0.$$

Writing P_1, P_2, P_3 and P_4 for the Galois translates of P so that they are permuted cyclically by σ_{17} , define

$$P_{g_\alpha} = P_1 - iP_2 - P_3 + iP_4, \quad P_{g_\beta} = P_1 + iP_2 - P_3 - iP_4,$$

following (76) and (75). Likewise, taking $u := -b^2$ as one of the two roots of $x^8 - x^7 + 9x^6 - 10x^5 + 102x^4 - 10x^3 + 9x^2 - x + 1$ in $\tilde{M}$ leads to a definition of the Stark units u_{g_α} and $u_{g_\beta} \in (\mathcal{O}_H^\times)_L^{\text{Ad}_g}$ attached to the two stabilisations of g .

The following identities were verified to 15-digits of 17-adic precision:

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_{37a} \cdot h &= \frac{4}{3} \cdot \mathcal{E}_\alpha \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\alpha})}{\log_{17}(u_{g_\alpha})}, \\ \int_{\gamma_{g_\beta}} f_{37a} \cdot h &= \frac{4}{3} \cdot \mathcal{E}_\beta \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\beta})}{\log_{17}(u_{g_\beta})}, \end{aligned}$$

with $\mathcal{E}_\alpha$ as in (77), and likewise for $\mathcal{E}_\beta$.

Consider next the elliptic curve $E = E_{629a}$. Here the generator for $E(\mathbb{Q})$ is chosen to be $P_\mathbb{Q} = (2, 2)$, and, just as above, the point

$$P = \left(\frac{-5b^7 - 11b^6 - 16b^5 - 31b^4 - 74b^3 - 47b^2 - 25b + 57}{36}, \frac{1083b^7 + 1784b^6 + 2366b^5 + 5569b^4 + 11572b^3 + 3647b^2 + 1907b - 3464}{1080} \right)$$

belongs to $E(\tilde{M})^-$. The points P_{g_α} and P_{g_β} , and the units u_{g_α} and u_{g_β} , are defined just as they were before. It was then checked to 15-digits of 17-adic precision that

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_{629a} \cdot h &= \frac{-2i - 2}{3} \cdot \mathcal{E}_\alpha \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\alpha})}{\log_{17}(u_{g_\alpha})}, \\ \int_{\gamma_{g_\beta}} f_{629a} \cdot h &= \frac{2i - 2}{3} \cdot \mathcal{E}_\beta \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\beta})}{\log_{17}(u_{g_\beta})}, \end{aligned}$$

with $\mathcal{E}_\alpha$ as in (77), and likewise for $\mathcal{E}_\beta$.

Finally, for the curve $E = E_{629d}$, let $P_\mathbb{Q} = (16, 47)$ be the Mordell-Weil generator for $E(\mathbb{Q})$ and let

$$P = \left(\frac{-1369b^7 + 259b^6 - 518b^5 - 4625b^4 - 5994b^3 + 12987b^2 - 17871b + 9656}{400}, \frac{-1258148b^7 + 64491b^6 - 649461b^5 - 5104594b^4 - 5377617b^3 + 14354298b^2 - 11310937b + 2516289}{40000} \right).$$

be the generator of $E(\tilde{M})^-$.

With the same definitions for the points P_{g_α} and P_{g_β} in terms of P as before, the following remarkably simple relation was checked to 15 digits of 17-adic precision:

$$\begin{aligned} \int_{\gamma_{g_\alpha}} f_{629d} \cdot h &= (-10i + 10) \cdot \mathcal{E}_\alpha \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\alpha})}{\log_{17}(u_{g_\alpha})}, \\ \int_{\gamma_{g_\beta}} f_{629d} \cdot h &= (-10i - 10) \cdot \mathcal{E}_\beta \cdot \frac{\log_{E,17}(P_\mathbb{Q}) \log_{E,17}(P_{g_\beta})}{\log_{17}(u_{g_\beta})}, \end{aligned}$$

with $\mathcal{E}_\alpha$ as in (77), and likewise for $\mathcal{E}_\beta$. Here $(-10i + 10) \cdot \mathcal{E}_\alpha = 160/17$ and $(-10i - 10) \cdot \mathcal{E}_\beta = -160/17$.

6. STARK POINTS ATTACHED TO ODD TWO-DIMENSIONAL ARTIN REPRESENTATIONS

This chapter considers the case where $g \in S_1(N, \bar{\chi})$ is a cuspidal newform and $h = E(1, \chi)$ is the Eisenstein series attached to the pair $(1, \chi)$. In that case,

$$V_{gh} = V_g \oplus V_{g^*}, \quad L(E, \varrho_{gh}, s) = L(E, \varrho_g, s)L(E, \bar{\varrho}_g, s).$$

The factorisation of L -functions guarantees that Hypothesis A of the introduction is automatically satisfied in this setting but Hypothesis B is only satisfied when $L(E, \varrho_g, 1) = 0$. Furthermore, the conditions formulated in Conjecture ES are fulfilled precisely when $L(E, \varrho_{gh}, s)$ has a *double zero* at $s = 1$, so that

$$r_{\text{an}}(E, V_g) = r_{\text{an}}(E, \bar{V}_g) = 1.$$

The eigenvalues of σ_p acting on V_g and on V_h respectively are given by

$$\alpha_g, \quad \beta_g, \quad \alpha_h = 1, \quad \beta_h = \chi(p) = (\alpha_g \beta_g)^{-1},$$

and therefore

$$\alpha_g \alpha_h = \alpha_g, \quad \alpha_g \beta_h = \bar{\beta}_g.$$

Let H be the Galois extension of $\mathbb{Q}$ through which ϱ_g (or, equivalently, $\bar{\varrho}_g$) factors. Assuming that V_g is not isomorphic to its contragredient, its underlying field L of coefficients is a CM subfield of a cyclotomic field, and the spaces $E(H)_L^{V_g}$ and $E(H)_L^{V_{g^*}}$ decompose as σ_p -modules as

$$E(H)_L^{V_g} = L \cdot P_{g,\alpha} \oplus L \cdot P_{g,\beta}, \quad E(H)_L^{\bar{V}_g} = L \cdot P_{g^*,\bar{\alpha}} \oplus L \cdot P_{g^*,\bar{\beta}},$$

where $P_{g,\alpha}$ is an eigenvector for σ_p with eigenvalue α_g , and similar conventions are adopted for the other points.

When g is regular at p and (12) is satisfied—and so Hypothesis C holds by Proposition 1.2—Conjecture ES of the introduction predicts that, for a suitable choice of test vectors

$$(\check{f}, \check{\gamma}_{g_\alpha}, h) \in S_2(N)_L \times S_1(N, \chi)_L^\vee[g_\alpha] \times M_1(N, \chi)_L,$$

(and likewise with α_g replaced by β_g) the following identities hold

$$\begin{aligned} \int_{\check{\gamma}_{g_\alpha}} \check{f} \cdot \check{h} &= \frac{\log_{E,p}(P_{g,\alpha}) \cdot \log_{E,p}(P_{g^*,\bar{\beta}})}{\log_p(u_{g_\alpha})}, \\ \int_{\check{\gamma}_{g_\beta}} \check{f} \cdot \check{h} &= \frac{\log_{E,p}(P_{g,\beta}) \cdot \log_{E,p}(P_{g^*,\bar{\alpha}})}{\log_p(u_{g_\beta})}, \end{aligned}$$

where u_{g_α} is precisely the same Stark unit as arose in Chapter 5.

The identities above turned out to be harder to check experimentally, because

- instances where $L(E, \varrho_g, s)$ acquires a simple zero at $s = 1$ are ostensibly sporadic in nature. When ϱ_g is not isomorphic to its contragredient representation, this L -function does not admit a sign in its functional equation which might be used to systematically predict the vanishing of $L(E, \varrho_g, 1)$, and it is expected that cases where $L(E, \varrho_g, 1) = 0$ are exceedingly rare (although very little data has been gathered to be able to make serious pronouncements in this direction);
- the field cut out by ϱ_g is larger than the field cut out by its adjoint Ad_g . More precisely, it is an abelian extension of it, of degree $t \geq 2$, where t is the order of the non-trivial central character attached to ϱ_g . In particular, the point $P_{g,\alpha}$ that arises in this section is entirely independent of the point denoted P_{g_α} in Chapter 5, is defined over a larger degree extension, and is therefore harder to produce numerically in practice, in addition to not arising in as systematic a manner.

The problem of producing numerical evidence becomes more tractable when $g = g^*$, which occurs precisely when g is induced from a ring class character of an imaginary quadratic field. Such examples are arguably less interesting since they also fall under the scope of Chapter 3. See the second half of Example 3.11 of Section 3.3 for a numerical verification of Conjecture ES in the case where g is further assumed to be self-dual.

7. STARK POINTS OVER CYCLOTOMIC FIELDS AND EISENSTEIN SERIES

7.1. Theoretical preliminaries. Consider the case where g and h are weight one Eisenstein series of the form

$$g = E_1(1, \chi^{-1}), \quad h = E_1(\chi_1, \chi_2), \quad \text{where } \chi_1 \chi_2 = \chi,$$

and where at least one of χ_1 or χ_2 is not quadratic, to exclude the case (already treated in Chapter 3) where both g and h are induced from genus characters of the imaginary quadratic field attached to the (quadratic) Dirichlet character χ . Let L be the field in which the characters χ_1 and χ_2 take values. The representations V_g , V_h , and V_{gh} decompose as direct sums of one-dimensional representations:

$$V_g = L \oplus L(\chi^{-1}), \quad V_h = L(\chi_1) \oplus L(\chi_2), \quad V_{gh} = L(\chi_1) \oplus L(\chi_2) \oplus L(\bar{\chi}_1) \oplus L(\bar{\chi}_2).$$

It is worth noting that the representation V_{gh} is isomorphic to its dual, but that its irreducible constituents are not all self-dual when χ_1 or χ_2 is non-quadratic. This phenomenon, which is specific to the case where at least one of the forms g or h is an Eisenstein series, makes the occurrence of situations where L -functions vanish somewhat more rare and sporadic, just as in the setting described in Chapter 6.

The adjoint representation Ad_g attached to V_g breaks up as a direct sum

$$\text{Ad}_g = L \oplus L(\chi) \oplus L(\chi^{-1}).$$

Let H_χ be the abelian extension of $\mathbb{Q}$ cut out by the character χ . In order for the p -adic iterated integral attached to (f, g, h) where f is a cusp form to have the possibility of not vanishing, it is necessary to make the assumption

$$\chi(p) = 1, \text{ i.e., } V_g \text{ is not regular and } \chi_1(p) = \chi_2(p)^{-1}.$$

Since χ is non-quadratic the L -vector spaces

$$(\mathcal{O}_{H_\chi}[1/p]^\times)_L^\chi, \quad (\mathcal{O}_{H_\chi}[1/p]^\times)_L^{\chi^{-1}}, \quad (\mathcal{O}_{H_\chi}[1/p]^\times)_L^{\chi+\chi^{-1}}$$

have dimensions 1, 1 and 2 respectively. Let u_{g_α} be the element of $(\mathcal{O}_{H_\chi}[1/p]^\times)_L^{\chi+\chi^{-1}}$ associated to g_α following the recipe of Section 1.2.

Let f be a weight two cusp form attached to an elliptic curve. Note that Hypothesis A is automatically satisfied in this setting, and (empirically) Hypothesis C holds as $\chi(p) = 1$. When $r_{\text{an}}(E, \chi_1 \oplus \chi_2) > 0$ (and so Hypothesis B holds) Conjecture ES predicts that the iterated integrals attached to (f, g, h) is non-vanishing if and only if $r_{\text{an}}(E, \chi_1 \oplus \chi_2) = 1$. Assume (after possibly re-ordering χ_1 and χ_2) that

$$r_{\text{an}}(E, \chi_1) = 1, \quad r_{\text{an}}(E, \chi_2) = 0.$$

Under this assumption, Conjecture ES predicts the existence of points $P_{\chi_1} \in E(H_{\chi_1})_L^{\chi_1}$ and $P_{\bar{\chi}_1} \in E(H_{\chi_1})_L^{\bar{\chi}_1}$ for which

$$\int_\gamma f \cdot h \stackrel{?}{=} \frac{\log_{E,p}(P_{\chi_1}) \log_{E,p}(P_{\bar{\chi}_1})}{\log_p(u_{g_\alpha})}.$$

A striking feature of this conjecture is that it involves global points over abelian extensions of $\mathbb{Q}$, whose occurrence seems sporadic, in contrast with the self-dual setting that is germane to the theory of Heegner and Stark-Heegner points.

7.2. Numerical evidence. Although Eisenstein series are plentiful, it is the very scarcity of points over odd degree abelian extensions, and the necessity of working at a prime p which splits completely in this abelian extension, which makes the numerical verification of Conjecture ES so challenging in this situation. Frequently, when such a point does exist, the common level of the Eisenstein series and cusp form f , and size of the prime p , pushes the example beyond the current reach of experimentation.

The authors were rather fortunate to encounter the following two examples (and in addition another cubic field example in which $r_{\text{an}}(E, \chi_1 \oplus \chi_2) = 2$ and the integral vanished, as predicted by Conjecture ES). In the first one we give for variety numerical details of the logarithms computed — these were suppressed elsewhere to lighten the burden of unilluminating long numbers in the paper.

Example 7.1. Let

$$E : y^2 + xy + y = x^3 + x^2 - 4x + 5$$

be the elliptic curve of conductor $2 \cdot 3 \cdot 7$ denoted 42a in the tables of Cremona. Let χ_1 be the (even) cubic character of conductor 19, and let χ_2 be the (odd) quadratic character of conductor 3, so that $\chi = \chi_1 \chi_2$ is an odd sextic character of conductor 57. As before, write $L = \mathbb{Q}(\zeta_3)$ for the field of coefficients generated by the values of χ_1 . A direct calculation reveals that

$$E(\mathbb{Q})_L = 0, \quad r(E, \chi_1) = r(E, \bar{\chi}_1) = 1, \quad r(E, \chi_2) = r(E, \bar{\chi}_2) = 0,$$

and likewise for the analytic ranks. More precisely, the cyclic cubic field cut out by χ_1 is given concretely by

$$H_{\chi_1} = \mathbb{Q}(\alpha), \quad \text{where } \alpha \text{ satisfies } \alpha^3 - \alpha^2 - 6\alpha + 7 = 0.$$

The Mordell-Weil group $E(H_{\chi_1})$ has rank two, and is generated as a $\text{Gal}(H_{\chi_1}/\mathbb{Q})$ -module by the point

$$P := (2\alpha^2 + 3\alpha - 8, 4\alpha^2 + 6\alpha - 10)$$

with coordinates in H_{χ_1} .

Fix the prime $p = 7$ and let ζ_3 be the unique cube root of unity in $\mathbb{Q}_7$ which is congruent to 2 modulo 7. Fix a primitive 6th root of unity $\zeta_6 = -\bar{\zeta}_3$, so $\zeta_3 = \zeta_6^2$. After letting σ be a generator of the Galois group $\text{Gal}(H_{\chi_1}/\mathbb{Q})$, let

$$\begin{aligned} P_{\chi_1} &= P + \bar{\chi}_1(\sigma)P^\sigma + \bar{\chi}_1(\sigma^2)P^{\sigma^2}, \\ P_{\bar{\chi}_1} &= P + \chi_1(\sigma)P^\sigma + \chi_1(\sigma^2)P^{\sigma^2}. \end{aligned}$$

A numerical evaluation of the p -adic logarithms shows that

$$\begin{aligned} \log_{E,7}(P_{\chi_1}) &= -199213220217977173722637405653124 \times 7 \bmod 7^{38}, \\ \log_{E,7}(P_{\bar{\chi}_1}) &= -61436332416679587407622380051451 \times 7^2 \bmod 7^{38}. \end{aligned} \tag{81}$$

Turning now to the Gross-Stark unit attached to g , the defining polynomial of the sextic field H_χ is chosen to be $x^6 + 2x^5 - 2x^4 - 14x^3 + 55x^2 + 19x + 343$, whose roots generate the group of 7-units in H_χ . Let u be the unique 7-adic root of this polynomial satisfying $\text{ord}_7(u) = 3$, and set

$$\begin{aligned} u_\chi &:= \sum_{\sigma \in \text{Gal}(H_\chi/\mathbb{Q})} \bar{\chi}(\sigma) \otimes u^\sigma, \\ u_{\bar{\chi}} &:= \sum_{\sigma \in \text{Gal}(H_\chi/\mathbb{Q})} \chi(\sigma) \otimes u^\sigma. \end{aligned}$$

An evaluation of the 7-adic logarithms of these units yields

$$\begin{aligned} \log_7(u_\chi) &= 793452521874901120340916121448 \times 7 \bmod 7^{38}, \\ \log_7(u_{\bar{\chi}}) &= 2536773629823487601064250408052 \times 7 \bmod 7^{38}, \end{aligned}$$

and

$$\log_7(u_\chi) + \log_7(u_{\bar{\chi}}) = 475746593099769817343595218500 \times 7^2 \bmod 7^{38}. \quad (82)$$

Let

$$g := E_1(1, \chi^{-1}), \quad h := E_1(\chi_1, \chi_2)$$

and take the stabilisation

$$\check{g} := (g(q) - g(q^7)) - (-\zeta_6^{-1} + 1)(g(q^2) - g(q^{14}))$$

in (classical) level $2 \cdot 7 \cdot 57$ with U_2 -eigenvalue 1. These are also 7-adic modular forms of level $114 = 2 \cdot 57$. Letting $\check{\gamma}$ be the element in the dual space which sends $\check{g}$ to 1 and the other stabilisation of $E_1(1, \chi^{-1})$ to zero, a computation of the p -adic iterated integral (by far the most resource intensive part of the experiment) reveals that

$$\int_{\check{\gamma}} f \cdot h = -1264003828062411821439581214808 \bmod 7^{36}. \quad (83)$$

A direct comparison of (83) with the invariants obtained in (81) and (82) shows

$$\int_{\check{\gamma}} f \cdot h = \frac{64}{7 \cdot 9} \frac{\log_{E,7}(P_{\chi_1}) \log_{E,7}(P_{\bar{\chi}_1})}{(\log_7(u_\chi) + \log_7(u_{\bar{\chi}}))} \bmod 7^{36}.$$

Replacing $\check{g}$ by the other stabilisation of $E_1(1, \chi^{-1})$, with U_2 -eigenvalue $-\zeta_6^{-1} + 1$, and defining $\check{\gamma}$ accordingly just multiplies the integral value by ζ_3^2 ; and likewise, taking $\check{g}$ to be a stabilisation of $E_1(\chi_1^{-1}, \chi_2^{-1})$ and $h := E_1(1, \chi)$ in the integral changes its value by at most a cube root of unity. This is as predicted by Conjecture ES.

Example 7.2. Consider the case where

$$E : y^2 + xy + y = x^3 - 5x - 8$$

is the elliptic curve of conductor $2 \cdot 13$ denoted $26a$ in the tables of Cremona. Write f for the newform of weight 2 attached to E . Let η_1 be an (even) cubic character of conductor 7, and let η_2 be the (odd) quadratic character of conductor 3. Define $\chi_1 := \eta_1 \eta_2$, an odd sextic character of conductor 21, $\chi_2 := 1$, and $\chi := \chi_1 \chi_2 = \chi_1$. (This rather roundabout notation is chosen just to align with that in Section 7.1.) Letting $g := E_1(1, \chi^{-1})$ and $h := E_1(\chi_1, \chi_2) = E_1(\chi, 1) = E_1(1, \chi)$, the representation V_{gh} is given by

$$V_{gh} = L \oplus L \oplus L(\chi) \oplus L(\bar{\chi})$$

where $L = \mathbb{Q}(\zeta_3)$ is the field generated by the values of χ . A descent calculation reveals that

$$E(\mathbb{Q})_L = 0, \quad r(E, \chi) = r(E, \bar{\chi}) = 1$$

and likewise for the analytic ranks. The cyclic sextic field cut out by χ is given by

$$H_\chi = \mathbb{Q}(\alpha), \quad \text{where } \alpha \text{ satisfies } \alpha^6 - \alpha^5 + 3\alpha^4 + 5\alpha^2 - 2\alpha + 1 = 0.$$

The Mordell-Weil group $E(H_\chi)$ has rank two, and is generated as a $\text{Gal}(H_\chi/\mathbb{Q})$ -module by the point

$$P := (2\alpha^5 - 6\alpha^4 + 8\alpha^3 - 10\alpha^2 + 4\alpha - 23, -50\alpha^5 + 38\alpha^4 - 134\alpha^3 - 36\alpha^2 - 242\alpha + 50)$$

with coordinates in H_χ . (This point is the image of one on the quadratic twist of E by -3 over the cyclic cubic field H_{η_1} cut out by η_1 .) The points P_χ and $P_{\bar{\chi}}$, and the units u_χ and $u_{\bar{\chi}}$, are now constructed in an analogous manner to Example 7.1.

We take $p = 13$. Noting here $a_2(g) = 1 + \zeta_6^{-1}$ with ζ_6 a primitive sixth root of unity, we replace g by its stabilisation

$$\check{g} := (g(q) - g(q^{13})) - (g(q^2) - g(q^{26}))$$

in level $2 \cdot 13 \cdot 21$, and let $\check{\gamma}$ be the element in the dual space which sends $\check{g}$ to 1 and the other stabilisation to zero. Embedding $\mathbb{Q}(\zeta_6)$ in $\mathbb{Q}_{13}$ so that $\zeta_6 \equiv 4 \pmod{13}$ we find

$$\int_{\check{\gamma}} f \cdot h = -1183909055963560023118938339892601 \times 13 \pmod{13^{30}}.$$

It was then checked to 30 digits of 13-adic precision that

$$\int_{\check{\gamma}} f \cdot h = \frac{4}{13 \cdot \zeta_6} \frac{\log_{E,13}(P_{\chi}) \log_{E,13}(P_{\bar{\chi}})}{(\log_{13}(u_{\chi}) + \log_{13}(u_{\bar{\chi}}))}$$

as predicted by Conjecture ES.

Remark 7.3. The experimental work in this paper fell into two phases—one of discovery and one of verification—between which Conjecture ES was gradually formulated and refined. Formula (44) and Examples 3.10, 4.1 and 7.1 belong to the former phase, and the remaining examples to the latter.

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